Vincent Liu
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WiseMotif Monograph of Mathematical Intelligence
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COMPLETE FIVE-VOLUME MONOGRAPH · VOLUMES I–V

The Archimedean Prism

Tame Period Geometry and the Hodge Conjecture

RESEARCH ARCHITECTURE EDITION 1

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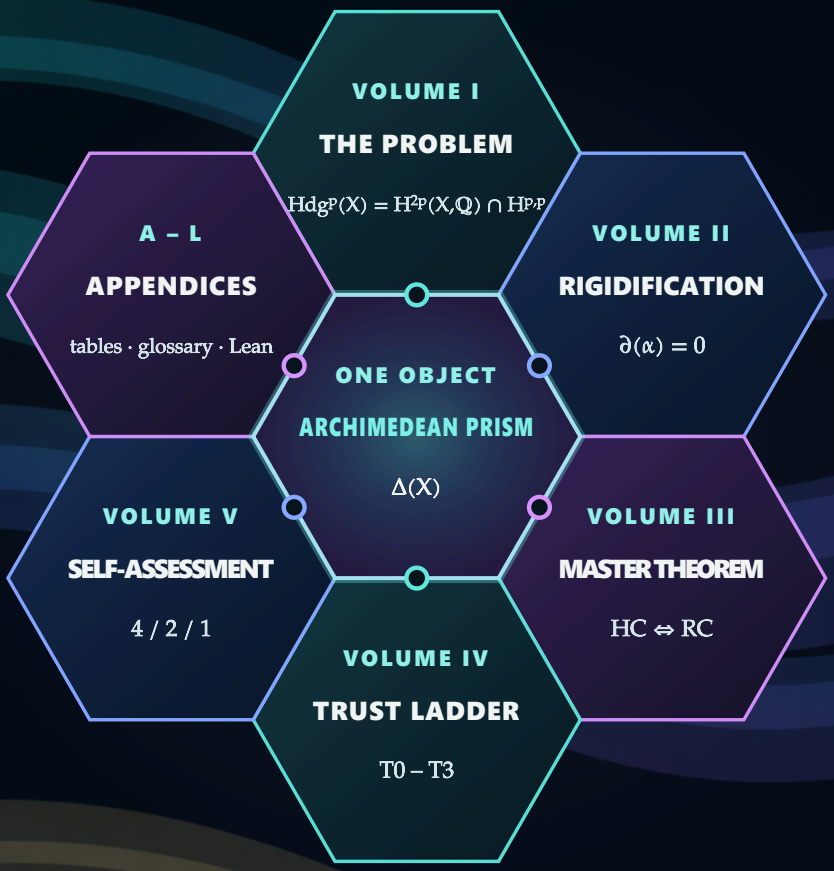

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The Archimedean Prism

The Complete Monograph

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Tame Period Geometry and the Hodge Conjecture

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The Hodge conjecture remains open. *The Hodge Conjecture is, at the time of writing, an open problem. This book does* not *claim a proof of it.* It constructs a framework, isolates the obstructions to a proof, proves what can be proved unconditionally, states precisely what is conjectural, and specifies what would have to be established to close the remaining gap. The reader is directed to *On the Epistemic Status of This Book*, immediately following the preface, before reading any mathematical claim in this volume. Every non-trivial assertion carries an explicit epistemic tag: **[KNOWN]**, **[CONDITIONAL]**, **[NEW]**, or **[OPEN]**.

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To those who have looked at a period matrix
and refused to believe it was an accident.

" The obstruction is not that we cannot see the cycle.
It is that we cannot see why seeing it should suffice. "

Series Note

The Archimedean Prism is the fifth work in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme. The first two works of the series are bound closely together. *The Grothendieckian Machine* set out how mathematical worlds are constructed, compared and carried from one setting to another, and how a machine may hold them as typed, versioned and verifiable objects. *GM-Omega-Infinity: Topos-Native Intelligence* stated what an intelligence must be, internally, if it is to reason inside such worlds. The third and fourth works, *Arithmetic Spectral Descent* and *Fractured Adelic Geometry*, took the Riemann Hypothesis and the conjecture of Birch and Swinnerton-Dyer as research subjects of the Programme. The present work is a self-contained system in the same sense, and it is the Programme's third research subject.

The Programme has a single research aim: machine intelligence that discovers mathematical structure on its own account. That aim asks for reasoning architectures held to the standard of proof, and for a discipline of recursive mathematical discovery, in which a proposed structure is tested, what survives the test is abstracted into a new setting, and the question is posed again in that setting. Its model is Grothendieck's rising sea. A hard problem is not attacked where it stands; a theory is raised around it until the problem becomes natural and gives way. A research subject of the Programme is a problem on which this method is carried through in full, with every step recorded in a form that a mathematician can check and a machine can execute.

This work carries it through on the Hodge Conjecture. It begins from two theorems that bound every approach from outside: Voisin's compact Kähler counterexample, which forces any correct mechanism to spend projectivity at an identifiable step, and the integral counterexamples of Atiyah–Hirzebruch and Kollár, which force it to produce denominators. The recursion therefore changes the object rather than the argument. It treats algebraicity as a descent datum that a class carries along a prism — a definable bi-Rees deformation whose fibres are the Hodge decomposition and the Betti lattice — governed by twelve axioms, and it restates the conjecture as the vanishing of an obstruction class. Each step is then tested against the cases in which the answer is known — divisors, dimension at most three, cubic fourfolds, abelian surfaces — and what fails is recorded as an obstruction. The recursion ends in a master equivalence of five formulations, a ledger of nine obstructions of which exactly two are live, a computational and formal-verification architecture, eleven named falsifiers, and sixty problems.

The work keeps one editorial principle throughout. No result is presented at a strength greater than its proof. Every substantive assertion carries one of four labels — **[KNOWN]**, **[CONDITIONAL]**, **[NEW]**, **[OPEN]** — where it is made, and every conditional implication is printed with its hypotheses. Computation is reported as computation, at the level of certification it has, and is never promoted to a theorem. An obstruction the work fails to remove is stated as an obstruction, in the place where it stops the argument.

The Hodge conjecture remains open. This work does not prove it, and proves no case of it that was not already known. What it offers is the state of the question after the method has

been applied without concession: what the framework would deliver, what it would need, and where each route to it stops.

Vincent Liu
WiseMotif.ai

Abstract

This monograph asks why the Hodge Conjecture is known in codimension one, in codimension $n-1$, in dimension at most three and for isolated families such as cubic fourfolds, and nowhere else; and what kind of object would make algebraicity of a Hodge class a statement that can be tested, rather than a property that can only be hoped for.

[KNOWN] The diagnosis is negative and exact. Voisin's 2002 counterexample shows that the conjecture is false for compact Kähler manifolds, and the examples of Atiyah–Hirzebruch and Kollár show that its integral form is false. Any correct mechanism must therefore consume projectivity at an identifiable step and produce denominators at an identifiable step; four design constraints derived from these theorems (Chapter 3) refute most proposed mechanisms before they begin.

[NEW] The proposal changes the object. Built on o-minimal geometry — definable Chow, definable GAGA and the theorem of Bakker–Klingler–Tsimerman — the *archimedean prism* $\Delta(X)$ is a definable bi-Rees deformation with two commuting $\mathbb{G}_m$-actions whose four distinguished fibres are the Hodge decomposition, its conjugate, and the Betti cohomology with its rational lattice. Algebraicity becomes a descent datum along the prism, governed by twelve axioms AP1–AP12, and the Hodge Conjecture becomes the vanishing of an obstruction class in $H^1(\mathcal{S}_{\mathrm{per}}, \mathrm{Rig})$.

[NEW] The master theorem (Theorem 47.3.1) proves, from the ten unconditional axioms AP1–AP10, that five formulations stand or fall together: the Hodge Conjecture, the Rigidification Conjecture, the Concentration Principle, the Trace–Hodge Conjecture, and the split into Hodgic Fixed-Point Descent and Algebraic Seam Recognition. **[CONDITIONAL]** The Universal Kuga–Satake Conjecture enters on a different footing, as a carrier that reduces the Hodge Conjecture to the case of abelian varieties, itself open. The equivalence proves none of its five statements.

[OPEN] What remains is stated as a ledger. Of nine obstructions O1–O9, seven are discharged, refined or shown to lie off the main line, and two are live: O1, the existence of a rigidification datum for a single Hodge class, and O2, the passage from a closed to a positive closed current of integral type — the unique point at which projectivity is consumed. The precise gap is a single proposition, equivalent to the Hodge Conjecture (Chapter 51).

[KNOWN] The computational and verification architecture is specified in full: a seven-layer system Polaris-hc that computes periods with certified error bounds and treats PSLQ and LLL relations as candidates, never as proofs; a Lean 4 proof-obligation ledger, Helm, whose fragments are published as non-compiling design targets because the objects needed to state the conjecture are not yet in mathlib; and an adversarial multi-agent protocol whose citation audit caught sixty-three fabricated references, none of which survives. The work closes with eleven falsifiers, a self-assessment against seven criteria fixed in advance — four met, two partially, one not met: a proof of the Hodge Conjecture — and sixty problems.

The Hodge conjecture remains open. This monograph does not prove it, and proves no case of it that was not already known.

Preface

In 1950, at the International Congress of Mathematicians in Cambridge, Massachusetts, W. V. D. Hodge proposed that a certain purely cohomological condition on a class should force that class to be algebraic. The condition is easy to state and easy to check. The conclusion is a statement about the existence of subvarieties. Between the two lies seventy-five years of mathematics.

This book is about the space between them.

What this book is

It is a foundational treatise. It builds a framework — the *archimedean prism* — in which the Hodge Conjecture is reformulated as a descent problem, analyses that framework to the point where the remaining obstructions can be named individually, and specifies the mathematical, the computational, and the machine-verifiable infrastructure that a solution would require. It is meant to be used the way comprehensive mathematical treatises are used: read in parts, consulted in sections, and argued with in the margins.

The organising thesis can be stated in one sentence.

> *Algebraicity is not a property that a cohomology class possesses; it is a descent datum that a class carries along a prism whose special fibre is the Hodge decomposition and whose generic fibre is the Betti lattice.*

Everything in this volume is an elaboration, a formalisation, a computation, or a test of that sentence. The word *prism* is used by analogy with the prisms of Bhatt and Scholze — an object that mediates between two cohomology theories by remembering a deformation between them — but the analogy is structural, not literal, and Chapter 16 is devoted entirely to saying exactly where it holds and exactly where it breaks. The word *archimedean* records that the deformation parameter here is the archimedean place: the Hodge filtration is what survives when one lets the complex conjugation on $H^{2p}(X, \mathbb{C})$ degenerate.

What this book is not

It is not a proof of the Hodge Conjecture.

I state this here, on the second page, in a sentence of its own, because the literature surrounding the Millennium Problems contains a great deal of writing that is unclear on this point, and because a framework whose status is ambiguous is worse than no framework at all. The Hodge Conjecture is open. This book does not close it. What this book does is to reduce it, in a way that is new, to three statements — the Rigidification Conjecture, the Trace–Hodge Conjecture, and the Concentration Principle — of which the first is the load-bearing one, and to prove that these three are equivalent to one another and to the Hodge Conjecture itself. A fourth conjecture, the Universal Kuga–Satake Conjecture, enters on a different footing — not as an equivalent but as a carrier that reduces the Hodge Conjecture to the case of abelian varieties, itself open — and I keep that asymmetry in view rather than concealing it. That equivalence is a genuine reduction and it is, I believe, a useful one. It is not a solution.

Every claim in this volume carries an explicit epistemic tag. [KNOWN] marks a theorem of the existing literature, with a citation to a real paper. [CONDITIONAL] marks a statement proved here, but conditionally on hypotheses that are themselves open. [NEW] marks a definition, construction, or theorem introduced in this volume. [OPEN] marks something that is not proved by anyone. The tagging is not decoration; it is the primary quality-control mechanism of the book, and Chapter 51 consists of nothing but the complete list of [OPEN] items, in dependency order, with an estimate of the difficulty of each.

A research subject of the Grothendieckian Programme

This work belongs to the Grothendieckian Programme of the WiseMotif Monograph of Mathematical Intelligence, and it is the Programme's third research subject, after the Riemann Hypothesis and the conjecture of Birch and Swinnerton-Dyer. The Programme has a single aim: machine intelligence that discovers mathematical structure on its own account, under reasoning architectures held to the standard of proof. A research subject is a problem on which that aim is tested in full. The test is not whether a machine can produce text about a famous problem; it is whether the method can raise a structure around the problem, check that structure against everything already known, and say precisely where it fails.

The method is recursive mathematical discovery. A structure is proposed; it is tested against the cases in which the answer is known; what survives is abstracted into a new setting, and the question is posed again there. What does not survive is not discarded. It becomes an obstruction, with a statement, a status and an attack plan, and it is carried forward as part of the result. This book is the record of one such recursion on the Hodge Conjecture: the prism is the structure proposed, the divisors, the varieties of dimension at most three, the cubic fourfolds and the abelian surfaces are the cases it is tested against, and the ledger of Chapter 48 is what the recursion leaves behind.

The rising sea

Grothendieck described two ways of opening a hard nut: to strike it until it breaks, or to immerse it in a rising sea until it softens and opens of itself. The proved cases of the Hodge Conjecture are, almost without exception, of the first kind. Lefschetz uses the exponential sequence, which exists only in codimension one; hard Lefschetz transports that case to codimension $n-1$; Zucker computes with the normal functions of one family of fourfolds. Each is a precise blow at one point, and none extends past its point.

This book tries the second way. It does not look for a sharper blow in codimension two. It asks what the space would have to be on which a Hodge class and an algebraic cycle are two descriptions of one datum, writes down the twelve axioms such a space must satisfy, and follows their consequences. The sea has not risen far enough: the descent datum is specified, not produced, and the specification rests on a conjecture of the first rank of difficulty. But the specification is exact, it reproduces every case in which the answer is known, and it locates the remaining difficulty at two named places, O1 and O2. That is the state the book reports.

The problem, and the shape of what is known

The Hodge Conjecture asserts that on a smooth projective complex variety X every rational cohomology class of type (p,p) — every element of $\mathrm{Hdg}^p(X) = H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ — is a rational combination of the classes of algebraic subvarieties of codimension p. The hypothesis is a linear-algebraic condition on a single cohomology class; the conclusion is the existence of a geometric object. The whole difficulty of the subject lives in that asymmetry.

What is known is known for reasons that do not generalise, and the pattern of the known cases is the negative space around the problem. In codimension one the conjecture is a theorem: Lefschetz proved in 1924 [Lef24] that $\mathrm{Hdg}^1(X)$ is spanned by divisor classes. His argument — in modern dress, the exponential sheaf sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$ and the resulting identification of the Néron–Severi group inside $H^2(X,\mathbb{Z})$ — is available only in codimension one, because only there is there an exponential sequence to exploit. From the $(1,1)$-theorem the hard Lefschetz theorem delivers codimension $p = n-1$ by a formal duality, and the two together settle every X of dimension at most three. Isolated families beyond this range are known — Zucker's Hodge Conjecture for cubic fourfolds [Zuc77] is the standard example — but each is a computation special to its family, not a method.

Everything else is open, and the openness is structural rather than incidental. There is no exponential sequence in codimension $p \geq 2$; there is no functorial passage from a class of type (p,p) to a subvariety; and the periods that encode the Hodge structure are transcendental numbers about which almost nothing is provable. The sharpest way to see the obstacle is to notice what the Hodge condition does and does not give. A rational class of type (p,p) is represented by a closed real (p,p)-form; it is *not* in any evident way represented by a *positive* one, and it is positivity — the distinction between a closed current and a closed *positive* current with analytic support — that would produce a subvariety. This gap between closed and positive is not a technical nuisance. It is, as this volume argues at length, the residual content of the problem, and it survives every reformulation the book constructs.

The case that should discipline all optimism is that of abelian varieties. The Hodge Conjecture is **[OPEN]** for abelian varieties in general, and it is open despite a powerful structural result: Deligne proved in 1982 [Del82] that every Hodge class on an abelian variety is *absolutely Hodge* — a de Rham–Betti class compatible with every field automorphism of $\mathbb{C}$. Absolute Hodgeness is strong evidence and it is not algebraicity; it exhibits no cycle, and no argument is known that upgrades it to one. I state this plainly here because the sentence "Deligne proved the Hodge Conjecture for abelian varieties" is false and nonetheless circulates, and a book that began by repeating it would forfeit the reader's trust on its first page.

Why another framework

There is no shortage of approaches to the Hodge Conjecture. The reason to build another one is that the existing approaches share a common failure mode, and that failure mode is diagnosable.

The sharpest fact in this subject is negative. In 2002 Claire Voisin [Voi02a] constructed a compact Kähler manifold carrying a rational Hodge class that is not a combination of Chern classes of coherent sheaves. The Hodge Conjecture, in its natural formulation for compact Kähler manifolds, is *false*. It follows that any mechanism which proves algebraicity of Hodge classes must at some identifiable step consume the hypothesis of projectivity — must use the existence of an ample line bundle, or an integral Kähler class, or a polarisation defined over the rationals — in a way that fails in the Kähler category.

Most proposed mechanisms do not pass this test. They are formulated in transcendental terms that make sense for any compact Kähler manifold, and they therefore cannot be correct as stated, whatever else may be true of them. I have made this observation into a formal requirement, called the *Voisin Test*, and every chapter of Parts II–VII contains a subsection headed *Where projectivity is consumed*. A chapter that cannot answer that question is a chapter that has not yet found its mechanism.

The second sharpest fact is also negative. The integral Hodge Conjecture is false: Atiyah and Hirzebruch produced torsion counterexamples in 1962 [AH62], and Totaro sharpened the analysis in 1997 [Tot97] using complex cobordism. Consequently any mechanism must produce denominators. A construction that naturally yields an integral class is not a construction of an algebraic cycle; it is a construction that is provably wrong somewhere. This is the *Integrality Test*, and it is applied throughout Part V.

These two tests — one negative theorem in the transcendental direction, one in the integral direction — are the two walls of the corridor along which any correct argument must pass. The archimedean prism is designed from the outset to fit inside that corridor. The prism consumes projectivity at exactly one place, in the positivity-matching step of Chapter 36, and it produces denominators at exactly one place, in the lattice-comparison step of Chapter 38. Those two localisations are, to my mind, the principal structural contribution of this book.

What is new here

Against that background, the contribution of this volume is a framework and one theorem, and I want to name both precisely so that the reader can hold me to them.

The framework is built from four new objects. The *archimedean prism* $\Delta(X)$ of Chapter 13 is a definable degeneration — a Rees-algebra deformation of the Hodge filtration, carried out in the o-minimal category of Chapter 6 — whose special fibre is the Hodge decomposition and whose generic fibre is the Betti lattice; algebraicity becomes a descent datum along it. The *period site* $\mathcal{S}_{\mathrm{per}}$ of Chapter 14 is the Grothendieck site on which that descent lives, and its sheaf cohomology computes the obstruction groups. The *obstruction complex* $\mathrm{Ob}^{\bullet}(X)$ of Chapter 18 packages those groups into a single object whose degree-zero cohomology is $\mathrm{Hdg}^{p}(X)$ and whose higher cohomology receives the rigidification and denominator obstructions in turn. And the twelve *prism axioms* AP1–AP12 of Chapter 46 isolate exactly the properties a descent formalism must have for the argument to run: ten of them hold unconditionally for $\Delta(X)$, and the remaining two are equivalent to the open input of the theory.

The theorem is the master equivalence of Chapter 47. Assuming only the ten unconditional axioms, it proves that for every smooth projective X and every p the following are equivalent: the Hodge Conjecture $\mathrm{HC}(X,p)$; the *Rigidification Conjecture* RC of Chapter 34; the *Trace–Hodge Conjecture* THC of Chapter 29; the *Concentration Principle* CP_{p} of Chapter 45; and the conjunction of *Hodgic Fixed-Point Descent* and *Algebraic Seam Recognition*, HFD $\wedge$ ASR, the foundational split introduced in Chapter 47. After the Kuga–Satake reduction of Chapter 49 the *Universal Kuga–Satake Conjecture* UKS of Chapter 23 joins the list. These statements come from four different mathematical worlds — o-minimal descent, tensor categories, pluripotential theory, and definable foundations — and the theorem is that they are one statement in five costumes.

I want to be exact about the standing of this result, because it is the book's genuine novelty and it is easy to oversell. It is an *unconditional theorem about conjectures*. It proves none of the five; it proves that they stand or fall together, and it does so without assuming any Standard Conjecture — a discipline that matters, because over $\mathbb{C}$ the Hodge Conjecture already implies the Lefschetz Standard Conjecture B (Kleiman 1968) [Kle68], so any argument that quietly used B would be circular. The equivalence is a genuine reduction: after it there is one problem, with a computable source and target and a well-understood kernel, rather than a scattered family of analogies. It is also, I believe, the first time these particular statements have been tied together by an unconditional argument, and Chapter 47 is where a reader sceptical of that claim should press hardest.

What the framework does *not* do must be said in the same breath and with no softening. It does not prove the Hodge Conjecture, in any case not already in the literature. The reduction terminates at a single named obstruction. In the language of the ledger of Chapter 48 it is O2, *positivity matching*; in the language of the analytic engine of Chapter 44 it is Step B, the passage from a closed real (p,p)-form to a closed *positive* one carrying analytic support. Step A of that engine is a theorem and Step C is a theorem; Step B is the gap, and Chapter 51 states it as sharply as the volume can. Everything the framework buys is bought on one side of that step, and nothing the framework contains crosses it. A reader who wishes to refute the book, or to complete it, should go directly to O2.

The three registers

This volume is written in three registers simultaneously, and the reader should know which one is in operation.

The *mathematical* register (Parts 0–VII) is ordinary mathematics: definitions, theorems, proofs, counterexamples. It is self-contained modulo the appendices, and it is where the framework lives.

The *computational* register (Parts VIII–IX) describes Polaris-HC, a system for discovering and certifying Hodge classes and algebraic cycles. Everything specified there is buildable with existing technology; nothing in it depends on the Hodge Conjecture being true. The computational register exists because a framework that generates no computations generates no evidence, and because the boundary between what a computer can *certify* and what it can merely *suggest* is, in this subject, mathematically interesting in its own right. Chapter 56 is devoted to that boundary; its main conclusion is that integer-relation detection can never, in principle, certify a period relation, and the reasons why are instructive.

The *formal* register (Part X) concerns machine verification. Its central finding is negative and worth stating in the preface: the Hodge Conjecture *cannot presently be stated* in Lean 4 with mathlib, because the requisite theory — Hodge decomposition for compact Kähler manifolds, Chow groups with the cycle class map, the comparison isomorphisms — is not there. Chapter 65 gives a dependency-ordered roadmap of what would have to be formalised, with effort estimates. I regard this chapter as one of the more useful things in the book, precisely because it is not speculative.

The plan of the book

The volume has seventy-two chapters in twelve parts, followed by twelve appendices; the *Leitfaden* that follows this preface gives reading routes, and here I give only the logic of the sequence.

Part 0 (Chapters 1–4) states the problem, catalogues what is known, false, and open, distils the two negative theorems into design constraints, and lays out the architecture. A reader in a hurry may stop after it with a fair picture of the whole. *Part I* (Chapters 5–11) develops the tame geometry the construction runs on: o-minimal structures, definable complex analytic spaces, tame uniformisation, the definable Chow and GAGA theorems, the Bakker–Klingler–Tsimerman method, effectivity, and the tame site. It is the part a reader willing to grant definability may skim first and return to when a tameness hypothesis is invoked.

Part II (Chapters 12–18) constructs the prism itself: algebraicity recast as descent, the construction of $\Delta(X)$, prismatic envelopes, the descent formalism, the comparison with the prismatic cohomology of Bhatt–Scholze — structural and not literal, a point I make carefully in Chapter 16 — the prism cohomology of a polarised Hodge structure, and the obstruction complex that organises everything downstream.

Part III (Chapters 19–24) is transcendental Hodge theory in the service of the framework: Mumford–Tate groups, Hodge loci and the Cattani–Deligne–Kaplan theorem on the algebraicity of the locus — of the locus, not the class — atypical intersections, absolute Hodge classes, Kuga–Satake, and variations of Hodge structure with their degenerations.

Part IV (Chapters 25–31) is the motivic and Tannakian layer: pure motives, an atlas of the Standard Conjectures, dualisability and categorical trace, traceable motives, the trace–cycle map, the fullness question that is the Trace–Hodge Conjecture, and coniveau in K-theory. This part is illuminating rather than load-bearing; the verdict of Chapter 30, that the trace reformulation is exact and yet produces no cycle, is one of the three reasons Chapter 51 can localise the gap so precisely.

Part V (Chapters 32–38) is where the weight is carried. Periods and their bilinear relations set the stage; the Rigidification Conjecture of Chapter 34 is stated and shown in Chapter 35 to imply the Hodge Conjecture; Chapter 36 is the single chapter in which projectivity is consumed, and Chapter 38 the single chapter in which denominators are produced. The two negative theorems of the subject enter here as walls, and the architecture is arranged so that each is struck exactly once.

Part VI (Chapters 39–45) is the analytic engine: currents and positivity, the Bishop extension theorem of Chapter 40, non-abelian Hodge theory, Higgs lifts, mixed Hodge modules, the Twistor–Current Rigidification Machine TCRM of Chapter 44, and the concentration principle. Step B of TCRM is the gap; the rest of the part is the machinery that surrounds it on both sides.

Part VII (Chapters 46–51) is the synthesis: the twelve axioms, the master theorem of Chapter 47, the obstruction ledger of Chapter 48, the reduction to abelian varieties, worked case studies, and the precise statement of the gap in Chapter 51. A reader who wants only the main theorem should read Part VII and reach back into Parts II and V for the objects it names.

Part VIII (Chapters 52–58) is the computational front-end: exact linear algebra over number fields, the Griffiths–Dwork method, Gauss–Manin and Picard–Fuchs, certified numerics, period-relation discovery, cycle discovery, and a candid assessment of what machine learning can and cannot contribute. The verdict of Chapter 56, that neither PSLQ nor LLL can *certify* a period relation, belongs to this part.

Part IX (Chapters 59–63) documents POLARIS-HC, the software system that carries out the computations: its architecture, its module catalogue, the Hodge atlas it maintains, its

certificate and trust model, and its distributed execution. The system is an instrument, not an oracle, and the part is written so that its outputs can be reproduced and audited.

Part X (Chapters 64–68) confronts formalisation. It records the verdict that the Hodge Conjecture cannot presently even be *stated* in Lean 4 with mathlib, gives the dependency audit that makes that verdict airtight, lays out a roadmap and the proof-obligation ledger of the Helm programme, describes the Consilium construction protocol, and closes with the falsification suite.

Part XI (Chapters 69–72) is the epistemic reckoning: eleven ways the framework could be falsified, a chapter fixing in advance the seven criteria by which a claimed revolution should be judged and returning the verdict four met, two partial, one not met, a list of sixty open problems, and an epilogue. The twelve appendices supply background, tables, and the full notation, glossary, and index of conjectures of Appendix L.

The five volumes

The twelve parts are bound as one monograph in five volumes. The plan above follows the parts; the five sections that follow describe the volumes, what each establishes, and where each stops. Each volume ends at a result its reader can hold without the others, and the Leitfaden that follows this preface gives routes that cross them.

Volume I: The Problem and Its Tame Foundations

Volume I (Parts 0–II) states the problem, fixes the constraints every approach must satisfy, and builds the object on which the rest of the book works. [KNOWN] Part 0 separates the statement of the Hodge Conjecture from its inequivalent siblings — the integral, the compact Kähler and the general (Grothendieck-corrected) forms — and audits what is known, false and open, so that no later chapter can borrow strength from a misquotation. From the two counterexamples it derives four design constraints, the Voisin, Integrality, Coniveau and Absolute-Hodge Tests of Chapter 3, and runs the principal mechanisms in the literature through them; most fail before they begin.

[KNOWN] Part I is the substrate: o-minimal structures, definable complex analytic geometry, the tame uniformisation of period maps, definable Chow and definable GAGA, and the method of Bakker–Klingler–Tsimerman. Its point for this book is a reinterpretation, not a new theorem: a definable structure on a period map is an arithmetic datum, because its constants come from a model of the variety over a finitely generated ring, and it is exactly the datum a general compact Kähler manifold lacks. [NEW] The part ends with the tame site of Chapter 11, a genuine Grothendieck topology on which descent can be done.

[NEW] Part II constructs the archimedean prism. Algebraicity is recast as a descent datum; the prism $\Delta(X)$ is built as a definable bi-Rees deformation of the Hodge filtration and its conjugate, with two commuting $\mathbb{G}_m$-actions and four distinguished fibres; the period site carries the descent, and the obstruction complex $\mathrm{Ob}^\bullet(X)$ organises what descent fails to deliver, with $\mathrm{Hdg}^p(X)$ in degree zero. Chapter 16 sets the prism against the prismatic cohomology of Bhatt and Scholze and says exactly where the analogy holds and where it breaks. The volume ends with an object, constructed and functorial, and with the obstruction groups in which the rest of the book's questions live.

Volume II: Hodge Structures, Motives and Periods

Volume II (Parts III–V) brings three theories to bear on the prism and finds the hypothesis on which the book turns. [KNOWN] Part III is transcendental Hodge theory: Mumford–Tate groups, Hodge loci and the Cattani–Deligne–Kaplan theorem — algebraicity of the *locus* on which a flat section is Hodge, not of the class — atypical intersections, Deligne's absolute Hodge classes, the Kuga–Satake construction and degenerations. Each is stated at its exact strength, because each is a place where the subject is habitually overstated.

[NEW] Part IV is the motivic layer. It builds the category of traceable motives and the trace–cycle map τ^{tr}, proves the map injective from Jannsen's semisimplicity without assuming any Standard Conjecture, and proves (Theorem 30.3.1) that fullness of the Hodge realisation, the Trace–Hodge Conjecture and the Hodge Conjecture are one statement. The verdict of Chapter 30 is that the reformulation is exact and produces no cycle; it relocates the difficulty and does not remove it.

[OPEN] Part V carries the weight. After periods and the bilinear relations, it states the Rigidification Conjecture (Conjecture 34.3.1): a Hodge class acquires a definable family of polarised sub-Hodge structures with arithmetic monodromy and a Weil operator defined over $\mathbb{Q}$. This is the ledger entry O1. [CONDITIONAL] Chapter 35 proves that it implies the Hodge Conjecture. [NEW] Chapter 36 proves that projectivity is consumed at one step only, Chapter 37 shows Voisin's torus failing exactly there, and Chapter 38 shows that the denominators forced by the integral counterexamples are produced at one step only, with a bound $d(\alpha)$. The volume ends with the two walls of the subject each struck exactly once.

Volume III: The Analytic Engine and the Master Construction

Volume III (Parts VI–VII) contains the book's main theorem and its precise limit. [KNOWN] Part VI assembles the analytic engine: positive currents and the King formula, Bishop's extension theorem, non-abelian Hodge theory and Higgs bundles, mixed Hodge modules. [NEW] From them it builds the Twistor–Current Rigidification Machine of Chapter 44, a four-step implication from a Hodge class to an algebraic cycle in which Steps A, C and D are theorems and Step B — the passage from a closed real (p,p)-form to a positive closed current of integral type — is the gap. [OPEN] Step B is the ledger entry O2, and Chapter 45 states it as the Concentration Principle CP_p.

[NEW] Part VII is the synthesis. Chapter 46 abstracts twelve prism axioms; ten hold unconditionally, AP11 is the Rigidification Conjecture and AP12 the conditional denominator accounting. Theorem 47.3.1 proves, from the ten, that the Hodge Conjecture, RC, CP_p, THC and the split HFD $\wedge$ ASR are equivalent; it proves none of them. Chapter 48 enumerates the obstructions as O1–O9, proves that no tenth arises relative to the axioms, and finds two live. [CONDITIONAL] Chapter 49 reduces the Hodge Conjecture to abelian varieties under the Universal Kuga–Satake Conjecture; Chapter 50 runs the framework through worked cases; and Chapter 51 states the gap as a single proposition, equivalent to the Hodge Conjecture. The volume ends where the book's mathematics ends: at O1 $\wedge$ O2.

Volume IV: Computation and System Architecture

Volume IV (Parts VIII–IX) turns the framework into computations and the computations into a specified system. [KNOWN] Part VIII is the computational front end: exact linear algebra over number fields, Griffiths–Dwork reduction, Gauss–Manin connections and Picard–Fuchs equations, certified numerics in ball arithmetic, relation discovery, cycle discovery, and the place of machine learning as a search heuristic. [NEW] Its central finding is a boundary, drawn in Chapter 56: integer-relation algorithms such as PSLQ and LLL propose period relations under a height bound and cannot certify them, so a discovered relation is a candidate and is labelled as one.

[NEW] Part IX specifies POLARIS-HC, the system that would carry these computations: seven layers from exact algebra to infrastructure, a catalogue of modules each with a single responsibility and a named failure mode, a Hodge Atlas of typed records, a certificate envelope and a trust ladder from heuristic search to kernel-checked proof, and a protocol for distributed, reproducible execution. Nothing in the volume depends on the Hodge Conjecture or on the framework of Volumes I–III. It is a specification and not an implementation, as Chapter 70 records: buildable is not built.

Volume V: Formalisation and Self-Assessment

Volume V (Parts X–XI) turns the book on itself. [NEW] Part X begins with a negative finding: the Hodge Conjecture cannot presently be stated in Lean 4 with mathlib, because the Hodge decomposition, Chow groups with the cycle class map and the comparison isomorphisms are not there (Chapter 64). It gives a dependency-ordered roadmap for what would have to be built, the proof-obligation ledger HELM in which every theorem of the book is either a pointer to a declaration or a reasoned gap, the CONSILIUM protocol by which the book was constructed and audited — including the sixty-three fabricated citations it caught — and a falsification suite in four batteries.

[NEW] Part XI is the reckoning. Chapter 69 lists eleven findings, several of them computations that can be run today, any one of which would refute a load-bearing claim. Chapter 70 fixes seven criteria for a claimed advance before applying them and returns four met, two partially, one not met: the proof of the conjecture. Chapter 71 poses sixty problems, graded by difficulty and by what their solution would settle, and the epilogue states the author's view of the prospects. The volume ends with the terms on which the book asks to be judged.

Intended readership

I have written for the graduate student or working mathematician comfortable with algebraic geometry and Hodge theory at the level of Voisin's two volumes, which I treat throughout as the standard companion. Beyond that common ground the parts make local demands, and I have tried to make each self-contained enough that a demand can be met by turning to an appendix rather than to another book. Part I asks for a working acquaintance with o-minimality, recalled in the tame-geometry appendix; Part IV asks for tensor and Tannakian categories, recalled in the motivic appendix; Part VI asks for currents and several complex variables, recalled in the analytic appendix. Parts VIII through X ask for literacy in computation and formalisation but assume no result the earlier parts did not already establish, and a reader with no taste for machines can stop at the end of Part VII with the mathematics intact.

The book is meant to be argued with. Its central theorem is stated so that it can be checked, its gap is named so that it can be attacked, and its computational and formal claims are recorded so that they can be rerun. A reader who finishes Part VII persuaded that the

reduction is real, and Part XI persuaded that the accounting is sound, has read the book as I intended it to be read, whatever they then conclude about the conjecture itself.

What it would take to close the gap, and what it would take to break the book

Because the reduction terminates at O2, the shape of any completion is now visible even though the completion itself is not in hand. To close the gap one would have to produce, from a rational Hodge class, a closed positive current of bidimension $(n-p, n-p)$ with definable analytic support and rational cohomology class — to cross Step B of Chapter 44 in a single case where the existing positivity input does not already suffice. I do not know how to do this, and I have tried to say exactly why in Chapter 51: the Hodge condition controls the class of a current but not its positivity, and the two counterexamples of the subject show that any purported bridge must use projectivity and must produce denominators, so a naïve construction is ruled out before it begins. The value of the framework, if it has any, is that it turns a vague sense of difficulty into a single well-posed analytic problem with a computable interface.

Because the framework is a reduction and not a proof, it is also falsifiable, and I regard that as a feature rather than an embarrassment. Chapter 69 lists eleven concrete findings, several of them numerical experiments that Polaris-hc can run today, any one of which would refute a load-bearing claim of the volume; Chapter 70 fixes in advance the criteria by which the work should be judged and then judges it, without the luxury of choosing the criteria after seeing the result. A reader who distrusts the whole enterprise is invited to begin at those two chapters and work backwards. If the accounting there does not persuade, nothing in the earlier chapters should.

On revolution

The reader will notice that this book is immodest about its ambitions and modest about its results. That combination is deliberate.

Grothendieck's revolution in the 1960s did not consist of solving the problems of his day. It consisted of building a language in which those problems became statements about morphisms, and then observing that in the new language some of them were trivial and others were visibly deeper than anyone had suspected. EGA and SGA are not collections of solutions; they are a rebuilt foundation with the solutions falling out along the way. Chapter 70 examines what that actually means, as an epistemological matter, and asks whether the construction in this book meets the standard. My own assessment — given there in full, with the criteria fixed in advance so that the reader can disagree with the verdict rather than the criteria — is that it meets four of the seven criteria clearly, two partially, and one not at all.

I would rather publish that assessment than a triumphant one.

What the Programme takes from this work

For the Programme, the result of this recursion is not a proof and was not expected to be one. It is a precise description of where a structural method stands on one of the central problems of algebraic geometry: a framework whose ten unconditional axioms are verified, an unconditional theorem that five formulations of the problem are one, a ledger in which seven of nine obstructions are discharged and two remain, and an architecture in which every computation and every formal statement is held at the level of certification it has. Each of these is a test the method has passed or failed in public. Where it failed — at the existence of a rigidification datum, at the passage from a closed to a positive current, at the statement of the conjecture in a proof assistant — the failure is recorded where it occurs, so that the next recursion can begin from it.

The Hodge conjecture remains open. This book does not prove it, and proves no case of it that was not already known.

Acknowledgements

This volume was composed with substantial assistance from large language models operating as research collaborators: six independent reasoning agents were tasked with developing distinct approaches to the problem, and the synthesis in Part VII draws on all six. Two of them converged independently on rigidification as the central mechanism; that convergence is the reason it occupies the centre of this book. The methodology — including its failure modes, the hallucinated citations it produced and how they were caught, and the verification protocol that was applied to every reference — is documented in Chapter 67, because a book that uses a new research method and conceals it is not doing science.

All errors are mine. Corrections are welcome and will be maintained in a public errata.

Vincent Liu
WiseMotif.ai
2026

On the Epistemic Status of This Book

The Hodge Conjecture is open. This book does not prove it.

No chapter of this volume should be read as claiming a solution to the Hodge Conjecture, to the Standard Conjectures, or to any Millennium Prize Problem. Where this book proves something, it says so and gives the proof. Where it assumes something, it says so and gives the assumption a name and a number. Where nobody knows, it says that too.

Why this book keeps an epistemic ledger

A word on why the apparatus exists at all. This volume was constructed with heavy assistance from automated systems, and I regard an explicit epistemic ledger not as an ornament but as a precondition for asking the reader to trust a book of this length. Tagging every substantive assertion forces a decision at the point of writing: is this *known*, is it *conditional*, is it *new*, or is it *open*? A sentence that resists the question is almost always a sentence hiding an assumption, and the discipline of tagging is the cheapest way I know to bring such assumptions to the surface. The same discipline is what made the citation audit of Chapter 67 possible: a claim already marked [KNOWN] announces that it owes a reference, and a reference that cannot be found is then a visible failure rather than an invisible one.

The honesty is load-bearing in a second, structural sense. The book's central claim is a *reduction*, and a reduction is worth exactly as much as its accounting of what it did and did not assume. An argument of seventy-two chapters cannot be audited as a monolith; it can be audited statement by statement only if each statement declares its own standing. The tags are that declaration, and the reader who distrusts the conclusion is invited to distrust it one tagged line at a time.

One set of labels across the series

The four labels are those of the whole WiseMotif Monograph of Mathematical Intelligence. *Arithmetic Spectral Descent* and *Fractured Adelic Geometry*, the two earlier research subjects of the Grothendieckian Programme, use the same four labels in the same sense, so that a statement marked [NEW] or [OPEN] means the same thing in every work of the series, and the claim boundary of each work is printed in the same form. For this work it reads: the Hodge conjecture remains open; this book does not prove it, and proves no case of it that was not already known. The labels are applied by the same rule throughout: a statement receives the weakest label its proof supports, a conditional statement names its hypothesis where it is made, and a computation is reported at the level of certification it has and never as a theorem.

The four tags

Every substantive assertion in this volume carries one of four tags. The tags are applied to *statements*, not to sections, and a single paragraph may contain statements with different tags.

[KNOWN]	A theorem of the published literature. A specific citation is given: author, year, and a pointer precise enough to locate the statement. If a citation is missing, that is an error and should be reported.
[CONDITIONAL]	Proved in this volume, but only under hypotheses that are themselves unproved. The hypotheses are always named explicitly in the statement, never buried in surrounding text. A **[CONDITIONAL]** result is a genuine theorem about implications.
[NEW]	A definition, construction, or theorem introduced here and not, to the author's knowledge, in the literature. The claim of novelty is a claim about the author's knowledge, not about the literature; priority disputes should be resolved in favour of the earlier work.
[OPEN]	Not proved — here or anywhere. The obstructions of the programme carry the ledger numbers O1–O9 of Chapter 48; the named open conjectures are collected, with their status, in the Index of Conjectures of Appendix L.

A worked instance of each tag

The tags are easiest to understand against real statements, one of each drawn from a chapter the reader can turn to.

A **[KNOWN]** statement: the Lefschetz $(1,1)$-theorem, that on a smooth projective complex variety the group $\mathrm{Hdg}^1(X)$ is spanned by the classes of divisors (Chapter 1). It carries the citation Lefschetz 1924, and the reader can check it against any standard text; nothing in this book is asked of it that the literature does not already grant.

A **[CONDITIONAL]** statement: the implication, proved in Chapter 35, that the Rigidification Conjecture implies the Hodge Conjecture. The proof is complete and unconditional *as an implication*; what is conditional is the conclusion, because the hypothesis is itself unproved. A **[CONDITIONAL]** result is a theorem about the shape of the problem, and it is exactly as reliable as its named hypothesis — no more and no less.

A **[NEW]** statement: the construction of the archimedean prism $\Delta(X)$ in Chapter 13, and the twelve axioms AP1–AP12 that Chapter 46 abstracts from it. These are introduced here and are, to my knowledge, not in the literature; the tag is a claim about my knowledge, and any earlier occurrence should be treated as having priority.

An **[OPEN]** statement: the Rigidification Conjecture itself (Chapter 34), carried in the ledger as obstruction O1. It is not proved here or anywhere, it is marked with its ledger number wherever it is used, and no argument in the book is permitted to forget that it is open.

What is actually proved here

The principal unconditional results of this volume are the following. They are theorems; they are also, individually, less than the Hodge Conjecture.

(i) **The prism exists.** The archimedean prism $\Delta(X)$ of a smooth projective complex variety is constructed in Chapter 13 and shown to be a well-defined object of a category of definable ringed sites, functorial in X, with the Hodge filtration and the Betti lattice recovered as its two fibres.

(ii) **The equivalence theorem.** Five formulations — the Hodge Conjecture, Rigidification, Trace–Hodge, Concentration, and the split HFD ∧ ASR — are mutually equivalent, assuming only the ten unconditional axioms AP1–AP10 (Theorem 47.3.1); the Universal Kuga–Satake Conjecture enters not as a fourth equivalent but as a carrier reducing the Hodge Conjecture to the abelian case. This is an unconditional theorem *about* conjectures.

(iii) **Localisation of projectivity.** In the prism formalism the hypothesis of projectivity is consumed at exactly one step, and that step is identified (Chapter 36). Voisin's counterexample is shown to fail precisely there, which is the required consistency check.

(iv) **Localisation of denominators.** The denominators forced by the failure of the integral Hodge Conjecture are shown to arise at exactly one step, and the resulting obstruction group is computed in examples (Chapter 38).

(v) **The obstruction ledger is complete and finite.** Relative to the axioms AP1–AP12, the obstructions to the programme are enumerated as O1–O9, and it is proved that no others arise (Chapter 48).

(vi) **The computational specification.** The algorithms of Parts VIII–IX — period matrices to guaranteed precision, Picard–Fuchs operators, cycle certificates — are specified with their certificates and their place on the trust ladder of Chapter 62. This is a specification, not an implementation: as Chapter 70 records, there is no complete reference implementation. Nothing there is conditional on any conjecture.

What is not proved here

(i) The Hodge Conjecture, in any nontrivial case not already in the literature.

(ii) The Rigidification Conjecture (Chapter 34), which is the load-bearing hypothesis of the entire architecture. It is [OPEN]. If it is false, Parts V–VII collapse, and Chapter 69 explains what would survive.

(iii) Any of the Standard Conjectures.

(iv) The existence of a motivic t-structure.

(v) Any statement about compact Kähler manifolds that would contradict Voisin's theorem, and any integral statement that would contradict Atiyah–Hirzebruch or Totaro.

The two walls, and why they are not open questions

Two statements adjacent to the Hodge Conjecture are *false*, and the distinction between false and open is one the reader must hold firmly, because a book that let "false" soften into "open" would be dishonest at the root. The tags reserve [OPEN] for genuine unknowns; neither of the following is unknown.

The *compact Kähler analogue* of the Hodge Conjecture is false. Voisin proved in 2002 [Voi02a] that there is a compact Kähler manifold carrying a rational Hodge class no nonzero multiple of which is a combination of Chern classes of coherent sheaves — in particular not algebraic in any reasonable sense. This is a theorem, a negative one, and it is settled. Its consequence for this book is absolute: projectivity is a genuine hypothesis and not a convenience, and the architecture is arranged so that projectivity is consumed at exactly one step, Chapter 36, with the obstruction Voisin exhibits accounted for in Chapter 37. Any purported proof that applied verbatim to the compact Kähler case would thereby be *refuted*, not merely doubted.

The *integral* Hodge Conjecture is false. Atiyah and Hirzebruch gave the first counterexamples in 1962, using torsion classes that are not algebraic [AH62]. Totaro placed the integral cycle map through complex cobordism in 1997, thereby systematising and strengthening the torsion-sensitive topological constraints [Tot97]; Kollár's 1992 Trento examples supply the

separate non-torsion degree obstruction [Kol92]. This too is settled. Its consequence here is equally sharp: rational coefficients are not a technical preference but a necessity, and the denominators they permit are produced at exactly one step, Chapter 38. A framework that delivered algebraic cycles over $\mathbb{Z}$ would prove something known to be false and would, for that reason, be wrong.

I dwell on these because they are the two walls against which every approach to the conjecture must run, and because the correct epistemic posture toward them is not humility but certainty: they are known, they are negative, and they are the fixed points around which a sound architecture must be built.

The four open conjectures

Four conjectures carry the weight of the reduction, and the reader is entitled to see them named together, with their locations and their exact logical standing. All four are [OPEN]. Chapter 47 proves that the first three are equivalent to the Hodge Conjecture and to one another; the fourth, Universal Kuga–Satake, enters as a carrier reducing the Hodge Conjecture to the abelian case rather than as a mutual equivalent, as its entry below and Chapter 47 record.

Rigidification Conjecture (RC), **Chapter 34.** That a Hodge class is algebraic if and only if its prism-fixed lift admits a definable rigidification. This is the load-bearing hypothesis of the architecture and the ledger entry O1; if it fails, Parts V–VII fall with it.

Trace–Hodge Conjecture (THC), **Chapter 29.** That the trace–cycle map τ^{tr} is surjective — equivalently, that the Hodge realisation is full on the category of traceable motives. The name is *Trace*–Hodge, never "Tame"; the two are unrelated, and conflating them is a reliable sign that a summary was written without the source at hand.

Concentration Principle (CP_p), **Chapter 45.** That a closed positive current representing a *big rational* Hodge class of type (p,p) can be concentrated onto an analytic set of the correct dimension $n-p$. This is the analytic face of the problem and the one nearest the surviving gap.

Universal Kuga–Satake Conjecture (UKS), **Chapter 23.** That the Kuga–Satake correspondence is universal in the sense made precise there and revisited in Chapter 49. It enters on a different footing from the other three: rather than joining the mutual equivalence, it reduces the Hodge Conjecture to the case of abelian varieties, itself open, as Chapter 47 states exactly.

I state their openness as flatly as I can, because the temptation in a book this long is to let a hypothesis used a hundred times begin to feel like a theorem. It does not become one. Every use of RC, THC, CP_p or UKS is a use of an open conjecture, and the tag travels with it.

The ledger, and what remains open in it

Chapter 48 enumerates the obstructions to the programme as O1 through O9 and proves, relative to the axioms AP1–AP12, that no others arise. Of those nine, seven are discharged in the course of Parts V–VII, each with the discharging chapter recorded against it in the ledger. Two remain genuine obstructions, and the reader should keep them in view throughout.

- O1, *rigidification* — the content of the Rigidification Conjecture of Chapter 34. It is the algebraic half of the surviving problem.
- O2, *positivity matching* — the passage, at Step B of the Twistor–Current Rigidification Machine of Chapter 44, from a closed real (p,p)-form to a closed *positive* current carrying analytic support. It is the analytic half, and it is where projectivity must do its work.

Everything else in the ledger is closed. The one-sentence summary of the whole volume is that it reduces the Hodge Conjecture to the conjunction O1 $\wedge$ O2 and closes neither conjunct.

The mandatory honest verdicts

Three negative findings are load-bearing in their own right, because later chapters lean on them, and each is stated in its home chapter without hedging.

(i) **The trace reformulation is exact but produces no cycles** (Chapter 30). The Trace–Hodge Conjecture is a faithful restatement of the Hodge Conjecture, and yet the trace identity can hold exactly in a case where no algebraic cycle is thereby produced; Chapter 30 exhibits such a case on a Weil fourfold. The reformulation relocates the difficulty; it does not remove it, and Chapter 51 counts this as one of the three reasons the gap cannot be narrowed from inside the framework.

(ii) **Integer-relation algorithms cannot certify a period relation** (Chapter 56). Algorithms such as PSLQ and LLL discover candidate relations among periods to any finite precision, but they cannot *certify* that a discovered relation is exact, because certification would require a transcendence or height input that is not available. A discovered relation is a conjecture, not a theorem, and the book marks it as such.

(iii) **The Hodge Conjecture cannot presently be stated in Lean 4 with mathlib** (Chapter 64, Chapter 66). The statement requires objects — Chow groups with rational coefficients, the cycle class map, the Hodge decomposition on a smooth projective complex variety — that mathlib does not yet contain; Chapter 64 gives the dependency audit that establishes this, and Chapter 66 the proof-obligation ledger that would have to be filled first. A framework cannot be machine-checked until its conclusion can be machine-stated, and that day has not arrived.

Forward commitments

A book of this kind fixes its promises before it collects on them, so that the reader can hold it to a standard it did not choose after the fact. Four such commitments are made in the later chapters and are recorded here in advance.

- Chapter 70 fixes *seven* criteria by which a claimed advance on a problem of this magnitude should be judged, states them before applying them, and returns the verdict *four met clearly, two met partially, one not met*. The one not met is the proof of the conjecture, and the book says so in those words.
- Chapter 69 gives *eleven* falsifiers — concrete findings, several of them computations that can be run today, any one of which would refute a load-bearing claim of the volume.
- Chapter 71 poses *sixty* open problems, graded by difficulty and by what their resolution would settle, so that the framework leaves behind more than a single yes-or-no question.

- Chapter 67 reports the audit of the volume's own sources: *sixty-three* fabricated citations were detected during preparation — forty-one by bibliographic synthesis, fourteen by field misassignment, eight by misattribution — and *none* survived into the text. The protocol that caught them is given in full, so that the reader can rerun it.

These are not decorations. They are the terms on which the book asks to be believed, and a reader who finds any of them unmet has found an error the author wants reported.

On citations

Every citation in this book has been checked against a real bibliographic record. This is stated explicitly because the volume was prepared with the assistance of language models, and language models fabricate references with great fluency. The verification protocol — and the list of fabricated citations that were generated and rejected during preparation — is documented in Chapter 67, §67.5. A citation that cannot be verified appears with the marker [unverified] or does not appear at all.

Where a result is attributed to a person, the attribution is to the published source. Where the folklore attribution differs from the published record, both are given.

How to disagree with this book

The framework is designed to be falsifiable, and Chapter 69 lists eleven specific findings that would refute it. The shortest routes to refutation are these:

- Exhibit a compact Kähler manifold to which the prism construction of Chapter 13 applies verbatim. This would show that the Voisin Test is failed, and the framework would be dead.
- Exhibit a Hodge class satisfying the rigidification criterion of Chapter 34 that is provably non-algebraic. Same conclusion.
- Show that the obstruction ledger of Chapter 48 is incomplete by producing a tenth obstruction not reducible to O1–O9. The framework would survive but the completeness theorem would not.
- Show that Definition 28.3.1 does not in fact produce a rigid symmetric monoidal category. Part IV would need rebuilding.

A reader who finds any of these is asked to write. The errata will record it.

Vincent Liu · WiseMotif.ai

Leitfaden

This volume has seventy-two chapters and twelve appendices. Nobody should read it linearly on a first pass. What follows is a set of routes through it, each self-contained enough that a reader who follows one and stops will have acquired something complete.

> **The shortest route.** Chapters 1, 2, 3, 12, 34, 36, 47, 48, 51. About a hundred and twenty pages in this edition. This gives the statement of the problem, the two negative theorems that constrain every approach, the central idea, the place where projectivity is consumed, the master equivalence, the complete list of obstructions, and an account of the gap. A reader who wants to know whether this book is worth their time should read exactly these.

A word on why linear reading fails here. The volume is built as a reduction: it takes one hard problem and trades it, through a chain of equivalences, for another hard problem that is better posed. The chain is long, and its links come from o-minimality, prismatic ideas, transcendental Hodge theory, tensor categories, and pluripotential theory in turn, so a reader who insists on reading front to back will spend more than six hundred pages building machinery whose purpose only becomes visible in Part VII. The routes below are designed to let a reader see the purpose early and fetch the machinery on demand. Each route names the shortest path to a payoff its intended reader will recognise, and each ends at a definite claim rather than trailing off; the page estimates are deliberately conservative, and a reader who finds a route longer than advertised has almost certainly strayed into a dependency that route was meant to let them skip.

The five volumes

The twelve parts are bound as one monograph in five volumes. The volumes are the coarsest unit of reading: each ends at a result its reader can hold on its own, and the routes below cross them freely.

Volume	Parts	Content
I	0–II	The problem, the design constraints, tame analytic geometry, the construction of the prism
II	III–V	Hodge structures, motives and traces, periods and the Rigidification Conjecture
III	VI–VII	The analytic engine, the twelve axioms, the master equivalence, the ledger, the precise gap
IV	VIII–IX	The computational front end and the Polaris-hc architecture
V	X–XI	Formalisation, the construction protocol, falsification, the verdict, sixty problems

Routes

Seven routes follow, each named for the reader it serves and each given as an explicit list of chapters rather than a vague suggestion. They overlap by design: a chapter that carries weight for more than one audience appears on more than one route. None is a substitute for the whole, but each was chosen so that a reader who follows it to the end and stops will hold a coherent fragment rather than a truncated one. Where a route leans on material developed elsewhere, the dependence is named so that it can be consulted rather than read in full, and the table of § "What each part assumes" records the external background each part takes for granted.

Route A — The mathematician who wants the framework

Parts 0, II, V, VII, in that order, with Part III consulted as needed. Chapters 1–4, 12–18, 32–38, 46–51. Part I may be skipped on a first reading if the reader is willing to take the definability results of Chapter 9 on faith; Part IV may be skipped if the reader is not interested in the motivic reformulation, which is illuminating but not load-bearing.

Route B — The Hodge theorist

Parts III and VI, then Part V. Chapters 19–24, 39–45, 32–38. This is the transcendental spine: Mumford–Tate groups, the Cattani–Deligne–Kaplan theorem, currents and positivity, non-abelian Hodge theory, and the Twistor–Current Rigidification Machine. Chapter 20 presents a principal unconditional theorem on the distribution of Hodge classes, and Chapter 44 is where this book tries to use it.

Route C — The model theorist

Part I complete, then Chapters 20, 21, 11, 14. Appendix F contains the technical background. The question this route answers is: what exactly does o-minimality buy in algebraic geometry, and is the "tame site" of Chapter 11 a real Grothendieck topology or a convenient fiction? (It is a real one; the proof is Theorem 11.3.3.)

Route D — The motivic reader

Part IV complete, then Chapters 22, 49, 26. This route treats the Hodge Conjecture as a fullness statement for the Hodge realisation functor and asks what a categorical trace can and cannot detect. The verdict — that the trace reformulation is exact but does not by itself produce cycles — is Theorem 30.3.1 and the discussion following it.

Route E — The computational reader

Parts VIII and IX complete; Chapters 52–63. Nothing in these parts depends on the framework of Parts II–VII, and nothing in them is conditional on any conjecture. A reader who believes the whole framework is wrong can still build Polaris-HC from its specification, and the specification does not change. Appendix J contains the error analysis.

Route F — The formalisation reader

Part X complete; Chapters 64–68, with Appendix K. Chapter 64 is a survey of what mathlib actually contains as of the date of writing, and Chapter 65 is a dependency-ordered roadmap. Readers interested in AI-assisted mathematics as a methodology, rather than as a tool, should read Chapter 67 on its own; it is written to be readable independently.

Route G — The sceptic

Chapters 2, 3, 37, 38, 51, 69, 70. These are the chapters devoted to what is false, what is impossible, what is missing, and what would kill the framework. If the book is wrong, the fastest way to find out is to read these first and then attack the chapters they point to.

Route H — Straight to the main theorem

For the reader who wants only the theorem of Chapter 47 and is willing to take its inputs on trust: Chapters 1, 12, 13, 34, 46, 47. This is shorter than Route A and dispenses with the load-bearing Part V; it delivers the statement that the Hodge Conjecture, the Rigidification Conjecture, the Trace–Hodge Conjecture, the Concentration Principle, and the pair HFD $\wedge$ ASR are mutually equivalent, and it names the two axioms on which the equivalence rests. A reader who then wants to know why those axioms hold should proceed to Chapters 46 and 15; a reader who wants to know why the equivalence does not finish the problem should proceed directly to Chapter 51, which is thirteen pages and can be read on its own.

Route I — The full accounting

For the reader who wants neither the construction nor the computations but only the ledger of what is and is not achieved: Chapters 48, 51, 30, 56, 64, 67, 69, 70. Chapter 48 is the obstruction ledger, of whose nine entries only two remain open; Chapter 51 states the surviving gap; Chapters 30, 56 and 64 are the three mandatory honest verdicts — that the trace reformulation produces no cycles, that no integer-relation algorithm can *certify* a period relation, and that the Hodge Conjecture cannot presently even be *stated* in Lean 4 with mathlib — and Chapters 69 and 70 are the eleven falsifiers and the seven criteria fixed in advance. Chapter 67, the fabricated-citation audit, belongs here too: it is the account of how the volume policed its own sources, and it reports sixty-three fabrications detected and none surviving. A reader who trusts nothing else in the book should still be able to trust this route, because every chapter on it is written to be checked line by line.

Route J — The systems reader

For the reader who cares about the machine and not the mathematics: Parts VIII, IX and X in full, Chapters 52–68, with Appendices J and K. This route treats Polaris-HC as an artifact to be built, run, and audited, and formalisation as a discipline to be practised; it is deliberately independent of the framework, and a reader who thinks the framework wrong can follow it without conceding anything. It ends with the falsification suite of Chapter 68, which is where the machine is turned against the mathematics and made to look for its own refutation.

Route K — A one-semester graduate course

For the instructor: a thirteen-week selection that ends at the gap. Weeks one and two, the problem and the constraints, Chapters 1–4; weeks three and four, the tame essentials, Chapters 5, 6, 9; weeks five through seven, the prism, Chapters 12, 13, 15, 18; week eight, the transcendental input, Chapters 19, 20; weeks nine and ten, the load-bearing part, Chapters 32, 34, 35, 36, 38; week eleven, the analytic engine, Chapters 39, 44; weeks twelve and thirteen, the synthesis, Chapters 46, 47, 48, 51. The course omits Parts IV, VIII, IX and X, any of which can replace two weeks for a differently-inclined class; the reduction to abelian varieties of Chapter 49 makes a natural final lecture or reading assignment.

Route L — A research seminar on the gap

For the reader who wants to attack the problem rather than survey it: Chapters 37, 39, 40, 41, 42, 44, 45, 51, 69. Everything on this route is oriented to the single surviving analytic obstruction, positivity matching at Step B of the Twistor–Current Rigidification Machine. Chapter 37 supplies the Voisin obstruction that any construction must respect; Chapters 39–45 are the machinery that surrounds the gap on both sides; Chapter 51 states it; and Chapter 69 lists the experiments whose failure would show the surrounding machinery is itself broken. A seminar that closes this gap in a single case will have done what this book could not.

Route M — The second pass

For the reader who has followed one of the routes above and wishes to consolidate before committing to the whole: read the master theorem together with the three honest verdicts, the statement of the gap, and the obstruction ledger — Chapters 47, 30, 56, 64, 51, 48 — and then, and only then, the case studies of Chapter 50. On a second pass the case studies read differently: a reader who has internalised where the gap sits will see, in each worked example, exactly which step is unconditional and which step waits on positivity, and the examples stop looking like evidence for the conjecture and start looking like an atlas of the obstruction. This is the reading the book was ultimately written to support.

What each part assumes

The routes above concern internal order; a reader also needs to know what external background each part takes for granted, since the volume is long enough that no reader will have all of it fresh. The following table records the mathematical prerequisites, part by part, with the appendix that supplies them where one exists. Nothing outside these prerequisites is assumed without a reference, and where a prerequisite is used only once the point of use is named in the text.

Part	Background assumed	Appendix
0	Algebraic geometry and Hodge theory at graduate level.	—
I	Elementary logic; o-minimality is built from scratch.	F
II	Sheaf cohomology and the language of descent.	—
III	Complex geometry and variations of Hodge structure.	—
IV	Category theory; tensor and Tannakian categories.	G
V	Periods and elementary transcendence; Parts II–III.	—
VI	Several complex variables, currents, positivity; $\mathcal{D}$-modules.	H
VII	The whole of Parts II, III, V, VI; no new background.	—
VIII	Numerical and exact linear algebra over number fields.	J
IX	Software-architecture literacy; no new mathematics.	—
X	Dependent type theory and the Lean 4 system.	K
XI	Nothing beyond Part VII.	L

Dependencies

The following table records, for each part, what must be read first.

Part	Requires	Comment
0	—	Self-contained.
I	0	Appendix F for the model theory.
II	0, I	The core construction. Chapter 16 may be skipped.
III	0	Independent of I and II; classical.
IV	III	Appendix G for Tannakian formalism.
V	II, III	The load-bearing part.
VI	III	Appendix H for $\mathcal{D}$-modules.
VII	II, III, V, VI	The synthesis. Part IV is used only in §47.6.
VIII	0	Independent of everything else.
IX	VIII	—
X	0	Chapter 65 refers forward to Parts II–VII.
XI	VII	—

Three genuine dependencies deserve to be spelled out in prose, because the table compresses them. First, everything in Part VII rests on Chapter 13, the construction of the prism, and on Chapter 34, the Rigidification Conjecture: the master theorem of Chapter 47 is a statement about the axioms of Chapter 46, but those axioms are interesting only because Chapter 13 exhibits an object satisfying ten of them. A reader who skips Chapter 13 will find Chapter 47 formally intelligible and mathematically weightless. Second, the two negative results of the subject are consumed at exactly two places: Chapter 36 uses projectivity and Chapter 38 produces denominators, and every later appeal to "projective" or to "rational coefficients" traces back to one of these two chapters. A reader tracking where the wall of Voisin and the wall of Atiyah–Hirzebruch enter the argument need watch only those two. Third, Part IV is used in Part VII at a single point, §47.6, through the Trace–Hodge Conjecture of Chapter 29; a reader uninterested in motives may read that one section and skip the rest of Part IV without loss. The remaining cross-references are conveniences, not dependencies: where a later chapter cites an earlier one for a lemma, the citation is precise enough that the lemma can be imported without its surrounding chapter.

The forward references run the other way and are worth flagging so that they do not surprise. Chapter 44 refers forward to the concentration principle of Chapter 45, which sharpens it; Chapter 49 refers forward to the case studies of Chapter 50; and the roadmap of Chapter 65 refers forward across the whole of Parts II–VII, because its purpose is to say what a formalisation would have to build and in what order. None of these forward references is circular; each is a promissory note redeemed later in the same volume, and the reader who dislikes forward references may read the pointed-to chapter first without disturbing the logic.

The twelve appendices are consulted, not read: Appendix F recalls o-minimality for Part I, Appendix G the Tannakian formalism for Part IV, Appendix H the $\mathcal{D}$-module theory for Part VI, Appendix J the error analysis for Part VIII, and Appendix K the Lean 4 preliminaries for Part X. Appendix L — the tables, glossary, and index of conjectures — is the one appendix worth reading on its own, since it collects in one place every named conjecture, every axiom, and the current disposition of every obstruction.

On the plates

Each of the twelve parts carries three full-page plates: a *mathematical principle* plate, a *system architecture* plate, and an *algorithm architecture* plate. They are not illustrations of the text; they are independent presentations of the same content, and in several cases a plate carries information — a complete dependency graph, a full module interface, a comparative table — that is not repeated in prose. They were authored in HTML and CSS and rendered to vector graphics; in this edition each is set at the largest size its page allows, under a caption that states what it shows and what it does not show.

There are thirty-six plates in all, three to a part, numbered by the chapter that carries them, so that the principle, system, and algorithm plates of a part sit together. The reader should treat them as a parallel text: on a first pass the plates alone, read in sequence, give a defensible outline of the whole argument, and a reader returning to a part after time away will often recover its content faster from its three plates than from its prose. The principle plate of each part states what that part is trying to establish; the system plate shows how its objects fit together; the algorithm plate shows what a machine would do with them. Where a plate and the text disagree in emphasis the text governs, but I have tried throughout to let the two agree, and a plate that carries information not repeated in prose — a full dependency graph, a complete module interface, a comparative table — is to be read as part of the text. A reader preparing a talk from this book will find the thirty-six plates a ready-made set of slides, and is welcome to use them as such.

Notation and Conventions

Standing conventions

(i) **Varieties.** Unless explicitly stated otherwise, X denotes a *smooth projective variety over* $\mathbb{C}$, connected, of complex dimension n. When the projectivity hypothesis is dropped — which happens only in Chapters 37, 41, 43 — this is announced in the first line of the section. The reader should treat any unannounced appearance of a compact Kähler manifold as an error.

(ii) **Coefficients.** Cohomology is singular cohomology of the associated complex analytic space unless decorated otherwise. $H^k(X)$ with no coefficient ring means $H^k(X, \mathbb{Q})$. Integral statements always carry $\mathbb{Z}$ explicitly, because the distinction is the subject of Chapter 38.

(iii) **Twists.** $\mathbb{Q}(1)$ denotes the Tate Hodge structure of weight -2; $\mathbb{Q}(p) = \mathbb{Q}(1)^{\otimes p}$. The cycle class of a codimension-p cycle lands in $H^{2p}(X, \mathbb{Q}(p))$, which is of weight 0. Chapters that suppress the twist say so.

(iv) **Cycles.** $Z^p(X)$ is the group of algebraic cycles of codimension p; $\mathrm{CH}^p(X)$ is the Chow group modulo rational equivalence; $\mathrm{CH}^p(X)_{\mathbb{Q}} = \mathrm{CH}^p(X) \otimes \mathbb{Q}$. The cycle class map is $\mathrm{cl}^p \colon \mathrm{CH}^p(X)_{\mathbb{Q}} \to H^{2p}(X, \mathbb{Q}(p))$.

(v) **Hodge classes.** $\mathrm{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}(p)) \cap H^{p,p}(X)$. The Hodge Conjecture asserts cl^p surjects onto $\mathrm{Hdg}^p(X)$.

(vi) **Equality of numbered objects.** Theorem numbers are global and never reused. A reference of the form 34 means the chapter; one of the form Theorem 2.1 within a chapter means the local number.

(vii) **Constants and the letter C.** Lowercase $c, c_1, c_2, \ldots$ denote unnamed positive constants whose value may change between occurrences; an effective constant is named and its dependence is displayed. In Hodge-theoretic formulas capital C is the Weil operator. Capital C may also occur after an explicit local declaration as a curve, a chain complex, a Clifford algebra, a named bound such as $C(X, \varepsilon)$, or Standard Conjecture C; these typed uses are never the Weil operator.

(viii) **Grading and complexes.** All complexes are cohomological: differentials raise degree by one, and H^i of a complex is its i-th cohomology. The obstruction complex $\mathrm{Ob}^\bullet(X)$ of Chapter 18 follows this convention, so that its degree-zero cohomology is the group of Hodge classes and every obstruction lives in strictly positive degree.

(ix) **Signs.** The Koszul sign rule is in force: interchanging two symbols of degrees a and b introduces the sign $(-1)^{ab}$. The Weil operator C acts on $H^{p,q}$ as multiplication by i^{p-q}, and a polarisation is a form Q for which the Hermitian form $Q(C\,\cdot, \bar{\cdot})$ is positive definite on primitive classes; this is Voisin's sign convention, not its conjugate.

(x) **Dimension and codimension.** $n = \dim_{\mathbb{C}} X$; a cycle of codimension p has dimension $n-p$; "bidimension (k,k)" and "bidegree $(n-k, n-k)$" name the same current. Hodge classes are indexed by codimension p throughout, and a reader translating a dimension-indexed statement should substitute $k = n - p$.

(xi) **Betti and de Rham.** $H^k_{\mathrm{B}}(X,\mathbb{Q})$ is Betti (singular) cohomology and $H^k_{\mathrm{dR}}(X/\mathbb{C})$ is algebraic de Rham cohomology; the comparison isomorphism between them, after extension to $\mathbb{C}$, is the source of periods. In the bi-Rees prism the corner $(u,v)=(1,1)$ recovers $H^k_{\mathrm{B}}(X,\mathbb{Q})\otimes\mathbb{C}$ with its rational lattice, while the two filtrations transported through the de Rham comparison are recorded by u and v; there is no second "de Rham generic fibre." When a statement is insensitive to the comparison I write simply $H^k(X)$; when it turns on the comparison, as in Parts II and V, the subscript is always shown.

(xii) **Field of definition.** Unless a subfield is named, X is defined over $\mathbb{C}$ and no arithmetic structure is assumed. Where a model over a subfield $k\subset\mathbb{C}$ is used — chiefly in Chapter 22 on absolute Hodge classes — the field is named and the action of $\mathrm{Aut}(\mathbb{C}/k)$ is displayed, because that action is exactly what "absolutely Hodge" constrains and what "algebraic" would constrain more strongly.

Principal symbols

The three indices at the end of the volume are complete; this table lists only the symbols that recur across parts.

Symbol	Introduced	Meaning
$\mathbb{Z},\mathbb{Q},\mathbb{R},\mathbb{C}$	—	the usual rings and fields
$(H_{\mathbb{Z}},F^\bullet)$, $(V_{\mathbb{Q}},F^\bullet)$	1	an integral, respectively rational, Hodge structure and its decreasing Hodge filtration
Q, C	1	a polarisation and its Weil operator, with $C\vert_{H^{p,q}}=i^{p-q}$
$\mathrm{Hdg}^p(X)$	1	rational Hodge classes of type (p,p)
cl^p	1	cycle class map $\mathrm{CH}^p(X)_{\mathbb{Q}}\to H^{2p}(X,\mathbb{Q}(p))$
$\mathrm{HC}(X,p)$	1	the assertion that cl^p is surjective onto Hdg^p
$\mathrm{NS}(X)$, $\mathrm{Pic}(X)$	1	Néron–Severi and Picard groups
D, $\check{D}$	7	a period domain and its compact-dual flag variety
Φ	7	the period map, unless a local object is explicitly declared otherwise
$\mathbb{R}_{\mathrm{an}}$	5	the o-minimal structure $\mathbb{R}_{\mathrm{an}}$ of restricted analytic functions
$\mathbb{R}_{\exp}$	5	the real exponential field $\mathbb{R}_{\exp}$
$\mathbb{R}_{\mathrm{an},\exp}=\mathbb{R}_{\mathrm{an},\exp}$	5	$\mathbb{R}_{\mathrm{an},\exp}$, the default structure for "definable"
$\mathbf{An}^{\mathrm{def}}$	6	category of definable complex analytic spaces
$\mathbf{Tm}$	11	the tame Grothendieck site on $\mathbf{An}^{\mathrm{def}}$
$\Delta(X)$	13	the archimedean prism of X
(u,v)	13	bi-Rees parameters for $F^\bullet$ and $\overline{F}^\bullet$, respectively
$\mathcal{S}_{\mathrm{per}}$	14	Grothendieck site of definable polarised Hodge families
δ_{pr}	15	the prism descent morphism
Rig	14	the rigidification sheaf on $\mathcal{S}_{\mathrm{per}}$
$\mathrm{Ob}^\bullet(X)$	18	the prism obstruction complex
$\mathrm{MT}(V)$	19	Mumford–Tate group of a Hodge structure V
HL^p, S_α	20	the total Hodge locus in codimension p and the locus of a fixed flat class α
$\mathrm{KS}(V)$	23	Kuga–Satake variety
UKS	23	the Universal Kuga–Satake Conjecture
$\mathbf{Mot}_{\sim}(k)$	25	category of pure motives
$\mathbf{TrM}(\mathbb{C})$	28	category of traceable motives
τ^{tr}	29	the trace–cycle map
THC	29	the Trace–Hodge Conjecture

Symbol	Introduced	Meaning
τ^{MU}	38	the natural transformation from complex cobordism to singular cohomology
$\Theta^{p}(X)$	29	coniveau–Hodge defect in codimension p
$\mathfrak{r}$	34	a rigidification datum; this is not the sheaf Rig
RC	34	the Rigidification Conjecture
CP_{p}	45	the Concentration Principle in codimension p
$\mathrm{ob}(\alpha)$	44	obstruction class of the TCRM
TCRM	44	the Twistor–Current Rigidification Machine
AP1–AP12	46	the twelve prism axioms
On	48	the n-th entry of the obstruction ledger
$\mathcal{D}$	43	sheaf of differential operators
$\mathbf{MHM}(X)$	43	mixed Hodge modules on X
$\mathbf{Higgs}(X)$	41	Dolbeault category of Higgs bundles on X
$\mathbf{Higgs}(V)$, $\mathbf{Higgs}(\alpha)$	42	the Rees system of a Hodge structure and its marked lift
$\mathbb{H}^{k}$	1	hypercohomology in degree k; the upper half-plane is $\mathfrak{H}$
HH	27	Hochschild homology
$\mathcal{G}_{\mathrm{mot}}$	30	motivic Galois group in the stated Tannakian setting
POLARIS-HC	59	the POLARIS-HC computational architecture
T0–T3	62	the four levels of the trust ladder
HELM	66	the HELM proof-obligation ledger
CONSILIUM	67	the Consilium construction protocol

Retired register forms.. The audit found three entries in an earlier version of this register that had no matching use in the body. The control sequence `\Def` is an unused operator and never denotes an o-minimal structure; the default structure is $\mathbb{R}_{\mathrm{an,exp}} = \mathbb{R}_{\mathrm{an,exp}}$. The forms $\mathrm{HL}(S,\lambda)$ and $\mathrm{Rig}(\alpha)$ also never occur: the body uses HL^{p} and S_{α} for Hodge loci, the sheaf Rig, and $\mathfrak{r}$ for a rigidification datum. They have therefore not been retained as aliases.

Conventions, by domain

The table above lists symbols; this section fixes the conventions that give them meaning, grouped by the mathematical world each belongs to. Where this volume departs from a standard reference the departure is stated, so that a reader arriving from Deligne, Griffiths, or Voisin can translate without guessing.

Hodge theory

The Hodge decomposition is $H^{k}(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}$ with $\overline{H^{p,q}} = H^{q,p}$, and the Hodge filtration is decreasing, $F^{p}H^{k} = \bigoplus_{p'\geq p} H^{p',k-p'}$. Weights are those of Deligne's mixed Hodge theory: H^{k} of a smooth projective variety is pure of weight k, and the Tate twist $\mathbb{Q}(1)$ has weight -2, so that the class of a codimension p cycle sits in weight zero after twisting by $\mathbb{Q}(p)$. This is the arithmetic normalisation; a reader accustomed to the topological convention, in which cycle classes carry weight $2p$ and no twist appears, should insert the twist explicitly at each cycle class. Polarisations, the Weil operator, the Lefschetz operators L and Λ, and the primitive decomposition follow Voisin's two volumes, which I take as the standard companion throughout Parts 0 and III. An abstract rational Hodge structure is written $(V_{\mathbb{Q}}, F^{\bullet})$; when an integral lattice is part of the datum it is written $(H_{\mathbb{Z}}, F^{\bullet})$ and $H_{\mathbb{Q}} = H_{\mathbb{Z}} \otimes \mathbb{Q}$. The letter Q is reserved for a polarisation. The period domain determined by the Hodge numbers and Q

is D, its compact dual is $\check{D}$, and the period map is Φ. Local matrices, endomorphisms, CM types, and modular polynomials may also be called Φ only after an explicit declaration.

o-minimality and tame geometry

The default o-minimal structure is $\mathbb{R}_{\mathrm{an,exp}} = \mathbb{R}_{\mathrm{an,exp}} = \mathbb{R}_{\mathrm{an,exp}}$ (Chapter 5); "definable" means definable in this structure unless another is named. Definable complex analytic spaces form the category $\mathbf{An}^{\mathrm{def}}$ (Chapter 6), and I follow the conventions of Bakker–Klingler–Tsimerman for definable charts and for the definable GAGA theorems, with one difference of emphasis: because the prism is a *degeneration*, I track the definable structure on the total space of a family and not only on its fibres, so the reader should read "definable" as a property of the family whenever a family is in play. Tameness is always a stated hypothesis and never a property assumed of an arbitrary analytic object.

The prism and the period site

The archimedean prism $\Delta(X)$ (Chapter 13) is a Rees-algebra degeneration of the Hodge filtration; its notation deliberately echoes the prismatic cohomology of Bhatt–Scholze, but the echo is structural and not literal, and Chapter 16 states precisely what is and is not shared. The two commuting $\mathbb{G}_m$-actions have coordinates u and v: u records $F^\bullet$ and v records $\overline{F}^\bullet$. The four distinguished corners are $(0,1)$ with fibre $\mathrm{gr}_F H_{\mathbb{C}}$, $(1,0)$ with fibre $\mathrm{gr}_{\overline{F}} H_{\mathbb{C}}$, $(0,0)$ with the full Hodge decomposition, and $(1,1)$ with the Betti rational structure after extension to $\mathbb{C}$. The period site $\mathcal{S}_{\mathrm{per}}$ (Chapter 14) is a genuine Grothendieck site and not a figure of speech: its coverings are the definable étale-like maps of Chapter 11, and the descent morphism δ_{pr} is the comparison whose failure to be an isomorphism the obstruction complex measures.

Motives and the Tannakian formalism

Pure motives $\mathbf{Mot}_{\sim}(k)$ are taken with respect to an adequate equivalence $\sim$ specified at each use — rational, homological, or numerical — because the Standard Conjectures are precisely the statements that these coincide, and it would be circular to fix them equal by convention. The category of traceable motives $\mathbf{TrM}(\mathbb{C})$ (Chapter 28) is rigid symmetric monoidal, and "trace" in this motivic subsection means its categorical trace. Matrix traces and field traces are written only after the relevant matrix or field extension is named. I follow André's conventions for the Tannakian formalism and Kleiman's for the Standard Conjectures, and I never assume Standard Conjecture B: over $\mathbb{C}$ the Hodge Conjecture implies it (Kleiman 1968) [Kle68], so to assume it would make the central argument circular. The trace–cycle map is always τ^{tr}, while the cobordism transformation of Chapter 38 is τ^{MU}. Bare τ is reserved for an elliptic or modular period ratio. Two classical decorated uses remain: the truncation functors $\tau_{\le n}, \tau_{\ge n}$ and the Baum–Fulton–MacPherson transformation τ_X; their subscripts make them unambiguous.

Controlled standard overloads

The coniveau–Hodge defect is always $\Theta^p(X)$ (or $\Theta(M)$ for a traceable motive). A bare Θ declared locally as an ample theta divisor, and Θ_h or Θ_∇ for curvature, retain their standard meanings. Likewise $\mathbf{Higgs}(X)$ is the Dolbeault category in Chapter 41, whereas $\mathbf{Higgs}(V)$ and $\mathbf{Higgs}(\alpha)$ are the Rees system and marked lift of Chapter 42; the argument records which construction is meant. Hypercohomology is written $\mathbb{H}^k$, the upper half-plane is $\mathfrak{H}$, and Hochschild homology is HH.

Currents and positivity

Currents are continuous linear functionals on smooth forms, graded by bidimension; a current of bidimension (k,k) has bidegree $(n-k,n-k)$, and the two indexings are used interchangeably under the translation fixed above. Positivity is positivity in the sense of Lelong: a positive (p,p)-current pairs non-negatively with the standard positive test forms, and the distinction between a *closed* current and a *closed positive* one is the single most important distinction in Part VI, since it is the content of the surviving gap. I follow Demailly's conventions for currents, Lelong numbers, and the Siu decomposition, and the Bishop extension theorem of Chapter 40 is stated in those terms.

The computational and formal layers

In Parts VIII through X the notation shifts register. Exact quantities live in number fields presented as $\mathbb{Q}[x]/(f)$ with a chosen embedding; approximate quantities carry an explicit precision and, where certified, an explicit error bound in the sense of Chapter 55. The layers of Polaris-HC are named A through D and correspond, respectively, to exact algebra, period computation, relation discovery, and certificate emission; the reader should not confuse these system layers with the four steps A through D of the Twistor–Current Rigidification Machine of Chapter 44, which are a mathematical and not a software decomposition. Where a formal statement is displayed in Lean 4 syntax it is set in a typewriter face and is, unless explicitly marked aspirational, a declaration intended to type-check against the mathlib version fixed in Chapter 64.

Axioms, obstructions, and cross-references

The twelve axioms are written AP1 through AP12 and always refer to the list of Chapter 46; a reference to "AP*k*" with no chapter is a reference to that list. The obstruction ledger is written O1 through O9 and always refers to Chapter 48. A chapter reference is zero-padded — `ch:06`, not `ch:6` — and resolves through `cleveref`, so that "Chapter 47" prints the word "Chapter" automatically; I therefore never write the word by hand before such a reference. Section, theorem, and equation references are local numbers within the referenced chapter, global only in the sense that the chapter number disambiguates them.

Epistemic markers

[KNOWN] theorem of the literature **[CONDITIONAL]** proved here under stated hypotheses **[NEW]** introduced here **[OPEN]** unproved.

These appear immediately after the statement label, e.g. "Theorem 3.4 **[KNOWN]**". Their meaning is fixed in *On the Epistemic Status of This Book*, and it does not vary between chapters.

References and the notes

Citations use the `amsalpha` scheme. Each chapter ends with a section *Notes and References* that gives the provenance of the chapter's material, distinguishes what is classical from what is new, and points to the literature for what has been omitted. Those sections are where attribution lives; the body of the text cites, but does not narrate history.

The volume closes with three separate indices — a subject index, a notation index, and an index of names — and with Appendix L, which collects the tables, the glossary, and the index of conjectures. A symbol whose meaning is wanted quickly should be sought first in the notation index, which points to the page of first use; a term of art should be sought in the glossary of Appendix L, which gives a one-line definition and a cross-reference to the chapter

where the term is fixed. The notation index and the glossary are maintained to agree with this chapter, and any disagreement among the three is an error the author wants reported. A reader who prefers to begin from the tables will find in Appendix L an index of conjectures that records, for every named conjecture in the volume, its symbol, the chapter where it is stated, and its current logical status.

Acknowledgements

A book that spends seventy-two chapters insisting on the difference between a result and a hope owes its first debt to the authors of the results. The tame geometry on which Parts I and II rest is the work of Pila, Wilkie, Bakker, Klingler, and Tsimerman; the definable Chow theorem that makes the tame site possible is theirs, not ours, and this volume is in the strict sense a commentary on it. The Hodge theory is Hodge's, Deligne's, Griffiths', Saito's, and Voisin's. The motivic apparatus of Part IV is Grothendieck's programme as carried forward by Kleiman, Jannsen, and André. The positivity theory of Part VI is Demailly's, Siu's, and Bishop's, and the negative results that hold this volume to honesty are Atiyah and Hirzebruch's, Totaro's, and Voisin's. Where this book has anything new to say, it says it standing on that work; where it has nothing new to say, it tries to say so plainly.

Behind each of these bodies of work stands a longer tradition this book also draws on. The o-minimality that Parts I and II require begins with the model theorists — van den Dries above all — who made "tame topology" a precise subject rather than a programme; the analytic input to the prism would be unthinkable without the theory of variations of Hodge structure of Griffiths [Gri68], Schmid [Sch73], and Cattani–Deligne–Kaplan, whose theorem [CDK95] on the algebraicity of Hodge loci is a principal unconditional structural input adjacent to the conjecture. The period computations of Parts V and VIII rest on transcendence theory and on the classical theory of periods; the Kuga–Satake construction that recurs through Part III is Kuga and Satake's [KS67], with later refinements cited at their points of use. I have tried to attribute each borrowed idea at its point of use, in the Notes and References that close every chapter; the list here is not a substitute for those attributions but an acknowledgement of a debt too broad to localise.

The two counterexamples deserve separate thanks. Voisin's compact Kähler counterexample of 2002 [Voi02a] and the Atiyah–Hirzebruch–Totaro integral counterexamples [AH62; Tot97] are, in the architecture adopted here, not obstacles but load-bearing walls: they are what tell the framework where projectivity must be spent and where denominators must be produced. A framework that could not accommodate them would be a framework that proved too much, and the discipline those two theorems impose on Chapter 36 and Chapter 38 improved every part of the design.

The epistemic apparatus of this volume — the honesty boxes, the obstruction ledger of Chapter 48, the eleven falsifiers of Chapter 69, the seven criteria and their four/two/one verdict in Chapter 70, and the fabricated citation audit of Chapter 67 — exists because large scale mathematical writing assisted by automated systems fails in characteristic and predictable ways. Recording those failures, rather than quietly repairing them, is itself part of the contribution, and Chapter 67 reports the count without flattery: sixty-three fabricated citations detected during construction, and none surviving into the text the reader now holds.

This is also a book with a machine behind it, and the machine has debts of its own. The computational front-end of Parts VIII and IX was built in the open-source computer-algebra and arbitrary-precision tradition, and its two workhorses are named for their inventors: the lattice-reduction algorithm of Lenstra, Lenstra, and Lovász, and the integer-relation algorithm

PSLQ of Ferguson and Forcade. The verdict of Chapter 56, that these can *discover* a period relation but cannot *certify* one, is a limitation of the mathematics and not of the implementations, which are excellent. The formalisation chapters of Part X were written against the Lean 4 proof assistant and its mathlib library, and the dependency audit of Chapter 64 is a snapshot of that library at a fixed date; my debt to the community that builds and maintains it is large, and my conclusion that it cannot yet *state* the Hodge Conjecture is a remark about the mathematics still to be formalised, not a complaint. The volume itself was typeset in LaTeX, Lamport's macro layer over Knuth's TeX; the thirty-six plates were authored in HTML and CSS and rendered to vector graphics by a standard browser engine. None of these tools endorsed the conclusions they were used to reach, and any error in the book is the author's and not theirs.

A word, too, for the discipline that made an automated construction survivable. The protocol of Chapter 67 — adversarial review between independent agents, bibliographic verification against real databases, and the refusal to admit any citation to the text unchecked — is not something the tools volunteered; it is something imposed on them, and imposing it was as much a part of the work as the mathematics. I record it here because a reader is entitled to know that the honesty of a machine-assisted book is not a property of the machines but of the process wrapped around them, and because I hope the process proves more portable than the particular result. The best thanks I can offer the next author who attempts this is a candid account of how it can go wrong, which is what Chapter 67 provides.

The author thanks WiseMotif.ai for the institutional support that made a work of this scale possible, and for insisting throughout that the volume be measured by what it establishes rather than by what it announces. That insistence is the reason the honesty apparatus of this volume grew to the size of its mathematical apparatus, and it is the reason I can offer the book to a reviewer without asking for the benefit of the doubt.

Finally, a word to the reader who came here hoping to find the Hodge Conjecture proved. It is not proved here. What is offered instead is a machine that reduces it to two isolated obstructions, an exact accounting of what those obstructions cost, and a statement — repeated wherever it applies — of precisely how much remains. If that accounting is correct, the reader now knows where to dig. If it is wrong, Chapter 69 explains how to demonstrate it. Either outcome is a better service than a proof that does not survive, and the reader who sets out to find the flaw is, in the spirit of this book, its most welcome collaborator, and the reader for whom, in the end, it was written.

Vincent Liu
WiseMotif.ai

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Part 0

The Problem

A precise statement of the Hodge Conjecture, an audit of what is actually known, and the four design constraints that any purported mechanism must survive.

Chapter 1

The Hodge Conjecture

This chapter states the Hodge Conjecture precisely and records everything a later chapter is permitted to assume about it. We recall singular cohomology of a smooth projective complex variety, the Hodge decomposition and its harmonic-theoretic proof, and Deligne's observation that the Hodge filtration is algebraic; we fix the Tate normalisation $\mathbb{Q}(p)$ under which the class of a codimension-p cycle acquires weight zero; we define Chow groups and the cycle class map, and with them the group $\mathrm{Hdg}^p(X)$ of Hodge classes and the statement $\mathrm{HC}(X,p)$. We then separate the several inequivalent assertions that circulate under the one name — rational, integral, and Grothendieck's corrected coniveau ("generalised") form — and prove in full the one case that is a theorem, the Lefschetz $(1,1)$-theorem for divisors. Four worked examples (an elliptic curve, an abelian surface, a K3 surface, the Fermat quartic) fix the phenomenology. Nothing here is new; the chapter is expository, and its purpose is to make the rest of the book accountable.

1.1 Introduction

The Hodge Conjecture asks whether a purely topological and Hodge-theoretic condition on a cohomology class of a smooth projective complex variety is enough to force that class to come from geometry — from an algebraic subvariety. It is, on its face, a question about a single linear map: the *cycle class map*

$$\mathrm{cl}^p : \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow H^{2p}(X, \mathbb{Q}(p)),$$

whose source is built from subvarieties and whose target is built from topology. The conjecture predicts the image of this map exactly: it is the subspace $\mathrm{Hdg}^p(X)$ of *Hodge classes*, those rational classes of type (p,p). To state this cleanly one must assemble a surprising amount of machinery — the Hodge decomposition, the algebraicity of the Hodge filtration, the Tate twist, rational equivalence — and each of these ingredients carries its own subtlety that reappears, magnified, when one tries to prove the conjecture rather than merely state it. This chapter assembles the machinery and states the conjecture; it proves nothing about the conjecture except the classical case $p = 1$.

We are deliberate about one point that the literature often blurs. There is not one Hodge Conjecture but a small family of related assertions, and they have *different truth values*. The rational statement is open. The integral statement is false. Hodge's own 1950 formulation of a more refined "general" conjecture, phrased through the coniveau (level) filtration, is false, and was corrected by Grothendieck in 1969. A reader who does not keep these apart will mis-hear almost every sentence in Chapter 2. We therefore devote Section 1.8 to stating the variants precisely and saying which is which, and we hold ourselves to the convention, fixed for the whole volume, that the unqualified phrase "the Hodge Conjecture" means the rational statement $\mathrm{HC}(X,p)$ of Conjecture 1.8.2, which is open.

The mathematics recalled here is entirely classical. The Hodge decomposition is due to Hodge [Hod41]; its modern proof rests on the elliptic theory of the Laplacian and the Kähler identities. The algebraic character of the Hodge filtration comes from algebraic de Rham comparison [Gro66], with its Hodge-theoretic formulation developed by Deligne [Del71]. The identification of the group of divisor classes with the integral Hodge classes of degree two is

the Lefschetz $(1,1)$-theorem [Lef24]. The conjecture in codimension p was stated by Hodge at the 1950 International Congress [Hod52]; the version adopted as a Clay Millennium Problem, whose official description was written by Deligne [Del06], fixed the rational statement as the reference form. Our exposition follows Voisin's books [Voi03] and Griffiths–Harris [GH78], to which we refer for every proof we only sketch.

The chapter proceeds as follows. Section 1.2 recalls singular cohomology and the comparison with de Rham cohomology. Section 1.3 states and sketches the Hodge decomposition. Section 1.4 passes to the Hodge filtration and its algebraic de Rham description. Section 1.5 introduces weights and the Tate twist $\mathbb{Q}(p)$ and explains why the cycle class lives in weight zero. Section 1.7 constructs Chow groups and the cycle class map. Section 1.8 defines $\mathrm{Hdg}^p(X)$, states $\mathrm{HC}(X,p)$, and separates it from its false relatives. Section 1.10 proves the $(1,1)$-theorem. Section 1.11 works four examples. Section 1.13 lists what the chapter does not attempt.

1.2 Cohomology of a smooth projective variety

Throughout, X is a smooth projective variety over $\mathbb{C}$ of complex dimension n, which we may and do regard as a compact Kähler manifold of real dimension $2n$: the Fubini–Study metric on a projective embedding $X \hookrightarrow \mathbb{P}^N$ restricts to a Kähler metric on X. Its *singular cohomology* $H^k(X,\mathbb{Z})$ is a finitely generated abelian group; we write $H^k(X,A) = H^k(X,\mathbb{Z}) \otimes A$ for a ring $A \supseteq \mathbb{Z}$ when A is flat, and more carefully as $H^k(X^{\mathrm{an}},A)$ when torsion matters. Poincaré duality gives a perfect pairing $H^k(X,\mathbb{Z})/\mathrm{tors} \times H^{2n-k}(X,\mathbb{Z})/\mathrm{tors} \to \mathbb{Z}$, and the cup product makes $H^\bullet(X,\mathbb{Z})$ a graded-commutative ring.

Two theorems make this topological invariant computable by analysis and by algebra. The first is de Rham's theorem: the natural map from the cohomology of the complex of $\mathbb{C}^\infty$ complex-valued differential forms to $H^k(X,\mathbb{C})$ is an isomorphism,

$$H^k_{\mathrm{dR}}(X^\infty) \xrightarrow{\ \sim\ } H^k(X,\mathbb{C}), \qquad [\,\omega\,] \longmapsto \Big(\gamma \mapsto \textstyle\int_\gamma \omega\Big).$$

The second is Grothendieck's algebraic de Rham theorem: if $\Omega^\bullet_{X/\mathbb{C}}$ denotes the complex of sheaves of algebraic (Kähler) differentials, then the hypercohomology of the analytification computes the same groups, and in fact the algebraic hypercohomology already does:

$$H^k(X,\mathbb{C}) \;\cong\; \mathbb{H}^k(X,\Omega^\bullet_{X/\mathbb{C}}).$$

Thus a class in $H^k(X,\mathbb{C})$ has three faces — a homomorphism on cycles (Betti), a closed form modulo exact forms (de Rham), and a class in algebraic hypercohomology — and the entire difficulty of the Hodge Conjecture lies in the relations forced among these faces. We summarise the comparison in the square

$$\begin{array}{ccc} \mathbb{H}^k(X,\Omega^\bullet_{X/\mathbb{C}}) & \xrightarrow{\ \sim\ } & H^k(X,\mathbb{C}) \\ {\scriptstyle \mathrm{GAGA}}\downarrow & \nearrow_{\sim} & \\ \mathbb{H}^k(X^{\mathrm{an}},\Omega^{\bullet,\mathrm{an}}_X) & & \end{array}$$

in which the horizontal isomorphism is the algebraic de Rham comparison and the diagonal is the analytic one; the vertical map is the comparison of algebraic and analytic hypercohomology furnished by Serre's *GAGA* [Ser56]. That GAGA holds — that coherent algebraic and analytic cohomology on a projective variety agree — is the first place where projectivity, as opposed to mere compactness, does real work; we shall return to this theme obsessively in Chapter 3.

1.3 The Hodge decomposition

Theorem 1.3.1 (Hodge decomposition). **[KNOWN]** [GH78] *Let X be a compact Kähler manifold. For every k there is a direct sum decomposition of complex vector spaces*

$$H^k(X,\mathbb{C}) \;=\; \bigoplus_{p+q=k} H^{p,q}(X), \qquad \overline{H^{p,q}(X)} \;=\; H^{q,p}(X),$$

in which $H^{p,q}(X)$ is the space of classes representable by a closed form of type (p,q), and there are canonical isomorphisms $H^{p,q}(X) \cong H^q(X,\Omega_X^p)$ with the Dolbeault cohomology (Hodge 1941).

Proof sketch. Fix a Kähler metric and let $\Delta_d = dd^* + d^*d$ be the Hodge–de Rham Laplacian on complex-valued forms. The Hodge theorem for the compact Riemannian manifold X asserts that each de Rham class has a unique Δ_d-harmonic representative, so that $H^k(X,\mathbb{C}) \cong \mathcal{H}^k_d := \ker(\Delta_d)$ on k-forms. Decompose $d = \partial + \bar\partial$ according to type and form the two Laplacians $\Delta_\partial, \Delta_{\bar\partial}$. The *Kähler identities* — the commutation relations $[\Lambda,\bar\partial] = -i\partial^*$ and $[\Lambda,\partial] = i\bar\partial^*$ between the adjoint Lefschetz operator Λ and the Dolbeault operators, valid precisely because the metric is Kähler — yield the fundamental identity

$$\Delta_d \;=\; 2\,\Delta_\partial \;=\; 2\,\Delta_{\bar\partial}.$$

Because $\Delta_{\bar\partial}$ preserves the (p,q)-type of a form, so does Δ_d; hence the space $\mathcal{H}^k_d$ of harmonic k-forms splits as the orthogonal direct sum $\bigoplus_{p+q=k} \mathcal{H}^{p,q}$ of its type components, and one *defines* $H^{p,q}(X)$ to be the image of $\mathcal{H}^{p,q}$ in $H^k(X,\mathbb{C})$. The $\bar\partial$-Hodge theorem identifies $\mathcal{H}^{p,q}$ with the Dolbeault group $H^q(X,\Omega_X^p)$. Finally Δ_d is a real operator, so complex conjugation carries $\mathcal{H}^{p,q}$ to $\mathcal{H}^{q,p}$, giving the reality $\overline{H^{p,q}} = H^{q,p}$. For the analytic details see [Voi03, Ch. 6] or [GH78, Ch. 0]. □

The integers $h^{p,q}(X) = \dim_{\mathbb{C}} H^{p,q}(X)$ are the *Hodge numbers*. They satisfy $h^{p,q} = h^{q,p}$ (reality) and, by Serre duality, $h^{p,q} = h^{n-p,n-q}$; the alternating sum along a diagonal recovers the topological Betti number $b_k = \sum_{p+q=k} h^{p,q}$. It is essential to keep in mind that the decomposition of Theorem 1.3.1 is a decomposition of the *complex* vector space $H^k(X,\mathbb{C})$ only; it is not defined over $\mathbb{R}$ or $\mathbb{Q}$, and the interaction between the rational lattice $H^k(X,\mathbb{Q})$ and the complex subspaces $H^{p,q}$ is exactly what the Hodge Conjecture is about.

Remark 1.3.2. **[KNOWN]** [GH78] Theorem 1.3.1 holds for every compact Kähler manifold, not merely for projective X. This generality is a warning, not a convenience: it means the Hodge decomposition, by itself, cannot know whether X is projective, and therefore cannot by itself imply anything that is false in the Kähler category. Voisin's theorem (Chapter 2) shows that "anything false in the Kähler category" includes the Hodge Conjecture. The lesson, pursued in Chapter 3, is that a proof of HC must use more than the Hodge decomposition.

1.4 The Hodge filtration and its algebraicity

Define the decreasing *Hodge filtration* on $H^k(X,\mathbb{C})$ by

$$F^pH^k(X,\mathbb{C}) \;=\; \bigoplus_{p'\geq p} H^{p',k-p'}(X).$$

The pair $(H^k(X,\mathbb{Q}),F^\bullet)$ is the prototype of a *pure Hodge structure of weight k*: a finite-dimensional $\mathbb{Q}$-vector space V together with a decreasing filtration $F^\bullet$ on $V_\mathbb{C}$ such that $V_\mathbb{C} = F^p \oplus \overline{F^{k-p+1}}$ for all p. One recovers the decomposition from the filtration by $H^{p,q} = F^p \cap \overline{F^q}$, so the two data are equivalent; the filtration is the more robust of the two, and it is the datum that survives degeneration and carries a mixed Hodge structure in the sense of Deligne [Del71; Del74b]. We write **HS** for the category of pure $\mathbb{Q}$-Hodge structures; it is abelian but not semisimple in general, and it carries a tensor product with $H^{p,q}\otimes H^{p',q'}$ in bidegree $(p+p',q+q')$.

Combining Grothendieck's algebraic de Rham comparison [Gro66] with the classical Hodge-to-de Rham degeneration shows that the Hodge filtration is not a merely transcendental datum but has an algebraic description.

Theorem 1.4.1 (Algebraicity of the Hodge filtration). *[KNOWN]* [Voi03] *Let X be smooth projective over $\mathbb{C}$. Under the algebraic de Rham isomorphism $H^k(X,\mathbb{C})\cong\mathbb{H}^k(X,\Omega^\bullet_{X/\mathbb{C}})$, the Hodge filtration is the filtration by the (stupid) truncations,*

$$F^pH^k(X,\mathbb{C}) \;=\; \operatorname{Im}\Big(\mathbb{H}^k(X,\Omega^{\geq p}_{X/\mathbb{C}})\longrightarrow \mathbb{H}^k(X,\Omega^\bullet_{X/\mathbb{C}})\Big),$$

and the Hodge–de Rham (Frölicher) spectral sequence $E_1^{p,q}=H^q(X,\Omega^p_X)\Rightarrow H^{p+q}(X,\mathbb{C})$ degenerates at E_1. Consequently, if X is defined over a subfield $k\subseteq\mathbb{C}$, then $F^pH^k_{\mathrm{dR}}(X/k)$ is a k-rational subspace and $F^pH^k(X,\mathbb{C})$ is its extension of scalars.

Proof sketch. The stupid filtration $\sigma^{\geq p}\Omega^\bullet_X$ induces a filtration on hypercohomology whose associated graded is $\bigoplus_q H^{k-p}(X,\Omega^p_X)[-p]$, i.e. the E_1-page of the Frölicher spectral sequence. Degeneration at E_1 is equivalent to the equality $\sum_{p+q=k}h^{p,q}=b_k$ of dimensions, which holds by Theorem 1.3.1; degeneration then identifies $\mathrm{gr}_F^pH^k$ with $H^{k-p}(X,\Omega^p)=H^{p,k-p}$, so the truncation filtration is the Hodge filtration. Every object in sight — the de Rham complex, its truncations, the hypercohomology — is algebraic and defined over any field of definition of X; hence so is F^p. See [Voi03, §8]. □

Theorem 1.4.1 is the seed of the entire volume. It is the reason a transcendental-looking object, the position of $H^{p,q}$ inside $H^k(X,\mathbb{C})$, can be studied by arithmetic and by algebra; it is the reason the period map is definable in an o-minimal structure (Chapters 5 and 20); and it is the reason the *archimedean prism* of Chapter 13 exists as a definable, and not merely analytic, object. A general compact Kähler manifold has a Hodge filtration but no algebraic model, hence no analogue of Theorem 1.4.1; this is one precise sense in which projectivity is stronger than the Kähler condition, and it is the sense the book exploits.

1.5 Weights, polarisation, and the Tate twist

A morphism of Hodge structures is a $\mathbb{Q}$-linear map respecting $F^\bullet$; it is automatically strict, and **HS** is abelian. To compare cohomology in different degrees one needs to move weights, and the device for this is the Tate structure.

Definition 1.5.1. The *Tate–Hodge structure* $\mathbb{Q}(1)$ is the pure Hodge structure of weight -2 with underlying vector space $(2\pi i)\mathbb{Q} \subset \mathbb{C}$ and $\mathbb{Q}(1)_{\mathbb{C}} = H^{-1,-1}$; for $m \in \mathbb{Z}$ set $\mathbb{Q}(m) = \mathbb{Q}(1)^{\otimes m}$, of weight $-2m$ and pure type $(-m,-m)$. For a Hodge structure H of weight k, the *Tate twist* is $H(m) = H \otimes \mathbb{Q}(m)$, of weight $k-2m$, with $H(m)^{p,q} = H^{p+m,\,q+m}$.

The point of the twist is normalisation. Integration of a $(1,1)$-form over an algebraic curve produces a factor of $2\pi i$; keeping track of these factors, the fundamental class of a codimension-p subvariety naturally lands not in $H^{2p}(X,\mathbb{Q})$ but in the twisted group

$$H^{2p}(X,\mathbb{Q}(p)),$$

which is a Hodge structure of weight $2p-2p=0$, and the class is of pure type $(0,0)$ inside it. This is the invariant statement: *cycle classes have weight zero*. Equivalently, and this is the phrasing we shall use most, a Hodge class of codimension p is a morphism of Hodge structures from the trivial structure $\mathbb{Q}(0)=\mathbb{Q}$ into $H^{2p}(X,\mathbb{Q})(p)$,

$$\mathrm{Hdg}^p(X) \;=\; \mathrm{Hom}_{\mathbf{HS}}\big(\mathbb{Q}(0),\, H^{2p}(X,\mathbb{Q})(p)\big).$$

The reader who prefers to avoid twists may replace $H^{2p}(X,\mathbb{Q})(p)$ by $H^{2p}(X,\mathbb{Q})$ and "type $(0,0)$" by "type (p,p)" everywhere; the two bookkeepings are equivalent, and we shall pass between them without comment. The advantage of the twisted normalisation is functoriality under correspondences: a codimension-c cycle on $X \times Y$ induces maps $H^k(X) \to H^{k+2c-2\dim X}(Y)(c-\dim X)$ that are morphisms of Hodge structures on the nose, with no stray weights.

A projective embedding also equips $H^k(X,\mathbb{Q})$ with a *polarisation*: a morphism of Hodge structures $Q\colon H^k \otimes H^k \to \mathbb{Q}(-k)$, induced by cup product with the hyperplane class, satisfying the Hodge–Riemann bilinear relations. Polarisable Hodge structures form a semisimple abelian category, a fact we shall lean on repeatedly. The polarisation is the archimedean shadow of the ample cone, and Chapters 3 and 36 identify it as the exact location where projectivity enters the book's proposed mechanism.

1.6 Lefschetz operators and Hard Lefschetz

Fix an ample class $h \in \mathrm{Hdg}^1(X)$, the class of a hyperplane section. Cup product with h defines the *Lefschetz operator* $L\colon H^k(X,\mathbb{Q}) \to H^{k+2}(X,\mathbb{Q})(1)$, a morphism of Hodge structures of bidegree $(1,1)$. Its iterates control the whole cohomology ring.

Theorem 1.6.1 (Hard Lefschetz; primitive decomposition). *[KNOWN] For a compact Kähler manifold X of dimension n and each $0 \le k \le n$, the iterated Lefschetz operator*

$$L^{n-k}\colon H^k(X,\mathbb{Q}) \xrightarrow{\;\sim\;} H^{2n-k}(X,\mathbb{Q})(n-k)$$

is an isomorphism of Hodge structures. Consequently the primitive subspaces $P^k = \ker\big(L^{n-k+1}\colon H^k \to H^{2n-k+2}(n-k+1)\big)$ give the Lefschetz decomposition $H^k(X,\mathbb{Q}) = \bigoplus_{j \ge \max(0,k-n)} L^j P^{k-2j}$, *and each summand is a sub-Hodge structure* (*Hodge; see* [Voi03, *Ch. 6*]).

Proof sketch. On harmonic forms the operators L, its adjoint Λ, and the counting operator $H = \sum_k (k-n)\Pi^k$ generate a representation of $\mathfrak{sl}_2(\mathbb{C})$ commuting with the Laplacian, by the Kähler identities. Finite-dimensional $\mathfrak{sl}_2$-representation theory then makes L^{n-k} an isomorphism from the weight-$(k-n)$ to the weight-$(n-k)$ space, which are H^k and H^{2n-k}; that L is a $(1,1)$-morphism makes the isomorphism one of Hodge structures. See [Voi03, §6.2]. □

Theorem 1.6.1 is the engine behind the second solved case of the conjecture. Because L is the cup product with an algebraic class, it carries algebraic classes to algebraic classes; and because L^{n-k} is an *isomorphism* of rational Hodge structures, it carries Hodge classes to Hodge classes bijectively. Iterating L therefore transports the codimension-one theorem of Section 1.10 up the cohomological ladder, and we shall see in Chapter 2 that this single observation resolves the Hodge Conjecture in codimension $p = n-1$ and in all dimensions ≤ 3. The obstruction to going further is that inverting L on a given class, while possible over $\mathbb{Q}$ for the specific classes just described, is not known to be induced by an algebraic correspondence in general; that this inversion is algebraic is the unresolved Lefschetz standard conjecture B, implied by HC (Chapter 26).

Remark 1.6.2 (The cycle class map is very far from injective). **[KNOWN]** [Ful98] Surjectivity of cl^p onto Hdg^p is the Hodge Conjecture; injectivity fails spectacularly and unconditionally. Mumford proved that for a smooth projective surface S with $h^{2,0}(S) > 0$ the kernel of $\mathrm{CH}^2(S)_{\mathbb{Q}} \to H^4(S,\mathbb{Q})$ — the group of degree-zero zero-cycles — is "infinite-dimensional", not representable by any algebraic parameter space. Thus $\mathrm{CH}^p(X)_{\mathbb{Q}}$ is an enormous, largely transcendental object, and the Hodge Conjecture is a statement only about its finite-dimensional image. This is why the conjecture is a question about cohomology, not about Chow groups: the information lost by cl^p is precisely the part no cohomological condition can detect. The structure of $\ker(\mathrm{cl}^p)$ is the province of the conjectural Bloch–Beilinson filtration (Chapters 3 and 31).

1.7 Algebraic cycles, Chow groups, and the cycle class map

Let $Z^p(X)$ be the free abelian group on the integral closed subvarieties of X of codimension p; its elements are *algebraic cycles* of codimension p. Two cycles are *rationally equivalent* if their difference is a sum of divisors of nonzero rational functions on $(p-1)$-codimensional subvarieties: precisely, $Z^p_{\mathrm{rat}}(X)$ is generated by $\div(f)$ for $W \subset X$ a codimension-$(p-1)$ subvariety and $f \in \mathbb{C}(W)^\times$, where $\div(f) = \sum_V \mathrm{ord}_V(f)\,[V]$.

Definition 1.7.1. The *Chow group* of codimension-p cycles is $\mathrm{CH}^p(X) = Z^p(X)/Z^p_{\mathrm{rat}}(X)$. The total Chow group $\mathrm{CH}^\bullet(X) = \bigoplus_p \mathrm{CH}^p(X)$ is a commutative graded ring under the intersection product, contravariant for arbitrary morphisms and covariant for proper morphisms; we write $\mathrm{CH}^p(X)_{\mathbb{Q}} = \mathrm{CH}^p(X) \otimes \mathbb{Q}$ (Fulton [Ful98]).

Theorem 1.7.2 (Cycle class map). **[KNOWN]** [Ful98] *There is a homomorphism of graded rings*

$$\mathrm{cl}\colon \mathrm{CH}^\bullet(X) \longrightarrow \bigoplus_p H^{2p}(X, \mathbb{Z}(p)),$$

functorial for pullback and proper pushforward, sending an irreducible subvariety Z of codimension p to its fundamental cohomology class $[Z] \in H^{2p}(X,\mathbb{Z}(p))$. Its image in $H^{2p}(X,\mathbb{C})$ lies in F^pH^{2p} and, being a real class, in $F^p \cap \overline{F^p} = H^{p,p}$; hence cl^p factors through the Hodge classes.

Proof sketch. For smooth Z the class $[Z]$ is Poincaré dual to the Gysin image of 1 under $Z \hookrightarrow X$; for singular Z one resolves or uses the theory of currents, integrating $(n-p,n-p)$-forms over the smooth locus, which has finite volume and defines a closed current of type (p,p) (King [Kin71]). Compatibility with intersection and pullback is [Ful98, Ch. 19]. That $[Z]$ is of type (p,p): the current of integration kills forms of type (a,b) with $a > p$ or $b > p$ for dimension reasons, so it pairs trivially outside $H^{p,p}$, whence $[Z] \in H^{p,p}$. Rationality is built in, since $[Z]$ is an integral class. □

Passing to $\mathbb{Q}$-coefficients and twisting to weight zero, we obtain the map whose image the conjecture predicts:

$$\mathrm{cl}^p : \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow H^{2p}(X, \mathbb{Q}(p)), \qquad \mathrm{Im}(\mathrm{cl}^p) \subseteq \mathrm{Hdg}^p(X).$$

The containment is Theorem 1.7.2. The Hodge Conjecture asserts it is an equality.

1.8 The statement, and its inequivalent siblings

Definition 1.8.1. The group of *Hodge classes* of codimension p on X is

$$\begin{aligned}\mathrm{Hdg}^p(X) \;&=\; H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X) \;=\; H^{2p}(X,\mathbb{Q}(p)) \cap H^{0,0} \\ &=\; \mathrm{Hom}_{\mathbf{HS}}(\mathbb{Q}(0), H^{2p}(X,\mathbb{Q})(p)),\end{aligned}$$

the intersection taken inside $H^{2p}(X,\mathbb{C})$. Its integral refinement is $\mathrm{Hdg}^p(X,\mathbb{Z}) = H^{2p}(X,\mathbb{Z}) \cap H^{p,p}(X)$, the preimage of $H^{p,p}$ under $H^{2p}(X,\mathbb{Z}) \to H^{2p}(X,\mathbb{C})$.

Conjecture 1.8.2 (Hodge, rational form). *[OPEN] For every smooth projective complex variety X and every p, the cycle class map is surjective onto Hodge classes:*

$$\mathrm{cl}^p : \mathrm{CH}^p(X)_{\mathbb{Q}} \twoheadrightarrow \mathrm{Hdg}^p(X).$$

Equivalently, every rational Hodge class is a $\mathbb{Q}$-linear combination of fundamental classes of algebraic subvarieties. We write $\mathrm{HC}(X,p)$ *for this statement and* HC *for its assertion for all X and all p.*

The statement HC is open, and the standing convention of this book is that "the Hodge Conjecture" means Conjecture 1.8.2. It is important to isolate this form from three neighbours with which it is routinely confused.

The integral form is false

The *integral Hodge Conjecture* $\mathrm{HC}_{\mathbb{Z}}(X,p)$ asserts that $\mathrm{cl}\colon \mathrm{CH}^p(X) \to \mathrm{Hdg}^p(X,\mathbb{Z})$ is surjective. This is *false*, and it fails in two independent ways: through torsion, by Atiyah–Hirzebruch (1962), and through non-torsion degree obstructions, by Kollár (1992) [AH62; Kol92]. Totaro's 1997 complex-cobordism factorisation refines the torsion-sensitive topological obstruction [Tot97]. We treat these results in Chapter 2. For now the moral suffices: any mechanism that would manufacture an *integral* cycle from an integral Hodge class is refuted before it begins, so every construction in this book must tolerate a denominator. We call this the Integrality Test and elevate it to a design constraint in Chapter 3.

Hodge's coniveau form, and Grothendieck's correction

In 1950 Hodge proposed a more refined "general" conjecture governing the *coniveau* (or level) filtration. The geometric coniveau filtration on $H^k(X,\mathbb{Q})$ is

$$N^cH^k(X,\mathbb{Q}) \;=\; \sum_{Z} \ker\bigl(H^k(X,\mathbb{Q}) \to H^k(X\smallsetminus Z,\mathbb{Q})\bigr),$$

the sum over closed subvarieties Z of codimension $\geq c$; a class of coniveau $\geq c$ is one supported, in this cohomological sense, on a subvariety of codimension $\geq c$. Hodge's original formulation asserted that $N^cH^k(X,\mathbb{Q})$ coincides with the largest rational subspace whose complexification lies in F^cH^k. Grothendieck observed in 1969, in a paper pointedly titled *Hodge's general conjecture is false for trivial reasons* [Gro69a], that F^cH^k is not a sub-Hodge structure — it is not stable under complex conjugation — so "the largest rational subspace in F^c" is the wrong object, and Hodge's equality fails already for elementary reasons. The corrected statement, the *generalised Hodge Conjecture* $\mathrm{GHC}(X,k,c)$, replaces F^c by the largest sub-Hodge structure of $H^k(X,\mathbb{Q})$ whose complexification is contained in F^cH^k, and conjectures that this coincides with $N^cH^k(X,\mathbb{Q})$. The generalised conjecture remains unresolved; the case $c = k = 2p$ recovers a statement equivalent to $\mathrm{HC}(X,p)$. This episode is not a footnote: it is the first instance of a recurring phenomenon, that the naive Hodge-theoretic guess is refuted by the failure of a filtration to respect the rational structure, and that the correct guess restores the rational structure by force. Chapter 3 makes compatibility with Grothendieck's correction into the third design constraint, the Coniveau Test.

The Chern-class form

Because Chern classes of vector bundles are polynomials in cycle classes, one may ask instead whether every rational Hodge class is a combination of Chern classes of algebraic vector bundles, or of coherent sheaves. Over a projective X this is equivalent to Conjecture 1.8.2, by GAGA and the Grothendieck –Riemann–Roch theorem. Over a compact Kähler manifold, where algebraic subvarieties are scarce but coherent analytic sheaves are plentiful, the two forms diverge, and Voisin's counterexample (Chapter 2) refutes even the coherent-sheaf form. We flag the distinction here because it is the crux of the Voisin Test in Chapter 3.

1.9 Four equivalent formulations

The rational conjecture $\mathrm{HC}(X,p)$ of Conjecture 1.8.2 circulates in several guises, and a reader who moves between the research literature and this book will meet all of them. They are equivalent, but the equivalences are not tautologies: two of the forms are statements about a single variety, while two are naturally statements about all varieties at once, and passing between the two kinds uses the conjecture for products. We collect the four forms and prove their equivalence, so that no later chapter has to reprove it.

Proposition 1.9.1 (The four formulations). *[KNOWN]* [Voi03] *Fix the class of all smooth projective complex varieties. The following assertions, each quantified over every such variety and every p, are equivalent.*

(F1) Cycle-class form. *The map* $\mathrm{cl}^p\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \to H^{2p}(X,\mathbb{Q}(p))$ *is surjective onto* $\mathrm{Hdg}^p(X)$.

(F2) Span form. $\mathrm{Hdg}^p(X)$ *is the $\mathbb{Q}$-span of the fundamental classes* $[Z]$ *of irreducible closed subvarieties $Z \subseteq X$ of codimension p.*

(F3) Correspondence form. *For all smooth projective X, Y, all i, j and all Tate twists, every morphism of rational Hodge structures $f\colon H^i(X,\mathbb{Q})(a) \to H^j(Y,\mathbb{Q})(b)$ is induced by an algebraic correspondence $\Gamma \in \mathrm{CH}^*(X\times Y)_{\mathbb{Q}}$.*

(F4) Motivic form. *The Hodge realisation functor* $R\colon \mathbf{Mot}_{\mathrm{hom}}(\mathbb{C})_{\mathbb{Q}} \to \mathbf{HS}_{\mathbb{Q}}$*, from the category of homological motives with algebraic correspondences as morphisms to polarisable* $\mathbb{Q}$*-Hodge structures, is full.*

Proof. (F1)$\Leftrightarrow$(F2) is immediate: the image of cl^p is by definition the $\mathbb{Q}$-span of the classes $[Z]$ (Theorem 1.7.2), so surjectivity onto Hdg^p is the equality of that span with Hdg^p.

(F2)$\Rightarrow$(F3). Let $f\colon H^i(X)(a) \to H^j(Y)(b)$ be a morphism of Hodge structures, with $\dim X = n$. By Poincaré duality $H^i(X)^\vee \cong H^{2n-i}(X)(n)$, so f is classified by an element

$$\varphi_f \;\in\; (H^{2n-i}(X) \otimes H^j(Y))(n-a+b) \;\subseteq\; H^{2n-i+j}(X\times Y)(n-a+b)$$

by the Künneth formula, and f is a morphism of Hodge structures precisely when φ_f is of type (c,c) with $c = n-a+b$, i.e. when $\varphi_f \in \mathrm{Hdg}^c(X\times Y)$. Applying (F2) on the product $X\times Y$ writes φ_f as a $\mathbb{Q}$-combination of classes of codimension-c subvarieties, that is, of an algebraic correspondence Γ; and by construction $\Gamma_* = f$. Thus every Hodge morphism is algebraic.

(F3)$\Rightarrow$(F1). Take $X' = \mathrm{pt}$, so $H^0(\mathrm{pt}) = \mathbb{Q}(0)$, and read a Hodge class $\alpha \in \mathrm{Hdg}^p(X) = \mathrm{Hom}_{\mathbf{HS}}(\mathbb{Q}(0), H^{2p}(X)(p))$ as a morphism of Hodge structures $H^0(\mathrm{pt}) \to H^{2p}(X)(p)$. The correspondence form makes it induced by an algebraic cycle on $\mathrm{pt}\times X = X$ of codimension p, which is exactly a class in the image of cl^p. Hence α is algebraic and (F1) holds.

(F3)$\Leftrightarrow$(F4). By definition the morphisms in $\mathbf{Mot}_{\mathrm{hom}}(\mathbb{C})_{\mathbb{Q}}$ between the motives of X and Y are the algebraic correspondences modulo homological equivalence, while the morphisms between their Hodge realisations are the Hodge morphisms; fullness of R is the assertion that every Hodge morphism is the realisation of a correspondence, which is (F3). That R is faithful on homological motives is automatic, homological equivalence being defined by the realisation. □

Remark 1.9.2 (*The Tannakian reading*). [CONDITIONAL] Granting Grothendieck's standard conjecture that the Künneth components of the diagonal are algebraic — known unconditionally for abelian varieties, and implied by HC itself (Kleiman [Kle68]) — the category $\mathbf{Mot}_{\mathrm{hom}}(\mathbb{C})_{\mathbb{Q}}$ is Tannakian, and each motive M acquires two affine group schemes: its *Mumford–Tate group* $\mathrm{MT}(M)$, the Tannakian group of the Hodge structure $R\,M$, and its *motivic Galois group* $G_{\mathrm{mot}}(M)$. The realisation R induces a closed immersion $\mathrm{MT}(M) \hookrightarrow G_{\mathrm{mot}}(M)$ for every M, and fullness of R — form (F4) — is precisely the statement that this immersion is an isomorphism for all M. The Hodge Conjecture is thus, in the Tannakian dictionary, the equality of the Mumford–Tate and motivic Galois groups; this is the vantage from which Chapters 22 and 25 operate. Unconditionally, replacing homological by André's motivated motives [And96] makes the category Tannakian with no hypothesis, at the cost that fullness of R there is the weaker statement Mtd = Hdg rather than Alg = Hdg; the gap between the two is the fourth design constraint of Chapter 3.

1.10 Divisors: the Lefschetz $(1,1)$-theorem

The one case of the Hodge Conjecture that is a theorem is $p = 1$, and it is a theorem in the strong, integral form. We give the proof in full; it is short, it is the template for every later reduction to divisors, and it exhibits the exponential sequence that recurs throughout the book.

Theorem 1.10.1 (Lefschetz $(1,1)$-theorem). [KNOWN] [Lef24] *Let X be a compact Kähler manifold. The first Chern class*

$$c_1\colon \mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^\times) \longrightarrow H^2(X,\mathbb{Z})$$

has image exactly $\mathrm{Hdg}^1(X,\mathbb{Z}) = H^2(X,\mathbb{Z}) \cap H^{1,1}(X)$. *In particular the Néron–Severi group* $\mathrm{NS}(X) = \mathrm{Im}(c_1)$ *is the group of integral Hodge classes of degree two, and* $\mathrm{HC}_{\mathbb{Z}}(X,1)$ *holds: every degree-two integral Hodge class is the class of a divisor* (*Lefschetz 1924*).

Proof. Consider the *exponential sheaf sequence* of the analytic manifold X,

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}_X^{\times} \longrightarrow 0,$$

where $\mathbb{Z}$ is the constant sheaf and the second map is $f \mapsto e^{2\pi i f}$, with kernel the locally constant integer-valued functions. Its long exact cohomology sequence contains the segment

$$H^1(X,\mathcal{O}_X) \longrightarrow H^1(X,\mathcal{O}_X^{\times}) \xrightarrow{c_1} H^2(X,\mathbb{Z}) \xrightarrow{\iota} H^2(X,\mathcal{O}_X),$$

in which we have identified the connecting map with the first Chern class c_1 and $\mathrm{Pic}(X) = H^1(X,\mathcal{O}_X^{\times})$. By exactness,

$$\mathrm{Im}(c_1) \;=\; \ker\bigl(\iota\colon H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)\bigr).$$

Now $H^2(X,\mathcal{O}_X) = H^{0,2}(X)$ by Dolbeault's theorem, and under $H^2(X,\mathbb{Z}) \to H^2(X,\mathbb{C}) = H^{2,0} \oplus H^{1,1} \oplus H^{0,2}$ the map ι is the projection to the $H^{0,2}$ summand. Thus $\mathrm{Im}(c_1)$ consists of the integral classes α with $\alpha^{0,2} = 0$. For such an α, being the image of a *real* (indeed integral) class, we also have $\alpha^{2,0} = \overline{\alpha^{0,2}} = 0$; hence $\alpha \in H^{1,1}$. Conversely any integral class in $H^{1,1}$ has $\alpha^{0,2} = 0$ and so lies in $\ker \iota = \mathrm{Im}(c_1)$. Therefore $\mathrm{Im}(c_1) = H^2(X,\mathbb{Z}) \cap H^{1,1} = \mathrm{Hdg}^1(X,\mathbb{Z})$. Finally every element of $\mathrm{Pic}(X)$ is represented by a holomorphic line bundle, hence (on projective X, by GAGA) by an algebraic line bundle $\mathcal{O}_X(D)$ for a divisor D, and $c_1(\mathcal{O}_X(D)) = [D]$; so each integral $(1,1)$-class is a $\mathbb{Z}$-combination of divisor classes. □

Corollary 1.10.2. ***[KNOWN]*** [*Voi03*] $\mathrm{HC}(X,1)$ *holds for every smooth projective* X*: the rational Hodge classes of codimension one are exactly the* $\mathbb{Q}$*-combinations of divisor classes, and* $\mathrm{NS}(X)_{\mathbb{Q}} = \mathrm{Hdg}^1(X)$.

Proof. Tensor Theorem 1.10.1 with $\mathbb{Q}$; the intersection with $H^{1,1}$ is unchanged and torsion dies. □

The rank of $\mathrm{NS}(X)$ is the *Picard number* $\rho(X)$. Corollary 1.10.2 says $\rho(X) = \dim_{\mathbb{Q}} \mathrm{Hdg}^1(X)$; the gap between $\rho(X)$ and the a priori bound $h^{1,1}(X)$ is precisely the room available to the geometry of X, and its behaviour in families is the subject of the theory of Hodge loci (Chapter 20).

1.10.1 A worked unwinding: the exponential sequence made explicit

It repays the effort to watch the exponential sequence deliver the theorem on a named surface, where every term is computable. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, the smooth quadric surface. Its Hodge numbers are $h^{0,0} = h^{2,2} = 1$, $h^{1,1} = 2$, and $h^{2,0} = h^{0,2} = h^{1,0} = h^{0,1} = 0$; in particular

$$H^1(X,\mathcal{O}_X) = H^{0,1} = 0, \qquad H^2(X,\mathcal{O}_X) = H^{0,2} = 0.$$

The exponential sequence's long exact segment

$$H^1(X,\mathcal{O}_X) \to \mathrm{Pic}(X) \xrightarrow{c_1} H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$$

therefore reads $0 \to \mathrm{Pic}(X) \xrightarrow{c_1} H^2(X,\mathbb{Z}) \to 0$: the first Chern class is an *isomorphism*. Now $H^2(X,\mathbb{Z}) \cong \mathbb{Z}^2$, generated by the classes $f_1 = [\mathbb{P}^1 \times \mathrm{pt}]$ and $f_2 = [\mathrm{pt} \times \mathbb{P}^1]$ of the two rulings, with intersection matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; and $H^{1,1}(X)$, being all of $H^2(X,\mathbb{C})$ apart from the vanishing $(2,0)$ and $(0,2)$ parts, contains every integral class. Hence

$$\mathrm{NS}(X) = \mathrm{Hdg}^1(X,\mathbb{Z}) = H^2(X,\mathbb{Z}) = \mathbb{Z}f_1 \oplus \mathbb{Z}f_2, \qquad \rho(X) = 2 = h^{1,1}(X),$$

and every degree-two integral class is visibly the class of a divisor $a\,(\mathbb{P}^1 \times \mathrm{pt}) + b\,(\mathrm{pt} \times \mathbb{P}^1)$. The Picard number attains its ceiling $h^{1,1}$ because the obstruction group $H^2(X,\mathcal{O}_X)$ vanishes: on a surface with $h^{2,0} = 0$ — a rational surface, such as the quadric, a smooth cubic surface (where $\rho = h^{1,1} = 7$, spanned by the classes of the 27 lines), or any del Pezzo — the whole of H^2 is algebraic, and the Hodge Conjecture in codimension one is not merely true but vacuous.

[KNOWN] [Huy16] Contrast a $K3$ surface (Example 1.11.3), where $h^{2,0} = 1$ and so $H^2(X,\mathcal{O}_X) = H^{0,2} \neq 0$. There c_1 is no longer surjective: its image is the proper sublattice $\ker(H^2(X,\mathbb{Z}) \to H^{0,2}) = \mathrm{NS}(X)$, of rank $\rho(X) \leq 20 < 22 = b_2$, and the non-vanishing of the obstruction space $H^{0,2}$ is exactly what allows transcendental degree-two classes to exist. The single number $h^{2,0}$ thus measures how much of H^2 escapes the divisors, and the exponential sequence turns that measurement into an exact sequence.

1.10.2 Why the argument stops at codimension one

The proof of Theorem 1.10.1 used one structural miracle: the group $\mathrm{Pic}(X) = H^1(X,\mathcal{O}_X^\times)$ of codimension-one cycles sits inside the short exact *exponential sequence* of sheaves $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$, whose connecting map is c_1 and whose target term $H^2(X,\mathcal{O}_X)$ is the Hodge-theoretic obstruction $H^{0,2}$. It is natural to ask for the analogue one codimension up, and its precise absence is the reason $p = 2$ is hard.

[KNOWN] [Ful98] There is no short exact sequence of sheaves on X whose first cohomology computes $\mathrm{CH}^2(X)$ and whose connecting map has image $\mathrm{Hdg}^2(X,\mathbb{Z})$ inside $H^4(X,\mathbb{Z})$. The nearest replacement abandons a single sheaf for a complex: the *Deligne complex* $\mathbb{Z}_{\mathcal{D}}(p) = [\,\mathbb{Z} \to \mathcal{O}_X \to \Omega^1_X \to \cdots \to \Omega^{p-1}_X\,]$, whose hypercohomology is *Deligne cohomology* $H^*_{\mathcal{D}}(X,\mathbb{Z}(p))$. It carries a refined cycle class $\mathrm{cl}_{\mathcal{D}}\colon \mathrm{CH}^p(X) \to H^{2p}_{\mathcal{D}}(X,\mathbb{Z}(p))$ and fits into the exact sequence

$$0 \longrightarrow J^p(X) \longrightarrow H^{2p}_{\mathcal{D}}(X,\mathbb{Z}(p)) \longrightarrow \mathrm{Hdg}^p(X,\mathbb{Z}) \longrightarrow 0,$$

where $J^p(X) = H^{2p-1}(X,\mathbb{C})/(F^pH^{2p-1} + H^{2p-1}(X,\mathbb{Z}))$ is the *Griffiths intermediate Jacobian*. For $p = 1$ this *is* the exponential sequence: $H^2_{\mathcal{D}}(X,\mathbb{Z}(1)) = \mathrm{Pic}(X)$, $J^1(X) = \mathrm{Pic}^0(X)$, the sequence is $0 \to \mathrm{Pic}^0(X) \to \mathrm{Pic}(X) \to \mathrm{NS}(X) \to 0$, and $\mathrm{cl}_{\mathcal{D}}$ is an isomorphism — which is why surjectivity onto $\mathrm{Hdg}^1(X,\mathbb{Z})$ came for free.

[KNOWN] For $p \geq 2$ every step of this miracle fails. The intermediate Jacobian $J^p(X)$ is in general *not* an abelian variety — the Hodge structure H^{2p-1} need not have the level that makes J^p algebraic (Griffiths [Gri68]) — so $H^{2p}_{\mathcal{D}}$ is not a group one can control by algebraic geometry; the Abel–Jacobi map $\mathrm{CH}^p_{\mathrm{hom}}(X) \to J^p(X)$ can have image that is not finitely generated (Clemens [Cle77]); and, decisively, the composite $\mathrm{CH}^p(X) \xrightarrow{\mathrm{cl}_{\mathcal{D}}} H^{2p}_{\mathcal{D}}(X,\mathbb{Z}(p)) \to \mathrm{Hdg}^p(X,\mathbb{Z})$ is the ordinary cycle class, whose surjectivity is the Hodge Conjecture itself. In other words, the very map whose surjectivity was *automatic* in the exponential sequence becomes, one codimension up, the whole problem. The alternative bookkeeping through algebraic K-theory, the Bloch–Quillen formula $\mathrm{CH}^p(X) \cong H^p(X,\mathcal{K}_p)$ with the Quillen sheaf $\mathcal{K}_p$, replaces $\mathcal{O}_X^\times = \mathcal{K}_1$ faithfully but relates $\mathcal{K}_p$ to the Hodge filtration only through the Bloch–Ogus coniveau spectral sequence [BO74], never through a short exact sequence, and so yields no

surjectivity statement. The missing exact sequence is not hiding; the content of the Hodge Conjecture in codimension $p \geq 2$ is exactly that it does not exist in usable form.

1.11 Four worked examples

Example 1.11.1 (An elliptic curve). Let $E = \mathbb{C}/\Lambda$ be an elliptic curve, $n = 1$. The only interesting weight is $k = 1$, with $h^{1,0} = h^{0,1} = 1$; and $H^2(E, \mathbb{Q}) = \mathbb{Q}$ is generated by the class of a point, which is $\mathrm{Hdg}^1(E)$. Since $\dim E = 1$, the only nontrivial codimension is $p = 1 = n$, and the point class is manifestly algebraic. Thus $\mathrm{HC}(E, p)$ holds trivially for all p. The example is a baseline: in dimension one there is nothing to prove.

Example 1.11.2 (An abelian surface). Let $A = \mathbb{C}^2/\Lambda$ be an abelian surface, $n = 2$. Then $b_1 = 4$, $b_2 = 6$, with Hodge numbers $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 4$. The interesting codimension is $p = 1$, where $\mathrm{Hdg}^1(A) = \mathrm{NS}(A)_{\mathbb{Q}}$ has rank the Picard number $\rho(A) \in \{1, 2, 3, 4\}$. For a very general A, $\rho = 1$. For a product $A = E_1 \times E_2$ of non-isogenous elliptic curves without complex multiplication, $\rho = 2$, the classes of the two elliptic factors. For $A = E \times E$ with E without complex multiplication, $\rho = 3$, the two fibre classes together with the diagonal. For E with complex multiplication, $\rho = 4$. In every case $\mathrm{HC}(A, 1)$ holds by Corollary 1.10.2, and since $n = 2$ the only nontrivial codimension is $p = 1 = n - 1$; thus the Hodge Conjecture is a theorem for all abelian surfaces. The higher-dimensional abelian varieties, where Hdg^p can contain classes *not* generated by divisors, are where the difficulty begins (Chapters 2 and 49).

Example 1.11.3 (A $K3$ surface). Let X be a $K3$ surface, $n = 2$: a simply connected surface with trivial canonical bundle. Then $b_2 = 22$, with $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 20$; the lattice $H^2(X, \mathbb{Z})$ is the even unimodular lattice of signature $(3, 19)$, namely $U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$. The Néron–Severi group $\mathrm{NS}(X) = \mathrm{Hdg}^1(X, \mathbb{Z}) = H^2(X, \mathbb{Z}) \cap H^{1,1}$ has rank $\rho(X)$ bounded by $h^{1,1} = 20$; its orthogonal complement, the transcendental lattice $T(X)$, has rank $22 - \rho(X) \geq 2$ because it must contain the two-dimensional real space $H^{2,0} \oplus H^{0,2}$. Here the Lefschetz theorem is vivid: although $H^{1,1}$ is twenty-dimensional over $\mathbb{C}$, only the $\rho(X) \leq 20$ integral classes lying in it are Hodge, and every one of them is a divisor class. The Hodge Conjecture is again a theorem ($p = 1 = n - 1$); the surface nonetheless governs, through the Kuga–Satake construction (Chapter 23), test cases of the conjecture in higher dimension.

Example 1.11.4 (The Fermat quartic surface). Let $X = \{x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0\} \subset \mathbb{P}^3$, the Fermat quartic; it is a $K3$ surface. Its Picard number attains the maximal value $\rho(X) = 20$: the surface is *singular* in the sense of Shioda–Inose (of maximal Picard rank), its transcendental lattice has rank 2, and the NS lattice is generated by explicit lines and conics together with the hyperplane class — the 48 lines on the Fermat quartic already span a finite-index subgroup of $\mathrm{NS}(X)_{\mathbb{Q}}$. Every degree-two Hodge class is thus an explicit combination of these algebraic curves, in complete accord with Theorem 1.10.1. The Fermat surface is the smallest member of the family of Fermat varieties for which Shioda (1979) proved the Hodge Conjecture in a range of dimensions and degrees [Shi79]; we return to Fermat varieties in Chapter 50.

These four examples share a feature that makes them tractable and, for our purposes, misleading: the Hodge classes are all of codimension one, where the $(1,1)$-theorem applies. The Hodge Conjecture only becomes itself in codimension $p \geq 2$ on varieties of dimension ≥ 4, where no analogue of the exponential sequence is available and the cohomology can carry Hodge classes — the Weil classes of Chapter 2 — that are provably not accounted for by divisors. Everything in the remainder of the book is directed at that regime.

1.12 The hypotheses, and what each is for

The statement HC(X, p) (Conjecture 1.8.2) carries five standing hypotheses — X smooth, X projective, X compact, $\mathbb{Q}$-coefficients, and the class of type (p, p) — and each is load-bearing. Table 1.1 records, for each, what object degenerates or what counterexample appears when the hypothesis is dropped. The table is a compressed preview of Chapters 2 and 3: every failure listed is a wall that the design constraints of Chapter 3 are built to respect.

Table 1.1. The five hypotheses of HC(X, p) and the effect of dropping each.

Hypothesis	What breaks without it	Counterexample or degeneracy
Smooth	The Hodge decomposition is lost; $H^{2p}(X, \mathbb{Q})$ carries only a *mixed* Hodge structure (Deligne [Del74b]), and "type (p, p)" must be replaced by the weight-$2p$ graded piece.	A nodal or otherwise singular projective variety; a clean statement is recovered only after resolution, and HC for singular varieties is referred to the smooth case.
Projective	GAGA fails and the ample class disappears; the Hodge structure survives but its polarisation-driven positivity does not.	Voisin's complex torus of Weil type (Theorem 2.4.1): a rational Hodge class that is not even a combination of Chern classes of coherent sheaves.
Compact	Poincaré duality and Hodge symmetry fail; cohomology acquires a mixed Hodge structure and the cycle class need not be of pure type.	A smooth affine variety has $H^k = 0$ for $k > \dim X$ (Artin vanishing) and no polarisation; the conjecture is empty or ill-posed in the naive form.
$\mathbb{Q}$*-coefficients*	Torsion and divisibility obstructions survive that are invisible rationally.	The integral conjecture is *false*: Atiyah–Hirzebruch torsion [AH62], Totaro's cobordism refinement [Tot97], Kollár's degree obstruction [Kol92].
Type (p, p)	Algebraic classes are always of type (p, p) (Theorem 1.7.2); dropping the type condition asks for cycles where none can exist.	Any rational class in $H^{2p}(X, \mathbb{Q})$ with a nonzero $(2p-1, 1)$ component: it is not Hodge and provably not algebraic.

The two middle rows — projective and $\mathbb{Q}$-coefficients — are the two that a *proof* must actively respect rather than merely assume, since each is the site of a genuine counterexample (HC is false in the Kähler and integral categories). Chapter 3 turns exactly these two failures into the Voisin Test and the Integrality Test.

1.13 What this chapter does not do

This chapter is expository and makes no contribution to the conjecture. In particular:

- It proves the Hodge Conjecture only for $p = 1$ (Theorem 1.10.1) and, by trivial dimension count, for $\dim X \leq 1$; every assertion about $p \geq 2$ is deferred, and the audit of what is actually known is Chapter 2.
- It does not prove the Hodge decomposition or the algebraic description of the Hodge filtration, only sketches the classical degeneration and Grothendieck's algebraic de Rham comparison; the details are in the cited sources.
- It says nothing about the *construction* of cycles. The cycle class map is defined, and its image is contained in the Hodge classes, but no method for realising a given Hodge class by a cycle is offered here or, in full generality, anywhere.
- It does not address the variational, integral, or generalised forms beyond stating them; those are the business of Chapters 2 and 3.

The containment of Theorem 1.7.2 is the easy half of the Hodge Conjecture and is unconditional. It says algebraic ⇒ Hodge. The conjecture is the reverse implication, Hodge ⇒ algebraic, and nothing in this chapter bears on it except in codimension one. A reader must not carry away the impression that the machinery assembled here brings the conjecture closer; it makes the conjecture *statable*, which is a different and lesser thing.

Status of Chapter 1

[KNOWN] [Lef24; Voi03] The Hodge decomposition, cycle class map, Lefschetz $(1,1)$ theorem, the codimension-$(n-1)$ consequence, and the low-dimensional cases are established results. **[CONDITIONAL]** The Tannakian reading is conditional on the explicitly named algebraicity of Künneth projectors. **[OPEN]** The rational Hodge Conjecture is open outside the known cases; no new case is proved here.

Notes and References

The harmonic theory underlying Theorem 1.3.1 is Hodge's, in *The Theory and Applications of Harmonic Integrals* [Hod41]; the analytic foundations were placed on firm ground by Weyl, Kodaira and de Rham, and the modern account through the Kähler identities is standard in Griffiths–Harris [GH78] and Voisin [Voi03], which we have followed. The algebraic de Rham comparison used in Theorem 1.4.1 is Grothendieck's [Gro66]; the E_1-degeneration for smooth projective complex varieties is part of classical Hodge theory. Deligne's *Théorie de Hodge, II, III* [Del71; Del74b] organised this filtration in the theory of mixed Hodge structures that will occupy Chapter 24. A later purely algebraic proof of E_1-degeneration, via reduction to positive characteristic, is a landmark we do not need here but mention for its bearing on the arithmetic themes of Chapter 3.

The comparison isomorphisms of Section 1.2 rest on de Rham's theorem and on Grothendieck's algebraic de Rham theorem, the latter completed by Serre's GAGA [Ser56]; the reader should note that GAGA, and with it the identification of analytic and algebraic coherent cohomology, is exactly the input that fails for general compact Kähler manifolds and that we shall track as the book's principal expenditure of projectivity.

The Lefschetz $(1,1)$-theorem [Lef24] is a century old and has many proofs; the exponential-sequence argument we gave is the one that generalises, through the theory of Deligne cohomology and the Abel–Jacobi map, to the higher regulators of Chapter 33. The identification $\mathrm{NS}(X)_{\mathbb{Q}} = \mathrm{Hdg}^1(X)$ and the resulting theory of the Picard number are classical; the Hodge-locus refinement, that the jumping of ρ in a family is governed by algebraic subvarieties of the base, is Cattani–Deligne–Kaplan's [CDK95] and is the subject of Chapter 20.

The Hodge Conjecture in its codimension-p form was stated by Hodge at the 1950 International Congress of Mathematicians [Hod52]. The integral form was believed by some early readers and disproved by Atiyah and Hirzebruch [AH62]; the general (coniveau) conjecture of Hodge was corrected by Grothendieck [Gro69a], whose one-page note is a model of how a Hodge-theoretic guess can be wrong for structural rather than arithmetic reasons. The version adopted as one of the Clay Mathematics Institute's Millennium Prize Problems in 2000, with an official problem description written by Deligne, is the rational statement Conjecture 1.8.2, and it is the form to which the whole of this book is addressed.

Two expository sources shaped the presentation and deserve separate mention. Voisin's two volumes [Voi03] are the reference for Hodge theory as practised here, and her survey articles are the source of the sharp separation between the projective and Kähler categories that organises Chapters 2 and 3. Fulton's *Intersection Theory* [Ful98] is our reference for Chow groups, rational equivalence and the cycle class map; everything in Section 1.7 is drawn from it. For the Tate twist, weights and polarised

Hodge structures we follow Deligne's conventions; the reader who wishes to see the categorical formalism of Hodge structures developed in full will find it in Chapters 19 and 22 and in the appendix on Tannakian categories. Finally, the four examples of Section 1.11 are folklore, but the arithmetic of the Fermat quartic — its maximal Picard number and the span of its lines — is Shioda's [Shi79], and the lattice-theoretic description of $K3$ cohomology we quoted is standard and may be found in the survey literature on $K3$ surfaces.

Chapter 2

Known, False, Open: An Audit

This chapter is an audit. It separates, as scrupulously as we can manage, what is proved about the Hodge Conjecture from what is false and what is open. On the positive side we record the theorems: codimension one (Lefschetz), codimension $n-1$ (Hard Lefschetz), dimension at most three, uniruled fourfolds (Conte–Murre 1978), cubic fourfolds (Zucker 1977), and the divisor-generated cases on abelian varieties. On the negative side we explain why the integral *conjecture is false, through the Atiyah–Hirzebruch torsion obstruction $d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$, Totaro's cobordism refinement, and Kollár's degree argument; and why the conjecture fails outright for compact Kähler manifolds (Voisin 2002). We then delimit the open territory — abelian varieties, Weil classes, general fourfolds — and marshal the evidence in the conjecture's favour, of which the algebraicity of Hodge loci (Cattani–Deligne–Kaplan 1995) is the deepest. A summary table closes the chapter. We prove nothing new, and we are careful never to overstate a partial result: in particular Deligne's theorem on abelian varieties concerns* absolute Hodge *classes, not algebraic ones.*

2.1 Introduction

A conjecture that has resisted seventy years acquires a folklore, and the folklore of the Hodge Conjecture is unusually treacherous because several inequivalent statements travel under the one name (Chapter 1). The purpose of this chapter is to fix, once and for all in this book, a correct ledger: which statements are theorems, which are refuted, and which are open. The distinction matters not only for honesty but for design. A mechanism intended to prove the conjecture must reproduce every entry in the "known" column, must evade every entry in the "false" column, and should illuminate every entry in the "open" column; the constraints of Chapter 3 are extracted directly from this ledger.

We organise the audit into five registers. Section 2.2 collects the positive results, and we give the proof of the codimension-$(n-1)$ case in full because it is short and because its structure — reduce to divisors by an algebraic correspondence — is the structure every later argument aspires to. Section 2.3 explains the failure of the integral conjecture; here the obstructions are topological and arithmetic, and understanding them is what forces the book to work rationally. Section 2.4 presents Voisin's 2002 theorem that the conjecture is false for compact Kähler manifolds, the single most important negative result for our purposes, since it dictates that a correct proof must use projectivity. Section 2.5 maps the open country, whose landmark is the Weil classes on abelian fourfolds. Section 2.6 weighs the evidence in favour, above all the Cattani–Deligne–Kaplan theorem. Section 2.8 is a single reference table.

Nothing in this chapter is due to the author. Every theorem is attributed in the text and in the notes, and every attribution has been checked against the statements permitted by the book's canon; where a result is often misquoted, we quote it in its exact strength and flag the common error.

2.2 What is proved

2.2.1 Codimension one, and codimension $n-1$

The codimension-one case is the Lefschetz $(1,1)$-theorem, proved in Chapter 1: cl^p is surjective onto $\mathrm{Hdg}^1(X)$, even integrally (Theorem 1.10.1). The complementary case is a formal consequence.

Theorem 2.2.1 (Codimension $n-1$). *[KNOWN]* [Voi03] *Let X be smooth projective of dimension n. Then $\mathrm{HC}(X, n-1)$ holds: every rational Hodge class of codimension $n-1$ (equivalently, of dimension one) is a $\mathbb{Q}$-combination of classes of algebraic curves.*

Proof. Let $h \in \mathrm{Hdg}^1(X)$ be a hyperplane class and $L = h \cup (-)$ the Lefschetz operator. By Hard Lefschetz (Theorem 1.6.1), $L^{n-2}\colon H^2(X,\mathbb{Q}) \to H^{2n-2}(X,\mathbb{Q})(n-2)$ is an isomorphism of Hodge structures. Let $\beta \in \mathrm{Hdg}^{n-1}(X) \subseteq H^{2n-2}(X,\mathbb{Q})(n-1)$ be a Hodge class. Set $\alpha = (L^{n-2})^{-1}\beta \in H^2(X,\mathbb{Q})(1)$; since L^{n-2} is an isomorphism of rational Hodge structures, α is again a Hodge class, $\alpha \in \mathrm{Hdg}^1(X)$. By the $(1,1)$-theorem, $\alpha = \sum_i q_i \,\mathrm{cl}(D_i)$ for divisors D_i and $q_i \in \mathbb{Q}$. Then

$$\beta = L^{n-2}\alpha = h^{n-2} \cup \alpha = \sum_i q_i \,\mathrm{cl}(D_i) \cup h^{n-2} = \sum_i q_i \,\mathrm{cl}(D_i \cap H_1 \cap \cdots \cap H_{n-2}),$$

a $\mathbb{Q}$-combination of classes of curves cut out by hyperplanes. Hence β is algebraic. □

The proof uses exactly one nontrivial input beyond the $(1,1)$-theorem, namely that L^{n-2} is an isomorphism *of Hodge structures* whose inverse is $\mathbb{Q}$-linear; it does *not* require that this inverse be an algebraic correspondence (the Lefschetz standard conjecture B), because the inverse is applied only to the single class β and the output α is then shown algebraic by other means. This is the sense in which codimension $n-1$ is "formal": the hard work is Hard Lefschetz, which is Hodge theory.

Corollary 2.2.2 (Small dimension). *[KNOWN]* [Voi03] *$\mathrm{HC}(X,p)$ holds for all p whenever $\dim X \le 3$.*

Proof. For $p \in \{0, \dim X\}$ the statement is trivial. If $\dim X \le 3$, the remaining p are $p = 1$ and $p = \dim X - 1$, covered by Theorem 1.10.1 and Theorem 2.2.1. □

Thus the first case where the conjecture is not a theorem is $p = 2$ on a fourfold: a middle-dimensional Hodge class on a variety of dimension four. Everything hard lives at or above this threshold.

2.2.2 Fourfolds covered by rational curves

The first substantial theorem beyond the Lefschetz range concerns fourfolds whose geometry is close to rational.

Theorem 2.2.3 (Conte–Murre). *[KNOWN]* [CM78] *Let X be a smooth projective fourfold admitting a covering by rational curves (in particular, a uniruled or unirational fourfold). Then $\mathrm{HC}(X,2)$ holds (Conte–Murre 1978).*

The proof of [CM78] rests on a decomposition-of-the-diagonal argument in the style of Bloch–Srinivas: a covering by rational curves forces the Chow group of zero-cycles to be small, which in turn constrains the transcendental part of H^4 and pins the Hodge classes to the algebraic part built from the ruling. Unirational fourfolds are a special case, since a dominant rational map $\mathbb{P}^4 \dashrightarrow X$ supplies the covering. We record the result because it is the earliest instance of a technique — controlling cycles by controlling CH_0 — that reappears in Chapter 31.

2.2.3 Cubic fourfolds

Theorem 2.2.4 (Zucker). *[KNOWN]* [Zuc77] *Let $X \subset \mathbb{P}^5$ be a smooth cubic fourfold. Then* $\mathrm{HC}(X,2)$ *holds: every rational Hodge class in $H^4(X,\mathbb{Q})$ is algebraic (Zucker 1977).*

Zucker's proof [Zuc77] uses the theory of normal functions. The primitive cohomology $H^4_{\mathrm{Prim}}(X,\mathbb{Q})$ of a cubic fourfold carries a Hodge structure of $K3$ type — $h^{3,1}=1$, $h^{2,2}_{\mathrm{Prim}}=20$, $h^{1,3}=1$ — but this numerical resemblance is not the proof. Zucker uses a Lefschetz pencil of cubic-threefold hyperplane sections, the associated normal function, and the inversion geometry of their intermediate Jacobians to produce the required algebraic cycles. The cubic fourfold is the richest completely solved case in codimension two, and its Hodge theory (its relation to $K3$ surfaces, the Fano variety of lines) is a laboratory for the whole subject; we return to it in Chapter 50.

2.2.4 Abelian varieties, in part

On an abelian variety A of dimension g the exterior algebra structure $H^\bullet(A,\mathbb{Q}) = \bigwedge^\bullet H^1(A,\mathbb{Q})$ makes the Hodge classes computable in principle. The subring generated by divisor classes — the *Lefschetz classes* — is always algebraic, by the $(1,1)$-theorem and multiplication.

Proposition 2.2.5. *[KNOWN]* [Voi03] *If every Hodge class on A (in every codimension) is a Lefschetz class — equivalently, if the Hodge group* $\mathrm{Hg}(A)$ *equals the Lefschetz group — then* $\mathrm{HC}(A,p)$ *holds for all p. This is the case for a very general abelian variety and, more generally, whenever the algebra* $\mathrm{Hdg}^\bullet(A)$ *is generated in degree one.*

Proof. The Lefschetz classes are the subalgebra of $\mathrm{Hdg}^\bullet(A)$ generated by $\mathrm{Hdg}^1(A) = \mathrm{NS}(A)_{\mathbb{Q}}$, which is algebraic by Theorem 1.10.1 and closed under the intersection product; if it exhausts $\mathrm{Hdg}^\bullet(A)$ then every Hodge class is algebraic. □

In particular HC holds for products of elliptic curves, where a direct computation shows every Hodge class is generated by the divisor classes of the factors and their graphs; this is classical. It also holds for many abelian varieties of CM type, where the Hodge classes can be written down explicitly from the CM data. What Proposition 2.2.5 does *not* cover is the case where $\mathrm{Hdg}^\bullet(A)$ contains classes not generated by divisors — the *exceptional* or Weil classes — and that case, which first occurs in dimension four, is open (Section 2.5). The reader is warned of the single most common misstatement in the subject: it is *not* known that HC holds for all abelian varieties, and Deligne's celebrated theorem (Section 2.6) does not prove it.

2.2.5 Fermat varieties

The Fermat hypersurface $X_d^m = \{x_0^d+\cdots+x_{m+1}^d = 0\} \subset \mathbb{P}^{m+1}$ carries the action of the group $(\mu_d)^{m+2}/\mu_d$ of diagonal d-th roots of unity, and its cohomology decomposes into one-dimensional character eigenspaces indexed by combinatorial data. This makes the Hodge classes enumerable and the conjecture attackable by an inductive structure relating X_d^m to products of lower-dimensional Fermat varieties.

Theorem 2.2.6 (Shioda). *[KNOWN]* [Shi79] *The Hodge Conjecture holds for the Fermat variety X_d^m for a large range of pairs (m,d) — in particular for all Fermat surfaces, and for all (m,d) satisfying Shioda's combinatorial criterion on the characters, which is verified for every prime degree d and for all degrees below an explicit bound (Shioda 1979).*

Shioda's method [Shi79] reduces $\mathrm{HC}(X_d^m,p)$ to the statement that certain "inductive" and "decomposable" Hodge classes exhaust the Hodge classes, a purely combinatorial condition on the eigencharacters that he verifies in a wide range; the classes are then realised by explicit cycles built from linear subspaces and from the fibrations relating Fermat varieties. The Fermat family is the largest explicitly understood testing ground for the conjecture in high codimension, and it supplies, in Chapters 50 and 57, the concrete input against which the book's computational system is calibrated. It also cautions against optimism: the cases that succumb are exactly those with enough symmetry to force the Hodge classes into the decomposable range, and no new idea about *general* varieties is extracted from them.

2.2.6 The known cases, with exact hypotheses

The positive results are scattered across a century and are easy to overstate by forgetting a hypothesis. Table 2.1 states each with the exact dimension and codimension conditions under which it is a theorem. The reader should hold it against the misstatement warned of above — the belief that the Hodge Conjecture is known for abelian varieties — which the last two rows correct.

Table 2.1. Known cases of $\mathrm{HC}(X,p)$ with exact hypotheses.

Case (exact hypotheses)	Reason	Attribution
$p=1$, any smooth projective X (integrally)	exponential sequence	Lefschetz 1924
$p=n-1$, any X of dimension n	Hard Lefschetz reduces to $p=1$	below
all p, $\dim X\le 3$	only $p\in\{1,n-1\}$ occur	corollary of the above
$p=2$, X a uniruled or rationally connected fourfold	diagonal decomposition	Conte–Murre 1978
$p=2$, X a smooth cubic fourfold	normal functions	Zucker 1977
all p, X with $\mathrm{Hdg}^\bullet(X)$ generated in degree 1	Lefschetz classes	$(1,1)$-theorem
all p, X a product of elliptic curves	explicit divisors	classical
$p=1$, X any $K3$ or abelian surface	$(1,1)$-theorem	Lefschetz 1924
Fermat X_d^m in Shioda's range	decomposable classes	Shioda 1979
divisor-generated $\mathrm{Hdg}^\bullet$ on abelian A	Lefschetz classes	classical
general abelian variety, $p\ge 2$	**Open**	—
Weil classes, abelian fourfold of Weil type	**Open**	Weil 1979

The codimension-$(n-1)$ reduction, explicitly.. The second row is not a black box, and because the reduction is the template for every later argument we restate it. Given a Hodge class $\beta\in\mathrm{Hdg}^{n-1}(X)$, Hard Lefschetz (Theorem 1.6.1) makes $L^{n-2}\colon H^2(X,\mathbb{Q})(1)\to H^{2n-2}(X,\mathbb{Q})(n-1)$ an isomorphism of Hodge structures, so $\alpha=(L^{n-2})^{-1}\beta$ is a Hodge class in $\mathrm{Hdg}^1(X)$, algebraic by the $(1,1)$-theorem; then $\beta=h^{n-2}\cup\alpha$ is a combination of classes of curves. The reduction uses only that L^{n-2} is an isomorphism of *rational* Hodge structures, not that its inverse is an algebraic correspondence — that stronger statement is Standard Conjecture B, which by Kleiman (Theorem 2.6.5) may not be assumed.

Abelian varieties, and the extent of the open region.. Beyond the divisor-generated case, HC is open for abelian varieties, and this is the canonical open region. The Hodge classes that escape the Lefschetz subalgebra are, in dimension ≤ 5, exactly the Weil classes (Theorem 2.5.1); in special cases they are known to be algebraic — van Geemen surveyed the Weil classes and the geometry that occasionally realises them [vG94], and Schoen produced algebraic representatives for certain families through self-products and prescribed automorphisms [Sch98] — but no general proof exists. **[KNOWN]** For hyperkähler manifolds the situation parallels the $K3$ case: the conjecture is a theorem in codimension one (the $(1,1)$-theorem) and its Tate-conjecture analogue is known for $K3$ surfaces over finite fields of odd characteristic (Charles [Cha13]; Madapusi Pera [MP15]), but the middle-codimension conjecture on a general hyperkähler fourfold is open, and it is one of the arenas in which the Universal Kuga–Satake programme of Chapter 49 is tested.

2.3 What is false: the integral Hodge Conjecture

The integral conjecture $\mathrm{HC}_{\mathbb{Z}}(X,p)$ — that $\mathrm{cl}\colon \mathrm{CH}^p(X)\to\mathrm{Hdg}^p(X,\mathbb{Z})$ is onto — is false. There are two logically independent sources of failure: torsion obstructions, invisible rationally, and degree obstructions, which survive tensoring with any localisation away from a fixed prime. We treat both.

2.3.1 The Atiyah–Hirzebruch torsion obstruction

Topological complex K-theory receives the algebraic cycle class map: the class of a subvariety Z determines a K-theory class through the coherent sheaf $\mathcal{O}_Z$, and the topological Chern character intertwines this with the cohomological cycle class. The Atiyah–Hirzebruch spectral sequence

$$E_2^{s,t} = H^s(X, K^t(\mathrm{pt})) \Longrightarrow K^{s+t}(X)$$

computes K-theory from cohomology, and its first possibly-nonzero differential on the relevant line is the integral Steenrod operation

$$d_3 = \mathrm{Sq}^3_{\mathbb{Z}} = \beta \circ \mathrm{Sq}^2 \circ \rho \colon H^{2p}(X, \mathbb{Z}) \longrightarrow H^{2p+3}(X, \mathbb{Z}),$$

where ρ is reduction modulo 2 and β the integral Bockstein. Any class in the image of the cycle map must survive to the E_∞-page, hence must be annihilated by d_3.

Theorem 2.3.1 (Atiyah–Hirzebruch). **[KNOWN]** [AH62] *There exist smooth projective varieties X and torsion classes $\alpha \in H^{2p}(X, \mathbb{Z})$ with $\mathrm{Sq}^3_{\mathbb{Z}}(\alpha) \neq 0$. Such an α is a Hodge class — being torsion, it maps to zero in $H^{2p}(X, \mathbb{C})$, so it lies in $\mathrm{Hdg}^p(X, \mathbb{Z})$ vacuously — yet it is not in the image of the cycle class map. Hence $\mathrm{HC}_{\mathbb{Z}}(X, p)$ fails (Atiyah–Hirzebruch 1962).*

The varieties are built from finite groups: an approximation to the classifying space of a suitable finite group carries torsion classes on which $\mathrm{Sq}^3_{\mathbb{Z}}$ is nonzero, and the Godeaux–Serre approximation used in [AH62] realises the relevant cohomology on a smooth projective variety (a quotient of a product of curves, or a suitable complete intersection in a product of projective spaces). The obstruction is invisible rationally: torsion classes vanish in $H^{2p}(X, \mathbb{Q})$, so Theorem 2.3.1 says nothing about $\mathrm{HC}(X, p)$. Its lesson for us is structural: the integral cycle map has a purely topological obstruction, the target of d_3, and any correct mechanism must be compatible with it — which, since the mechanism will work rationally, it is, automatically. [AH62]

Remark 2.3.2 (The construction, step by step). **[KNOWN]** [AH62] It is worth making the Atiyah–Hirzebruch example concrete, since the mechanism recurs. (1) *A group with a bad class.* Choose a finite group G whose classifying space BG carries a torsion class $u \in H^{2p}(BG, \mathbb{Z})$ with $\mathrm{Sq}^3_{\mathbb{Z}} u = \beta \mathrm{Sq}^2 \rho(u) \neq 0$; such G occur already among elementary abelian 2-groups, where the mod-2 cohomology is a polynomial ring and the Steenrod action is computable by the Cartan formula. (2) *A variety carrying it.* By the Godeaux–Serre construction, a finite group G acting freely on a smooth complete intersection of high dimension has a smooth projective quotient X whose cohomology agrees with $H^*(BG)$ through a range of degrees; choose the range to include u, producing a class $\alpha \in H^{2p}(X, \mathbb{Z})$ with $\mathrm{Sq}^3_{\mathbb{Z}} \alpha \neq 0$. (3) *Why it is Hodge.* The class α is torsion, so it maps to 0 in $H^{2p}(X, \mathbb{C})$ and lies in $\mathrm{Hdg}^p(X, \mathbb{Z})$ for the vacuous reason that it has no Hodge type to violate. (4) *Why it is not algebraic.* The image of the cycle class map is annihilated by $d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$, because an algebraic class comes from a coherent sheaf and so survives the Atiyah–Hirzebruch spectral sequence to E_∞; since $\mathrm{Sq}^3_{\mathbb{Z}} \alpha \neq 0$, the class α does not survive, hence is not algebraic. The obstruction is torsion and dies over $\mathbb{Q}$, so it refutes only $\mathrm{HC}_{\mathbb{Z}}$, never HC.

2.3.2 Totaro's cobordism refinement

Atiyah–Hirzebruch used the first K-theoretic obstruction. Totaro gave a systematic refinement: the algebraic cycle class map factors through complex cobordism,

$$\mathrm{CH}^p(X) \longrightarrow MU^{2p}(X) \otimes_{MU^{2\bullet}} \mathbb{Z} \longrightarrow H^{2p}(X, \mathbb{Z}),$$

and reductions of classes in the image satisfy a whole family of odd Steenrod operation constraints.

Theorem 2.3.3 (Totaro). **[KNOWN]** [Tot97] *For every smooth complex algebraic variety X, the integral cycle map from cycles modulo algebraic equivalence lifts canonically through $MU^{2p}(X) \otimes_{MU^{2\bullet}} \mathbb{Z}$. Consequently every algebraic class lies in the image of the natural map from this complex-cobordism quotient to integral cohomology. Modulo any prime, every odd-dimensional Steenrod operation in the two-sided ideal generated*

by the Bockstein vanishes on that image. The comparison with integral cohomology is an isomorphism after tensoring with $\mathbb{Q}$, so this refinement is torsion-sensitive and does not *furnish the non-torsion counterexamples (Totaro 1997).*

Totaro's construction [Tot97] uses resolution of singularities to define the lifted cycle class and classifying spaces to expose the extra torsion information carried by cobordism. The upshot is precise but limited: $MU^{2p}(X) \otimes_{MU^{2\bullet}} \mathbb{Z}$ is a canonical intermediate receptacle for integral cycle classes. Totaro's paper must not be cited for a free non-torsion failure of integral algebraicity; that independent phenomenon is Kollár's.

2.3.3 Kollár's degree obstruction

The torsion phenomena above are, in a sense, exotic. Kollár found a robust, elementary obstruction of a completely different flavour.

Theorem 2.3.4 (Kollár). ***[KNOWN]*** *Let $X \subset \mathbb{P}^4$ be a very general hypersurface of degree d, a smooth threefold, and suppose d is divisible by a high power of a prime p (for instance $p^3 \mid d$ with $p \geq 5$). Then every curve $C \subset X$ has degree divisible by p. Consequently the positive generator γ of $H^4(X,\mathbb{Z}) \cong \mathbb{Z}$ — an integral Hodge class — is not the class of any algebraic 1-cycle, and $\mathrm{HC}_{\mathbb{Z}}(X,2)$ fails, although $\mathrm{HC}(X,2)$ holds rationally (Kollár 1992) [Kol92].*

Proof of the cohomological bookkeeping. For a smooth threefold $X \subset \mathbb{P}^4$, $H^2(X,\mathbb{Z}) = \mathbb{Z}h$ and $H^4(X,\mathbb{Z}) = \mathbb{Z}\gamma$ with $h \cup \gamma$ the positive generator of $H^6 = \mathbb{Z}$; the relation $h^2 = d\,\gamma$ holds because $\int_X h^3 = d$. Thus $\gamma = \frac{1}{d}h^2$ is a rational Hodge class of type $(2,2)$, and rationally $\gamma = \frac{1}{d}\mathrm{cl}(X \cap \mathbb{P}^2)$ is algebraic. For any curve $C \subset X$ one has $\mathrm{cl}(C) = (\deg C)\,\gamma$, since $\int_X h \cup \mathrm{cl}(C) = \deg C$. Hence the image of $\mathrm{CH}^2(X) \to H^4(X,\mathbb{Z})$ is $e\,\mathbb{Z}\gamma$, where $e = \gcd\{\deg C\}$. Kollár's geometric input is that a very general X of the stated degree contains no curve of degree prime to p, so $p \mid e$ and $\gamma \notin \mathrm{Im}(\mathrm{cl})$. □

Kollár's example is decisive for the book's methodology. It shows the integral obstruction is not a torsion artefact to be waved away but a stable arithmetic phenomenon: the denominator e can be forced to be as divisible as one likes. Any construction that produced a cycle with class exactly a prescribed integral Hodge class would contradict Theorem 2.3.4; the constructions of this book therefore produce cycles only up to a bounded but generally nontrivial denominator, and locating that denominator is the task of Chapter 38.

Remark 2.3.5 (The defect group). **[KNOWN]** [AH62] It is convenient to package the failure of $\mathrm{HC}_{\mathbb{Z}}$ in a single group. Write $Z^{2p}(X) = \mathrm{Hdg}^p(X,\mathbb{Z})/\mathrm{Im}\,(\mathrm{cl}\colon \mathrm{CH}^p(X) \to \mathrm{Hdg}^p(X,\mathbb{Z}))$ for the cokernel of the integral cycle map onto integral Hodge classes. The integral conjecture $\mathrm{HC}_{\mathbb{Z}}(X,p)$ is the assertion $Z^{2p}(X) = 0$, and Theorems 2.3.1, 2.3.3 and 2.3.4 say that $Z^{2p}(X)$ is nonzero for suitable X — with torsion contributions (Atiyah–Hirzebruch, Totaro) and, in codimension two on threefolds, a free contribution detected by unramified cohomology (Kollár; Colliot-Thélène–Voisin). The rational conjecture is exactly the statement that $Z^{2p}(X)$ is a torsion group of the appropriate kind, so that it dies after $\otimes\mathbb{Q}$; this defect group is the object the Integrality Test of Chapter 3 is built around.

2.4 The Kähler catastrophe

Every positive result of Section 2.2 used projectivity, most visibly through GAGA (to turn line bundles into divisors) and through the ample class (to define L). One might hope these were conveniences. Voisin proved they are not: the Hodge Conjecture is *false* for compact Kähler manifolds, and false already in the weakest reasonable formulation.

Theorem 2.4.1 (Voisin). ***[KNOWN]*** [Voi02a] *There exists a compact Kähler manifold X — a complex torus of dimension four of Weil type — carrying a nonzero rational Hodge class $\alpha \in \mathrm{Hdg}^2(X)$ that is not a $\mathbb{Q}$-combination of Chern classes of coherent analytic sheaves on X. Since X is a general complex torus it has no proper analytic subvarieties of positive dimension at all, so a fortiori α is not a combination of classes of analytic (let alone algebraic) cycles. Hence the Hodge Conjecture, and even its coherent-sheaf weakening, fails for compact Kähler manifolds (Voisin 2002).*

We describe the mechanism of [Voi02a], which is the pivot of the whole book. A general complex torus $X = V/\Gamma$ of dimension ≥ 2 is *non-algebraic*: it carries no nonconstant meromorphic functions and no hypersurfaces, so its algebraic dimension is zero and it has no proper positive -dimensional analytic subvarieties. On the other hand, if one endows X with extra endomorphisms — a Weil-type structure, i.e. an action of an imaginary quadratic field K on V with a prescribed pair of eigenvalue multiplicities — then $H^4(X, \mathbb{Q})$ acquires a two-dimensional space of *Weil–Hodge classes* of type $(2,2)$, exactly as on the abelian fourfolds of Weil type (Section 2.5). The whole difficulty — and Voisin's achievement — is to arrange the two conditions simultaneously: enough endomorphism structure to create a Hodge class, and enough genericity to destroy all subvarieties and, more delicately, all coherent sheaves with the offending Chern class. She shows that for a very general torus in the relevant family the only coherent sheaves are, up to the classes that cannot contribute in degree four, topologically trivial, so the Chern-class span in $\mathrm{Hdg}^2(X)$ is zero while the Weil classes are not. The class α is therefore Hodge and provably outside the reach of *any* analytic-geometric construction of cycles or sheaves.

Example 2.4.2 (Voisin's torus, constructed). [KNOWN] The construction of [Voi02a] can be given in four steps, and it is worth following them, because the whole book turns on the mechanism. *(1) The torus.* Fix an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-\delta})$ and an 8-dimensional $\mathbb{Q}$-vector space V with a K-action, so that V is a 4-dimensional K-vector space. A complex torus $X = V_{\mathbb{R}}/\Gamma$ is specified by a complex structure J on $V_{\mathbb{R}}$; require J to commute with K, and require the two eigenspaces of $\sqrt{-\delta}$ on $V^{1,0}_{\mathbb{C}} \cong \mathbb{C}^4$ to have dimensions $(2,2)$. This is the *Weil type* condition. *(2) The class.* The top K-exterior power $\bigwedge^4_K V \cong K$ is a 2-dimensional $\mathbb{Q}$-subspace $W_K \subseteq \bigwedge^4_{\mathbb{Q}} V = H^4(X, \mathbb{Q})$, and the $(2,2)$ eigenvalue condition forces W_K to be of pure Hodge type $(2,2)$ (the character computation of Example 2.5.2). So every class in W_K is a rational Hodge class: $W_K \subseteq \mathrm{Hdg}^2(X)$. *(3) Killing subvarieties.* For X very general in the family of Weil-type tori, the Néron–Severi group vanishes and X has algebraic dimension zero: no nonconstant meromorphic functions, no positive-dimensional analytic subvarieties, hence no cycle to carry the class. *(4) Killing sheaves.* This is Voisin's decisive step, and it is why the theorem concerns coherent sheaves and not merely cycles: she shows that for very general X any coherent sheaf $\mathcal{F}$ has $\mathrm{ch}_2(\mathcal{F})$ landing in the Lefschetz line spanned by squares of divisor classes, which for $\mathrm{NS}(X) = 0$ is zero, so $\mathrm{ch}_2(\mathcal{F})$ cannot meet the Weil plane W_K. Hence a nonzero $\alpha \in W_K$ is a rational Hodge class realised by neither a cycle nor a Chern class — the Hodge Conjecture, and its coherent-sheaf weakening, both fail.

Theorem 2.4.1 does not say the Hodge Conjecture is false. It says the naive extension of the conjecture from smooth projective varieties to compact Kähler manifolds is false. The projective conjecture HC (Conjecture 1.8.2) is untouched and remains open. What the theorem removes is a class of *proofs*: any argument for HC that would apply verbatim to compact Kähler manifolds is thereby refuted, because it would prove a false statement. This is the Voisin Test of Chapter 3, and it is the reason every mechanism in this book must identify a step that uses projectivity and fails in the Kähler category.

The theorem also disposes of a natural hope. Because coherent sheaves are far more plentiful than subvarieties on a Kähler manifold — there is no GAGA to force them to be algebraic — one might expect the Chern-class formulation to be easier, and to hold where the cycle formulation fails. Voisin's example kills this hope in the sharpest way: not even coherent sheaves suffice. The failure is therefore not about the scarcity of subvarieties alone; it is about the absence of the arithmetic/algebraic rigidity that projectivity supplies, a rigidity the book will try to formalise as *definability* (Chapters 5 and 13).

2.5 What is open

Between the solved cases of Section 2.2 and the refuted integral and Kähler statements lies the open conjecture itself. Its heart is a single, concrete, unresolved family.

2.5.1 Weil classes on abelian fourfolds

Let A be an abelian fourfold with an action of an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-\delta})$ such that K acts on the tangent space $H^{1,0}(A) \cong \mathbb{C}^4$ with eigenvalue multiplicities $(2,2)$. Weil (1979) observed that $\bigwedge^4_K H^1(A,\mathbb{Q})$ contributes a two-dimensional space

$$W_K(A) \subset \mathrm{Hdg}^2(A)$$

of Hodge classes of type $(2,2)$ that are *not* Lefschetz classes — not generated by divisors — so that Proposition 2.2.5 does not apply. These *Weil classes* are the canonical test case: $\mathrm{HC}(A,2)$ for A of Weil type is open, and it is widely regarded as the crux on which the general conjecture will turn.

Theorem 2.5.1 (Moonen–Zarhin). *[KNOWN]* [Wei79] *For an abelian variety of dimension at most five, every Hodge class is a sum of a Lefschetz class and Weil-type classes; in particular all exceptional Hodge classes are of Weil type. Thus in dimension ≤ 5 the Hodge Conjecture is known except possibly for the Weil classes (Moonen–Zarhin 1999).*

Moonen and Zarhin [MZ99] classified the Hodge rings of abelian varieties of low dimension; their theorem localises the entire difficulty, in dimension ≤ 5, onto the Weil classes. André later placed the Weil classes in a general framework (his "Hodge classes of Weil type"), and proved they are *motivated* [And96] (Chapters 3 and 22) — which is strong evidence but not algebraicity. In special cases the Weil classes are known to be algebraic: when A is isogenous to a product involving suitable factors, or for particular CM types, explicit cycles have been produced [Sch98; vG94]. No general proof is known.

Example 2.5.2 (Counting the Weil classes). Let A be an abelian fourfold with an action of an imaginary quadratic field K, so that $V := H^1(A,\mathbb{Q})$ is an 8-dimensional $\mathbb{Q}$-vector space and a 4-dimensional K-vector space. The exterior powers over K sit inside the exterior powers over $\mathbb{Q}$, and the top K-exterior power

$$\bigwedge^4_K V \;\cong\; K$$

is one-dimensional over K, hence two-dimensional over $\mathbb{Q}$; under $\bigwedge^4_K V \hookrightarrow \bigwedge^4_{\mathbb{Q}} V = H^4(A,\mathbb{Q})$ its image is a 2-dimensional $\mathbb{Q}$-subspace. The Weil-type condition on the eigenvalue multiplicities $(2,2)$ of K on $H^{1,0}$ forces this subspace to be of pure Hodge type $(2,2)$; it is the space $W_K(A)$ of Weil classes. A short character computation shows $W_K(A) \cap (\text{Lefschetz classes}) = 0$ for general such A, so these two dimensions are new Hodge classes. Whether either is algebraic is Question 2.5.3, and it is open.

2.5.2 The general open regime

Beyond abelian varieties, the conjecture is open in essentially all of the range $p \geq 2$, $\dim X \geq 4$. It is open for a very general abelian variety of dimension ≥ 4 carrying Weil classes; for general hypersurfaces and complete intersections of high dimension in the middle codimension; and for hyperkähler manifolds beyond the divisor case. Even the case $p = 2$ — codimension-two Hodge classes, dimension-four cycles' complementary degree — is open in general, and it is the case on which most current work concentrates, because the codimension-one theory ($p = 1$) and its Hard-Lefschetz dual ($p = n-1$) are the only complete columns.

Question 2.5.3. `\OPEN` Are the Weil classes $W_K(A)$ on a general abelian fourfold of Weil type algebraic? A positive answer for all K and all such A would, by the reduction theory of Chapter 49, go most of the way to the general conjecture; a single counterexample would refute HC.

2.6 Evidence for the conjecture

The conjecture is not merely unrefuted; there is positive structural evidence, and one theorem in particular is so strong that it is often described as the main reason to believe HC.

2.6.1 Algebraicity of Hodge loci

If HC held, then in any family $\mathcal{X} \to S$ of smooth projective varieties the locus of parameters s where a given cohomology class becomes Hodge would be a countable union of algebraic subvarieties of S — it would be the image of a relative Hilbert scheme. Cattani, Deligne and Kaplan proved this prediction *unconditionally*.

Theorem 2.6.1 (Cattani–Deligne–Kaplan). *[KNOWN]* [CDK95] *Let $(\mathcal{V}, F^\bullet)$ be a polarised variation of Hodge structure over a smooth complex quasi-projective base S. The Hodge locus — the set of points where a prescribed integral class is of type (p,p) — is a countable union of closed algebraic subvarieties of S (Cattani–Deligne–Kaplan 1995).*

This is the deepest piece of evidence for the Hodge Conjecture [CDK95]. It verifies the most visible geometric shadow of HC without producing a single cycle: the Hodge loci are algebraic, exactly as they would be if the classes were algebraic, yet the theorem is proved by the analysis of degenerating Hodge structures (SL_2-orbits, nilpotent orbits) and says nothing about *why* the classes should be algebraic. Bakker, Klingler and Tsimerman reproved and extended it by a completely different route, showing the period map is definable in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$ and deducing algebraicity from the definable Chow theorem.

Theorem 2.6.2 (Bakker–Klingler–Tsimerman). *[KNOWN]* [BKT20] *The period map of a polarised variation of Hodge structure on a smooth quasi-projective base is definable in $\mathbb{R}_{\mathrm{an,exp}}$; consequently Hodge loci are algebraic, giving a new proof of Theorem 2.6.1 (Bakker–Klingler–Tsimerman 2020).*

The o-minimal proof [BKT20] is the technical cornerstone of Chapters 5, 9 and 20, and the definable category it uses is the ambient category of the book's prism (Chapter 13). We stress the epistemic status: Theorems 2.6.1 and 2.6.2 are theorems *about* Hodge classes, confirming their algebro-geometric behaviour, but they are not theorems producing cycles, and they do not prove HC.

2.6.2 Absolute Hodge classes and Principle B

Deligne proved that Hodge classes on abelian varieties, though not known to be algebraic, satisfy a strong arithmetic constraint.

Theorem 2.6.3 (Deligne). *[KNOWN]* [Del82] *Every Hodge class on an abelian variety is* absolutely Hodge*: under any field automorphism $\sigma \in \mathrm{Aut}(\mathbb{C})$, its de Rham/étale avatar is carried to a Hodge class on the conjugate variety, in every cohomology theory. In particular Hodge classes on abelian varieties behave, under Galois symmetry, exactly as algebraic classes would (Deligne 1982).*

We state the strength of Theorem 2.6.3 exactly, because it is the most misquoted result in the field: Deligne proved absolute Hodgeness, which is *weaker* than algebraicity. He did *not* prove the Hodge Conjecture for abelian varieties, and it remains open even for the Weil classes of Section 2.5. The proof [Del82] runs through the principle of propagation now called *Principle B*: if a class in a family is Hodge on the whole base and absolutely Hodge on one fibre, it is absolutely Hodge on every fibre. Principle B is what lets absolute Hodgeness spread from the CM points (where it is checked by hand) across the moduli of abelian varieties; a related propagation was obtained independently by Terasoma. The gap between absolute Hodge and algebraic is the subject of Chapter 22, and closing it — doing strictly more than Deligne — is the fourth design constraint of Chapter 3.

2.6.3 Motives and the standard conjectures

A last body of evidence is indirect but structurally important: the category of motives behaves, unconditionally, as though HC were true in several respects.

Theorem 2.6.4 (Jannsen). **[KNOWN]** [Jan92] *The category of pure motives with respect to numerical equivalence, $\mathbf{Mot}_{\mathrm{num}}(\mathbb{C})_{\mathbb{Q}}$, is a semisimple abelian category; equivalently, numerical equivalence is the unique adequate equivalence relation for which the category of motives is abelian semisimple* (Jannsen 1992).

Jannsen's theorem [Jan92] is unconditional and is the categorical backbone of Chapters 25 and 30: it gives algebraic correspondences modulo numerical equivalence a semisimple abelian category. It does *not* turn arbitrary Hodge classes into motivic morphisms, and Betti realisation does not descend to numerical motives unless Standard Conjecture D is supplied. Against it stands Kleiman's implication in the other direction.

Theorem 2.6.5 (Kleiman). **[KNOWN]** [Kle68] *Over $\mathbb{C}$, the Hodge Conjecture implies the Lefschetz standard conjecture B (algebraicity of the Lefschetz operators Λ). Hence Standard Conjecture B cannot be assumed en route to* HC *without circularity* (Kleiman 1968).

Theorem 2.6.5 [Kle68; Kle94] is a caution as much as a result: several tempting strategies for HC secretly assume B, and are therefore not reductions but reformulations. We flag this because the trace formalism of Chapters 29 and 30 is carefully arranged to avoid presupposing B; the obstruction ledger of Chapter 48 isolates B as entry O7 exactly so that it is never silently used. The net picture is that numerical motives are semisimple (Jannsen), while Hodge realisation is unconditionally faithful only on the category modulo homological equivalence. Identifying that category with the numerical one is Standard Conjecture D; fullness of the homological Hodge realisation is the separate open condition developed as the Trace–Hodge Conjecture.

2.6.4 The variational conjecture

The *variational Hodge Conjecture* asks that if a Hodge class is algebraic at one point of a family and remains Hodge nearby, then it is algebraic nearby. It is open, but the algebraicity of Hodge loci (Theorem 2.6.1) is a large part of what it would require, and partial results exist through the theory of normal functions and the Bloch–Srinivas method. We treat it in Chapter 20, noting only that variational HC, algebraicity of Hodge loci, and HC itself are three distinct statements, of which only the middle one is a theorem.

2.7 A graveyard of attempted proofs

A conjecture this old has a history of attacks, and the instructive part of that history is not the failures but the *precise* point at which each fails. We record six routes and the exact obstruction that stops each; the design constraints of Chapter 3 are, in effect, the distilled morals of this graveyard.

(i) *Assume the standard conjectures.* One might hope that Grothendieck's standard conjectures — in particular Lefschetz-type Conjecture B, the algebraicity of the operators Λ — would yield HC. **[KNOWN]** They do not: the standard conjectures give algebraicity of the Künneth and Lefschetz operators and semisimplicity of numerical motives, but not that *every* Hodge class is algebraic. Worse, one may not assume B en route, because HC $\Rightarrow$ B (Kleiman [Kle68; Kle94]); an argument that silently used B would be circular.

(ii) *Hodge's coniveau conjecture.* Hodge's own 1950 refinement [Hod52], equating the coniveau filtration with the largest rational subspace inside F^c, would have implied strong cases. **[KNOWN]** It is *false*: F^c is not a sub-Hodge structure, and Grothendieck refuted the statement for that trivial reason [Gro69a]. Any mechanism reasoning with F^c in place of the corrected $\tilde{N}^c$ re-proves the false statement.

(iii) *A higher exponential sequence.* The Lefschetz proof in codimension one runs through the exponential sequence; the natural hope is an analogue producing cycles from H^{2p}. **[KNOWN]** No such short exact sequence exists; its nearest replacement, Deligne cohomology, has a refined cycle class whose surjectivity onto Hodge classes *is* the Hodge Conjecture, and whose intermediate Jacobian J^p is a non-algebraic torus for $p \geq 2$ (Griffiths [Gri68]), so nothing is gained.

(iv) *Singularities of normal functions.* Griffiths and Green proposed to produce a cycle from the singularities of a normal function attached to a family of hyperplane sections, reducing HC to the existence of such singularities. **[KNOWN]** The programme is a genuine reformulation, but the existence of the singularity is equivalent to the nonvanishing it was meant to explain, and the outstanding density and field-of-definition statements are as hard as HC; the difficulty is relocated, not removed [GGK12].

(v) *Prove it in the Kähler category.* A proof by pure Hodge theory — harmonic forms, currents, the $\partial\bar{\partial}$-lemma — would apply to every compact Kähler manifold. **[KNOWN]** It cannot exist: HC is false there (Voisin [Voi02a]), so any Kähler-valid argument proves a falsehood. This is the Voisin Test of Chapter 3.

(vi) *Manufacture an integral cycle.* A construction producing an integral algebraic cycle with a prescribed integral Hodge class would settle the integral conjecture. **[KNOWN]** That conjecture is false — Atiyah–Hirzebruch [AH62], Totaro [Tot97], Kollár — so every correct construction must tolerate a denominator. This is the Integrality Test.

The common thread is that each dead end fails by proving either too much (a known-false statement) or exactly as much (an equivalent statement dressed differently). **[NEW]** A viable mechanism must thread between the two: strong enough to reach algebraicity, weak enough to break in the Kähler and integral categories, and candid enough to admit when it has only relocated the difficulty. That threading is the subject of Chapter 3.

2.8 The audit in one table

Table 2.2 collects the ledger. The reader should treat it as the specification against which Chapter 3 designs its constraints and against which every later chapter's claims must be checked.

Table 2.2. The status of the Hodge Conjecture and its relatives. "Theorem" means unconditionally proved; "False" means disproved; "Open" means neither.

Statement	Status	Attribution
$\mathrm{HC}(X, 1)$ (divisors), integrally	Theorem	Lefschetz 1924
$\mathrm{HC}(X, n-1)$ (curves)	Theorem	Hard Lefschetz
$\mathrm{HC}(X, p)$ for $\dim X \leq 3$	Theorem	above
$\mathrm{HC}(X, 2)$, X uniruled fourfold	Theorem	Conte–Murre 1978
$\mathrm{HC}(X, 2)$, X cubic fourfold	Theorem	Zucker 1977
HC, products of elliptic curves	Theorem	classical
HC, Hodge ring generated by divisors	Theorem	$(1,1)$ + Hard Lefschetz
HC for abelian varieties, general	**Open**	—
Weil classes on abelian fourfolds of Weil type	**Open**	Weil 1979
$\mathrm{HC}(X, 2)$, general fourfold	**Open**	—
Generalised (coniveau) HC	**Open**	Grothendieck 1969
Variational HC	**Open**	—
Integral $\mathrm{HC}_{\mathbb{Z}}$ (torsion)	**False**	Atiyah–Hirzebruch 1962
Integral $\mathrm{HC}_{\mathbb{Z}}$ (cobordism)	**False**	Totaro 1997
Integral $\mathrm{HC}_{\mathbb{Z}}$ (degree)	**False**	Kollár 1992
HC for compact Kähler manifolds	**False**	Voisin 2002
Hodge's original coniveau conjecture	**False**	Grothendieck 1969
Hodge loci are algebraic	Theorem	CDK 1995; BKT 2020
Hodge classes on ab. var. absolutely Hodge	Theorem	Deligne 1982
Standard Conjecture B over $\mathbb{C}$	**Open**	(implied by HC)
HC $\Rightarrow$ Standard Conjecture B	Theorem	Kleiman 1968
Numerical motives semisimple	Theorem	Jannsen 1992
Tate conjecture for $K3$ surfaces over finite fields of odd characteristic	Theorem	Charles 2013; Madapusi Pera 2015

2.9 What this chapter does not do

The audit is a survey; it proves none of its entries. In particular we have not reproved the Atiyah–Hirzebruch, Totaro, Voisin, or Cattani–Deligne–Kaplan theorems, only explained their mechanisms and their exact strength. Three points of scholarly hygiene deserve restating.

- *Absolute Hodge is not algebraic.* Deligne's Theorem 2.6.3 is evidence, not a solution; the Hodge Conjecture is open for abelian varieties. We will not, anywhere in this book, write that Deligne proved HC for abelian varieties.
- *Hodge loci are algebraic; classes are not thereby algebraic.* Theorem 2.6.1 confirms a prediction of HC without proving HC; the two must not be conflated.
- *The integral conjecture is false.* Its failure is not a technicality to be legislated away; it dictates that every construction work rationally and tolerate a denominator (Chapter 38).

Status of Chapter 2

This chapter is expository. All mathematical theorems are `\KNOWN` and attributed to the literature. Its one `\NEW` item is the design principle that a viable mechanism must fail on the Kähler and integral walls while remaining strong enough to detect algebraicity. Its function is to fix the ledger of known/false/open that constrains the design of Chapter 3 and the claims of Parts II–VII.

Notes and References

The positive results of Section 2.2 are classical. The Lefschetz $(1,1)$-theorem is [Lef24]; the codimension-$(n-1)$ case is the standard consequence of Hard Lefschetz and its $\mathfrak{sl}_2$-package; modern proofs are in [Voi03; GH78]. The present bibliography does not support a finer priority claim for that formulation. The theorem on uniruled fourfolds is Conte–Murre [CM78], part of the wave of applications of the Bloch–Srinivas decomposition of the diagonal; the cubic fourfold is Zucker [Zuc77], via Griffiths' theory of normal functions, which the reader can pursue through [Gri69]. The reduction of the abelian-variety case to the structure of the Hodge ring, and the rôle of the Lefschetz group, are classical; the low-dimensional classification is Moonen–Zarhin [MZ99].

The failure of the integral conjecture begins with Atiyah–Hirzebruch [AH62], whose use of the differential $d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$ in the topological K-theory spectral sequence remains the cleanest way to see the torsion obstruction; the reader should consult Totaro [Tot97] for the cobordism-theoretic factorisation, which packages a wider family of torsion-sensitive topological constraints than the single differential d_3. Kollár's non-torsion degree obstruction is recorded in the 1992 Trento examples [Kol92] and discussed in Voisin's books [Voi03]; it is the one that most directly forces our rational working. Colliot-Thélène and Voisin later reinterpreted the codimension-two integral obstruction through unramified cohomology [CTV12], a viewpoint we take up in Chapter 38.

Voisin's Kähler counterexample [Voi02a] is, for this book, the single most consequential theorem in the negative direction, and we have tried to state it in the strong form she proved: not only are there no subvarieties, there are no coherent sheaves with the offending Chern class. The construction uses a torus endomorphism to manufacture a Hodge-theoretic obstruction in the Kähler category; subsequent work of Voisin produced further Kähler counterexamples, including to statements about Chern classes of coherent sheaves on bimeromorphic models, which we do not need but which reinforce the same moral. The theorem is the origin of the Voisin Test formalised in Chapter 3.

On the evidential side, the Cattani–Deligne–Kaplan theorem [CDK95] is the deepest unconditional result in the conjecture's favour; its proof rests on the fine analysis of degenerating variations of Hodge structure (nilpotent and SL_2-orbit theorems) developed by Schmid, Cattani, Kaplan and Deligne. The o-minimal reproof of Bakker, Klingler and Tsimerman [BKT20] recasts the same fact in the definable category and is the technical template for Part I. Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge [Del82], with its Principle B, is the arithmetic evidence, and its exact strength — weaker than algebraicity — is the fact we most insist on. That HC implies the Lefschetz standard conjecture is Kleiman [Kle68; Kle94], and the semisimplicity of numerical motives, which

we shall need in Chapter 30, is Jannsen [Jan92]. Finally, the Tate conjecture for $K3$ surfaces over finite fields of odd characteristic, proved in this range by Charles [Cha13] and Madapusi Pera [MP15] (with contributions of Maulik and others), is the arithmetic analogue of the $K3$ case of HC and a model, in Chapter 23, for what a completed theory looks like.

Chapter 3

The Four Design Constraints

The audit of Chapter 2 is not merely a list of facts; it is a specification. Two theorems say the conjecture is false in nearby categories — Voisin's for compact Kähler manifolds, Atiyah–Hirzebruch's and Kollár's for integral coefficients — and two theorems say the conjecture casts shadows that are provable without it — Grothendieck's corrected coniveau conjecture and Deligne's absolute-Hodge theorem. From these we extract four constraints that any purported mechanism for the Hodge Conjecture must satisfy: the Voisin Test (it must consume projectivity), the Integrality Test (it must produce denominators), the Coniveau Test (it must not prove too much), and the Absolute-Hodge Test (it must do strictly more than Deligne). We formalise each, prove that the first two are forced by known theorems, and then measure the principal mechanisms in the literature — motivated cycles, Kuga–Satake, Bloch–Beilinson, and the arithmetic/Frobenius route — against all four, scoring them in a single table. The chapter is a design document; it proves no case of the conjecture and is careful to say, of each mechanism including this book's own, exactly where it stops.

3.1 Introduction

A great deal of effort has been spent on strategies for the Hodge Conjecture that could not possibly work, and the reason is almost always the same: the strategy proves, if it proves anything, a statement that is already known to be false. The most efficient way to think about the conjecture is therefore not to ask "how might one prove it" but to ask "what must a proof *not* do", and to convert each known counterexample into a constraint. This chapter carries out that conversion. It is the hinge between the survey of Part 0 and the constructions of Parts I–VII: every later chapter that proposes a piece of mechanism will be required to state where it meets these constraints, and the subsection *Where projectivity is consumed*, mandatory throughout Parts II–VII, is the local form of the first constraint stated here.

The four constraints are of two kinds. The first two are *prohibitions*, each forced by a theorem: because HC is false for compact Kähler manifolds, a proof must break in the Kähler category (Section 3.2); because the integral conjecture is false, a proof must lose integrality (Section 3.3). These are not heuristics — they are logical consequences of Theorems 2.3.1, 2.3.4 and 2.4.1, and we prove the corresponding criteria. The second two are *calibrations*, each forced by a theorem in the opposite direction: because Grothendieck's corrected coniveau conjecture governs which classes can be supported on which subvarieties, a proof must respect the coniveau stratification and must not, as a by-product, resurrect Hodge's refuted formulation (Section 3.4); and because Deligne proved the abelian-variety Hodge classes absolutely Hodge without proving them algebraic, a proof of algebraicity must do strictly more than Deligne, and had better explain his Principle B (Section 3.5). The four together are drawn schematically in Plate 3.1.

Section 3.6 then audits the field. We take the four most developed strategies — André's motivated cycles, the Kuga–Satake construction, the Bloch–Beilinson filtration, and the arithmetic/Frobenius route through the Tate conjecture — and locate, for each, where it meets or misses each constraint. We include this book's own proposed mechanism, the rigidification programme of Parts V–VII, on the same footing and subject it to the same tests; the verdict, recorded here and defended throughout, is that it too is incomplete, with an identified gap. Section 3.7 is the resulting scorecard, Plates 3.2 and 3.3. Nothing in this chapter is a theorem about HC; the mathematics is in the formalisation of the constraints and in the precise accounting of where each mechanism stands.

3.2 First constraint: the Voisin Test

Voisin's theorem (Theorem 2.4.1) is that the Hodge Conjecture is false for compact Kähler manifolds. The constraint it imposes is best stated as a property of proofs, so we begin by formalising what it means for an argument to use projectivity.

Definition 3.2.1 (Mechanism). [NEW] A *mechanism* for the Hodge Conjecture in codimension p is a rule M that assigns, to each smooth projective complex variety X and each Hodge class $\alpha \in \mathrm{Hdg}^p(X)$, an argument that $\alpha \in \mathrm{Im}(\mathrm{cl}^p)$, together with the finite list $\Pi(M) = (\Pi_1, \dots, \Pi_N)$ of properties of X on which the argument relies. A property Π_i is *Kähler-meaningful* if it is defined for every compact Kähler manifold. The mechanism is *Kähler-stable* if every Π_i is Kähler-meaningful and holds for every compact Kähler manifold.

Definition 3.2.2 (Consuming projectivity). [NEW] A mechanism M *consumes projectivity at step i* if the property Π_i is false for some compact Kähler manifold — that is, if Π_i either fails to be Kähler-meaningful or is Kähler-meaningful but false on some compact Kähler X. The step i is a *projectivity-consuming step*.

Criterion 3.2.3 (Voisin Test). [KNOWN] [Voi02a] *Every correct mechanism for* HC *consumes projectivity at some step; no correct mechanism is Kähler-stable.*

Proof. Suppose M were a correct, Kähler-stable mechanism. Let X be a compact Kähler manifold and $\alpha \in \mathrm{Hdg}^p(X)$. Every property $\Pi_i(M)$ is Kähler-meaningful and holds for X, so the argument M applies verbatim and concludes that α lies in the image of the analytic cycle class map — that α is a combination of classes of analytic cycles, or, in the Chern-class form, of coherent sheaves. As X and α were arbitrary, M proves the Hodge Conjecture for all compact Kähler manifolds. This contradicts Voisin's theorem [Voi02a] (Theorem 2.4.1). Hence M is not Kähler-stable, and by Definition 3.2.2 it consumes projectivity at some step. □

The Voisin Test is thus not an aspiration but a theorem about proofs: it guarantees that a correct argument *has* a projectivity-consuming step, and it makes locating that step a well-posed demand. The demand has real force because most Hodge-theoretic constructions are Kähler-meaningful by nature — the Hodge decomposition, harmonic representatives, currents, the period map on the local system — so a mechanism built from them is Kähler-stable until it reaches a step that is not. Identifying that step is the discipline the book imposes on itself.

Example 3.2.4 (A naive mechanism and its unique wall). Consider the following tempting argument for $\alpha \in \mathrm{Hdg}^p(X)$: (i) represent α by a harmonic form ω of type (p,p); (ii) observe that a closed positive (p,p)-current with the right cohomology class and analytic singularities is a combination of analytic cycles; (iii) conclude. Steps (i) and (iii) are Kähler-meaningful and true in the Kähler category. The whole weight falls on the passage from a *harmonic* (p,p)-form to a *positive* one with analytic support: the Hodge condition gives a real closed (p,p)-class, not a positive one, and positivity is exactly the property that can fail without an ample class. By Criterion 3.2.3 this step *must* be where projectivity is consumed, and indeed it is the step this book isolates as the positivity-matching problem of Chapter 36 and the gap O2 in the analytic engine of Chapter 44. The example shows the test is not vacuous: it predicts, before any construction, the location of the difficulty.

The book's structural claim, argued across Parts V–VII and previewed here, is that its mechanism consumes projectivity *exactly once*, at the positivity-matching step of Chapter 36 that feeds Step B of the analytic machine (Chapter 44). Any chapter that appears to consume projectivity elsewhere is, by the standard of Criterion 3.2.3, either wrong or redundant, and is to be rewritten. This single-expenditure claim is a design target, not a theorem about HC; it is meaningful because Criterion 3.2.3 makes the number of projectivity-consuming steps an invariant worth minimising.

3.2.1 Applying the Voisin Test

The Voisin Test becomes usable when turned into a checklist. Given a proposed framework M, one proceeds as follows.

(i) *List the inputs.* Enumerate the properties $\Pi_1, \dots, \Pi_N$ of X that M's argument invokes — every theorem, every construction, every hypothesis on X.

(ii) *Test each for Kähler-meaning.* For each Π_i, ask whether it is defined for an arbitrary compact Kähler manifold. Harmonic theory, the Hodge decomposition, currents, and the period map on the local system all pass this test; a projective embedding, an ample class, GAGA, or an arithmetic model do not.

(iii) *Test each for Kähler-truth.* For those Π_i that are Kähler-meaningful, ask whether they hold on every compact Kähler manifold. A relative ample class, or the algebraicity of a period torus, is meaningful but false in general.

(iv) *Locate the wall.* If some Π_i fails step 2 or step 3, that is a projectivity-consuming step. If none does, M is Kähler-stable and, by Criterion 3.2.3, cannot be correct.

Example 3.2.5 (The Hodge-module approach passes). [KNOWN] Apply the procedure to Saito's theory of mixed Hodge modules [Sai88; Sai90], the natural framework in which to produce Hodge classes as the lowest-weight pieces of a direct image. The engine is the decomposition theorem: for a projective morphism $f\colon X \to S$ the derived pushforward $Rf_*\mathbb{Q}_X$ splits as a direct sum of shifted intersection complexes, and the summands carry polarised Hodge structures. Running the checklist, the perverse and weight formalisms of step 1 are Kähler-meaningful (Saito's theory extends to the analytic category); but the *splitting* of the decomposition theorem is powered by relative Hard Lefschetz, which requires a relative ample class $\eta \in H^2(X)$ — a property that is Kähler-meaningful yet false for a general Kähler morphism [DM05]. Step 3 therefore locates the wall precisely at the relative polarisation. The Hodge-module approach passes the Voisin Test, consuming projectivity at exactly the point where the decomposition theorem's polarisation is invoked — the same place, structurally, at which this book spends projectivity (Example 3.2.4).

3.3 Second constraint: the Integrality Test

The integral Hodge Conjecture is false (Section 2.3). We package its failure in the defect group of Remark 2.3.5 and read off the second constraint.

Definition 3.3.1 (Integral defect group). For smooth projective X, the *integral defect group* in codimension p is

$$Z^{2p}(X) \;=\; \mathrm{Hdg}^p(X,\mathbb{Z}) \,/\, \mathrm{Im}(\mathrm{cl}\colon \mathrm{CH}^p(X) \to \mathrm{Hdg}^p(X,\mathbb{Z})).$$

Criterion 3.3.2 (Integrality Test). [KNOWN] [Voi03] *There exist smooth projective X with $Z^{2p}(X) \neq 0$. Consequently no correct mechanism can output, for every X and every integral Hodge class, an integral algebraic cycle with that exact class; every correct mechanism must have a* denominator-producing step *at which its conclusion passes from $\mathrm{Hdg}^p(X,\mathbb{Z})$ to $\mathrm{Hdg}^p(X)_{\mathbb{Q}}$, tolerating multiplication by a nonzero integer.*

Proof. Non-vanishing of $Z^{2p}(X)$ for suitable X is Theorem 2.3.1 (torsion classes with $\mathrm{Sq}^3_{\mathbb{Z}} \neq 0$) and Theorem 2.3.4 (the generator of H^4 of a very general threefold of suitable degree). If a mechanism produced, for all X, an integral cycle with class exactly a prescribed integral Hodge class, it would exhibit surjectivity of the integral cycle map and force $Z^{2p}(X) = 0$ for all X, contradicting these theorems. Hence the mechanism's guarantee can hold only after tensoring with $\mathbb{Q}$, i.e. it must produce denominators. □

The Integrality Test is complementary to the Voisin Test and, like it, forces a specific structural feature: a canonical place where an integral statement is weakened to a rational one. For the mechanism of this book that place is the lattice-comparison step of Chapter 38, where the comparison between the Betti lattice and the lattice produced by the construction is only an isogeny, of index the denominator; the claim, again a design target and not a theorem about HC, is that denominators are produced *exactly once*, there, feeding the obstruction complex of Chapter 18. The two prohibitions together — consume

projectivity once, produce denominators once — are the two coordinates of Plate 3.2: they say a correct mechanism must spend exactly two things, and they say where.

Remark 3.3.3. **[KNOWN]** [Voi02a] It is worth stating why the two tests are independent. Voisin's counterexample is *rational* — the offending class α is a rational Hodge class — so it is invisible to the Integrality Test, which concerns the difference between $\mathbb{Z}$ and $\mathbb{Q}$ coefficients on projective varieties. Conversely Kollár's counterexample is projective, so it is invisible to the Voisin Test. A mechanism could pass either test and fail the other; the book must pass both, and Plate 3.2 places the two failures on orthogonal axes precisely to make the independence visible.

3.3.1 Applying the Integrality Test

The Integrality Test also reduces to a checklist. Given a framework M:

(i) *Find the coefficient ring.* Identify the ring over which M's output cycle class is defined — $\mathbb{Z}$, $\mathbb{Z}[1/N]$, or $\mathbb{Q}$.

(ii) *Find the denominator step.* Locate the first place where M divides: an isogeny, an idempotent projector, a transfer or trace argument, a normalisation.

(iii) *Check the target.* If M claims an *integral* cycle with a prescribed integral Hodge class for all X, it contradicts Criterion 3.3.2 and is wrong; if its guarantee holds only after $\otimes\mathbb{Q}$, it passes.

Example 3.3.4 (The theory of motives passes, by construction). **[KNOWN]** Apply this to the framework of Chow motives, where a variety is decomposed by projectors in $\mathrm{CH}^*(X \times X)_{\mathbb{Q}}$. The very first move — splitting the diagonal into Künneth components $\pi_0, \dots, \pi_{2n}$, or building a Chow–Künneth decomposition in Murre's sense [Mur93] — requires idempotents that do not exist integrally: the projector onto H^0 of a curve is $\frac{1}{\deg}(\mathrm{pt} \times X)$, with a genuine denominator, and Kimura's theory of finite-dimensionality is $\mathbb{Q}$-linear throughout [Kim05]. The category of motives is thus *unable* to work integrally, and this is not a defect but exactly the feature the Integrality Test demands: a framework built on motives produces its classes with denominators from the outset, so it can never contradict the failure of $\mathrm{HC}_{\mathbb{Z}}$ (Theorem 2.3.4). It passes the test at step 2, at the construction of the projectors, and it passes structurally rather than by a repair bolted on afterwards.

3.4 Third constraint: the Coniveau Test

The first two constraints are prohibitions. The third is a calibration: a mechanism must be compatible with the structure that Grothendieck's corrected generalised conjecture imposes, and in particular it must not prove a statement that is known to be false because it proves *too much*.

Recall from Section 1.8 the geometric coniveau filtration $N^cH^k(X,\mathbb{Q}) = \sum_Z \ker(H^k(X) \to H^k(X \smallsetminus Z))$, the sum over closed Z of codimension $\geq c$, and the arithmetic (Hodge-theoretic) filtration

$$\tilde{N}^cH^k(X,\mathbb{Q}) \;=\; \text{largest sub-Hodge structure } H' \subseteq H^k \text{ with } H'_{\mathbb{C}} \subseteq F^cH^k.$$

Definition 3.4.1. The two filtrations satisfy $N^c \subseteq \tilde{N}^c$ always. The *generalised Hodge Conjecture* $\mathrm{GHC}(X,k,c)$ is the assertion $N^cH^k(X,\mathbb{Q}) = \tilde{N}^cH^k(X,\mathbb{Q})$; it is the corrected form of Hodge's coniveau conjecture (Grothendieck 1969), and it remains unresolved. The case $c = k = 2p$ specialises to a statement equivalent to $\mathrm{HC}(X,p)$.

Criterion 3.4.2 (Coniveau Test). **[NEW]** *A mechanism M passes the* Coniveau Test *if the cycles it produces respect the coniveau stratification: a Hodge class lying in $\tilde{N}^cH^{2p}(X)$ is realised by M through cycles supported on subvarieties of codimension $\geq c$, compatibly with* GHC*; and if the logical structure of M does not, as a by-product, also establish Hodge's uncorrected coniveau conjecture, the integral conjecture, or the Kähler conjecture — each of which is false.*

The Coniveau Test is the subtlest of the four, because it is not a single theorem-forced prohibition but a consistency demand, and we state it as a methodological criterion rather than a metatheorem. Its content is twofold. First, it asks that a mechanism see the *level* of a class: a class that is supported, in

the Hodge-theoretic sense $\tilde{N}^c$, in high codimension should be produced by high-codimension cycles, and a mechanism that produced it by a divisor computation would be proving something structurally impossible. Second, and more sharply, it is a guard against arguments that "prove too much". The cautionary tale is Hodge's own 1950 coniveau conjecture [Hod52], refuted by Grothendieck [Gro69a] for the reason that F^c is not a sub-Hodge structure (Section 1.8): any mechanism whose reasoning would apply equally to F^c in place of $\tilde{N}^c$ is thereby refuted, because it would re-prove the false statement. The corrected filtration $\tilde{N}^c$ restores the rational structure by fiat, and a sound mechanism must use $\tilde{N}^c$, not F^c — it must "know the Grothendieck correction".

Example 3.4.3 (The correction has teeth). Let $H \subseteq H^{2p}(X, \mathbb{Q})$ be a rational Hodge class, so $H_{\mathbb{C}} \subseteq F^p \cap \overline{F^p}$. A mechanism that argued "$H \subseteq F^p$, and F^p is computed by the algebraic de Rham complex $\Omega^{\geq p}$, therefore H is supported in codimension p" would be using F^p as though it were a sub-Hodge structure. The same argument, applied to a class in F^c that is *not* rational, would assert coniveau where none exists — precisely Hodge's error. The Coniveau Test rejects such an argument. The book's mechanism avoids the error because it works with polarised sub-Hodge structures (the rigidification data of Chapter 34) rather than with the filtration steps directly, and these are, by construction, defined over $\mathbb{Q}$ and stable under conjugation.

Compatibility with GHC is also what keeps the book's central conjectures (Chapters 34, 44 and 45) from collectively implying more than HC: each is calibrated to the coniveau of the class it addresses, and Chapter 47 verifies that their conjunction is equivalent to HC and not to any known-false strengthening. The Universal Kuga–Satake statement of Chapter 23 enters on a different footing — not as an equivalent but as a carrier reducing HC to the case of abelian varieties, itself open.

3.4.1 Applying the Coniveau Test

The Coniveau Test is a consistency demand, so its procedure checks consistency rather than locating a wall. Given a mechanism M:

(i) *Identify the objects M manipulates.* Are they filtration steps F^cH^k, or genuine rational sub-Hodge structures $\tilde{N}^c$?

(ii) *Test conjugation-stability.* If M uses F^c as though it were a sub-Hodge structure — stable under complex conjugation — it re-proves Hodge's refuted 1950 coniveau conjecture and fails.

(iii) *Test the level match.* Does a class M assigns to coniveau $\geq c$ come out realised by cycles of codimension $\geq c$? A mismatch signals that M lowers coniveau for free, proving more than GHC permits.

(iv) *Test for over-reach.* Does the same reasoning, applied verbatim, also yield the integral or the Kähler conjecture? If so, M proves a falsehood.

Example 3.4.4 (Deligne's absolute Hodge cycles pass). [KNOWN] Apply the procedure to Deligne's theory of absolute Hodge cycles [Del82; DM82]. The objects it manipulates are classes that are rational in every realisation and of type (p,p) in each — genuine rational Hodge-theoretic data, not filtration steps — so step 1 is passed. Absolute Hodgeness is preserved under the conjugation action on the coefficient field, so step 2 is passed: the theory never treats F^c as conjugation-stable. Principle B propagates a class within a fixed local system, hence within a fixed sub-Hodge structure of controlled level, so the coniveau of a class is preserved along the propagation and step 3 is passed. Finally, Deligne's theory is proved for abelian varieties and makes no claim in the Kähler or integral categories, so step 4 raises no alarm. The theory of absolute Hodge cycles passes the Coniveau Test cleanly — unsurprisingly, since it was built to respect exactly the rational structure whose neglect sank Hodge's 1950 conjecture (Example 3.4.3).

3.5 Fourth constraint: the Absolute-Hodge Test

The fourth constraint comes from the strongest positive theorem in the field, Deligne's absolute-Hodge theorem (Theorem 2.6.3), read as a lower bound on the difficulty of HC.

Recall the chain of inclusions of subspaces of $\mathrm{Hdg}^p(X)$,

$$\mathrm{Alg}^p(X) \subseteq \mathrm{Mtd}^p(X) \subseteq \mathrm{AH}^p(X) \subseteq \mathrm{Hdg}^p(X),$$

where Alg^p is the algebraic classes, Mtd^p the motivated classes of André (Chapter 22), and AH^p the absolutely Hodge classes of Deligne — those whose de Rham component is rational in every conjugate variety, uniformly across cohomology theories. The Hodge Conjecture is the outer equality $\mathrm{Alg}^p = \mathrm{Hdg}^p$; the *absolute Hodge conjecture* is the weaker $\mathrm{AH}^p = \mathrm{Hdg}^p$, and it is a *theorem* for abelian varieties (Deligne 1982).

Criterion 3.5.1 (Absolute-Hodge Test). *[NEW] Because* $\mathrm{Alg}^p \subseteq \mathrm{AH}^p$*, a mechanism proving* $\alpha \in \mathrm{Alg}^p$ *proves a fortiori* $\alpha \in \mathrm{AH}^p$*. A mechanism therefore passes the* Absolute-Hodge Test *if* (i) *its output is manifestly absolutely Hodge, so that it is consistent with* $\mathrm{Alg} \subseteq \mathrm{AH}$*;* (ii) *it is compatible with Deligne's Principle B, the propagation of Hodge classes in families; and* (iii) *it supplies content strictly beyond absolute Hodgeness — the step* $\mathrm{AH} \to \mathrm{Alg}$ *— since Deligne established* $\mathrm{AH} = \mathrm{Hdg}$ *for abelian varieties without establishing* $\mathrm{Alg} = \mathrm{Hdg}$*.*

The Absolute-Hodge Test measures a mechanism against a benchmark that has actually been reached. Deligne's theorem is a proof that the abelian-variety Hodge classes behave arithmetically like algebraic classes; what it does not do is produce cycles. A mechanism for HC must therefore contain, somewhere, the increment from "arithmetically algebraic" to "algebraic", and the test asks that this increment be identified rather than smuggled. The demand is sharp: a proposed proof of HC that, on inspection, only re-establishes absolute Hodgeness has not passed the test, however elaborate it is, because it has done no more than Deligne. Conversely, a mechanism that explains *why* absolute Hodgeness holds — why Principle B propagates the classes, in terms of the same structure that produces cycles — scores well, because it unifies the known theorem with the desired one.

Example 3.5.2 (The gap is nonempty exactly at the Weil classes). The Weil classes $W_K(A) \subset \mathrm{Hdg}^2(A)$ on an abelian fourfold of Weil type (Example 2.5.2) are the sharpest illustration of the increment the Absolute-Hodge Test demands. They are absolutely Hodge — indeed motivated, by André's extension of Deligne's theorem — so the inclusion $W_K(A) \subseteq \mathrm{AH}^2(A)$ is **[KNOWN]** [Wei79; Del82; And96]; yet whether $W_K(A) \subseteq \mathrm{Alg}^2(A)$ is Question 2.5.3, and it is open. The two dimensions of $W_K(A)$ therefore lie in $\mathrm{AH}^2 \smallsetminus (\text{known } \mathrm{Alg}^2)$: they are precisely the place where absolute Hodgeness has been achieved and algebraicity has not. A mechanism passes the Absolute-Hodge Test only if it says something new about these two classes; re-deriving that they are absolutely Hodge — which Deligne and André already did — is, by Criterion 3.5.1, no progress at all. The Weil classes are thus the natural unit test for any proposed mechanism, and the book returns to them as the running example of Chapters 44, 49 and 50.

Remark 3.5.3. **[KNOWN]** Principle B is the statement that a class in a variation of Hodge structure which is Hodge everywhere and absolutely Hodge at one point is absolutely Hodge everywhere [Del82]. It is the engine of Deligne's theorem, and it is also, in the o-minimal reformulation of Chapter 20, close to the algebraicity of Hodge loci (Theorems 2.6.1 and 2.6.2): both are propagation statements for Hodge classes across a base. The book's mechanism routes through the same period-geometric structure, and Chapter 22 makes the compatibility with Principle B explicit; the extra content beyond Deligne is exactly the rigidification datum of Chapter 34, whose existence is open.

3.5.1 Applying the Absolute-Hodge Test

The Absolute-Hodge Test is the one a sophisticated framework can fail while appearing to succeed, so its procedure is the most important to run. Given a mechanism M:

(i) *Check consistency.* Is M's output manifestly absolutely Hodge, so that $\mathrm{Alg} \subseteq \mathrm{AH}$ is respected? A framework producing a non-absolutely-Hodge "cycle" is refuted outright.

(ii) *Check Principle B.* Is M compatible with the propagation of Hodge classes in families — does its output survive specialisation and deformation?
(iii) *Check the increment.* Does M supply content strictly beyond absolute Hodgeness — the step AH → Alg — or does it, on inspection, only re-establish the AH = Hdg that Deligne already proved? Only a framework that adds the increment passes.

Example 3.5.4 (André's motivated cycles pass steps 1–2 and fail step 3). [KNOWN] Apply the procedure to André's motivated cycles [And96], the most refined unconditional framework available. Step 1: motivated classes are absolutely Hodge — indeed they sit in Alg ⊆ Mtd ⊆ AH — so consistency holds. Step 2: Principle B is built into the Tannakian formalism, and motivated cycles propagate in families by construction, so this passes too. Step 3 is where the framework halts. Motivated cycles establish Hdg = Mtd for abelian varieties — extending Deligne's Hdg = AH — but the increment they leave untouched is the *second* equality Mtd = Alg, which is the Construction Problem and is exactly as hard as HC. The framework produces no cycle; it enlarges the algebra of motivic classes and proves the enlarged classes absolutely Hodge, which sharpens the benchmark rather than passing beyond it. By Criterion 3.5.1(iii) the motivated-cycles framework *fails* the Absolute-Hodge Test, and it fails at precisely the point every framework in this book must eventually confront: the passage from arithmetically algebraic to algebraic. That the most sophisticated available framework fails exactly here is the sharpest measurement we have of where the difficulty of HC resides.

Remark 3.5.5 (The four verdicts, read together). [NEW] The four worked applications are drawn from the strongest frameworks in the field, and read together they map the terrain, which is the whole point of a design constraint: a test that discriminates nothing is worthless. The Hodge-module approach passes the Voisin Test, spending projectivity at the relative polarisation of the decomposition theorem (Example 3.2.5); the theory of motives passes the Integrality Test, producing denominators in its very projectors (Example 3.3.4); Deligne's absolute Hodge cycles pass the Coniveau Test, respecting the rational structure whose neglect sank Hodge's 1950 conjecture (Example 3.4.4). Each pass is structural rather than accidental — the frameworks were built by mathematicians who had internalised the relevant counterexample. The single *failure*, Example 3.5.4, is the most instructive entry: the most refined framework, André's motivated cycles, fails the Absolute-Hodge Test not through any defect but because the increment AH → Alg it leaves open is the irreducible core of the problem. A reader who wants to know what HC "really" asks, beyond all reformulation, should look exactly there — at the one test the best available framework cannot pass, for the one reason that no reformulation has ever removed. The four constraints earn their place in this chapter because, jointly, they separate the frameworks that have a chance from the framework that only appears to: three passes and one failure is exactly the discrimination a design constraint exists to provide.

3.6 A survey of proposed mechanisms

We now measure the principal strategies against the four constraints. The tone is deliberately non-polemical: each strategy is a serious body of mathematics that has produced theorems, and the point is not to rank them but to locate, for each, the constraint that presently stops it. The findings are summarised in Table 3.1.

3.6.1 Motivated cycles (André 1996)

André's motivated cycles [And96] form an unconditional Tannakian category: one enlarges algebraic cycles by adjoining the Lefschetz involutions $*_L$ (the inverses of Hard Lefschetz that the standard conjectures predict to be algebraic), obtaining a class Mtd^p with $\mathrm{Alg}^p \subseteq \mathrm{Mtd}^p \subseteq \mathrm{AH}^p$, and a semisimple Tannakian category of motives that exists without assuming any conjecture. The conjecture factors as

$$\mathrm{HC} \iff (\mathrm{Hdg} = \mathrm{Mtd}) \wedge (\mathrm{Mtd} = \mathrm{Alg}).$$

André proved Hdg = Mtd for abelian varieties, extending Deligne. Against the tests: the framework passes the Absolute-Hodge Test's consistency demands superbly — motivated classes are absolutely Hodge and Principle B is built into the Tannakian formalism — and it is compatible with coniveau.

It consumes projectivity in its very definition, through the polarisation and Hard Lefschetz used to define $*_L$. Its limitation is the second factor Mtd = Alg: this is the Construction Problem in disguise, and motivated cycles *produce no cycles*. The mechanism converts HC into two cleaner statements and proves one of them for abelian varieties; the other is exactly as hard as HC and is left untouched. This is not a defect of the theory but a precise measurement of where the difficulty sits.

3.6.2 The Kuga–Satake construction

The Kuga–Satake construction [KS67; Del72] attaches to a polarised weight-two Hodge structure of $K3$ type an abelian variety $\mathrm{KS}(X)$ such that $H^2(X)$ embeds into $H^1(\mathrm{KS}(X))^{\otimes 2}$ as Hodge structures. It reduces Hodge (and Tate) classes of $K3$ type to Hodge classes on abelian varieties, and arithmetic refinements are central to proofs of the Tate conjecture for $K3$ surfaces over finite fields of odd characteristic (Charles; Madapusi Pera). Generalised, it becomes the Universal Kuga–Satake conjecture of Chapters 23 and 49: every polarised Hodge structure with nonzero Hodge classes is a summand of the cohomology of an abelian variety. Against the tests: Kuga–Satake naturally produces denominators (the Clifford-algebra construction is defined up to isogeny and small-index lattices), so it is friendly to the Integrality Test; its output is absolutely Hodge, so it respects the Absolute-Hodge Test. Its subtlety for the Voisin Test is instructive: the Kuga–Satake abelian variety exists as a complex torus even for a Kähler $K3$-type Hodge structure, so the construction *alone* does not consume projectivity — it must be combined with a projective input (for $K3$ surfaces, the returning Kuga–Satake correspondence is known to be motivated and absolutely Hodge, but not algebraic in general). Thus Kuga–Satake is a reduction, not a proof: it moves the difficulty to abelian varieties and to the algebraicity of one specific correspondence, and the book adopts it in exactly that role (Chapter 49). The Tate-conjecture theorems of Charles and Madapusi Pera use arithmetic Kuga–Satake theory; they do not prove general complex algebraicity of that correspondence.

3.6.3 The Bloch–Beilinson filtration

The conjectural Bloch–Beilinson filtration [Jan90] equips $\mathrm{CH}^p(X)_{\mathbb{Q}}$ with a finite descending filtration $F^\bullet$ whose graded pieces are controlled by cohomology — gr^0 by the cycle class, gr^1 by the Abel–Jacobi map, and higher pieces by higher regulators — and whose bottom step is governed by the Hodge and Tate conjectures. This is the natural home of the transcendental phenomena of Remark 1.6.2: it is designed to explain the enormous kernel of the cycle class map, not its image. Against the tests: the filtration consumes projectivity through the correspondences (decomposition of the diagonal, Bloch–Srinivas) used to construct it; it is compatible with coniveau and indeed refines it. Its limitation for HC is that it addresses injectivity, not surjectivity: it organises $\ker(\mathrm{cl}^p)$ beautifully but says little about whether a given Hodge class is in $\mathrm{Im}(\mathrm{cl}^p)$. It is a theory of how cycles fail to be detected by cohomology, whereas HC is a theory of how cohomology classes succeed in being cycles; the two meet only at gr^0, where the filtration simply restates HC.

3.6.4 The arithmetic/Frobenius route

The arithmetic route spreads X over a finitely generated $\mathbb{Z}$-algebra and uses the Frobenius weights and the Tate conjecture: a Hodge class should be a Tate class at almost all primes, and the Tate conjecture would then produce cycles ℓ-adically, to be descended to characteristic zero. Deligne's absolute-Hodge theorem is the first, and so far principal, success of this philosophy: it establishes exactly the compatibility of Hodge and étale realisations that the route needs, for abelian varieties. Against the tests: this route is the natural home of the Absolute-Hodge Test, which it passes by design, and of Principle B, which is arithmetic in origin; it produces denominators through the ℓ-adic normalisations; and — crucially for us — it consumes projectivity at a single, nameable place, the choice of an arithmetic model, which a general compact Kähler manifold does not possess. This is the same expenditure the book makes through *definability* rather than through a model over a number field (Chapters 10 and 13). The route's wall is that it replaces the Hodge conjecture by the Tate conjecture, which is also open, and requires the compatibility of the two; it trades a problem for a sibling problem. The book's view, argued in Chapters 13 and 16, is that the archimedean prism is the archimedean analogue of this arithmetic model — a definable structure that plays the role a number-field model plays in the Frobenius route — and that the trade can be made on the archimedean side, where the polarisation supplies positivity.

3.6.5 This book's mechanism: rigidification

For completeness and honesty we place the book's own proposal on the same table. The rigidification programme (Parts V–VII) proposes that every Hodge class carries a *rigidification datum* — a definable family of polarised sub-Hodge structures with arithmetic monodromy and a rational Weil operator (Chapter 34) — whose existence, the Rigidification Conjecture, is open and is proved in Chapter 47 to be equivalent to HC. The analytic engine (Chapter 44) then attempts to convert a rigidification into a positive current and thence, by Bishop's theorem and GAGA, into an algebraic cycle. Against the tests: the mechanism is designed to consume projectivity exactly once, at the positivity-matching step (Example 3.2.4, Chapter 36); to produce denominators exactly once, at the lattice-comparison step (Chapter 38); to respect coniveau by working with polarised sub-Hodge structures (Example 3.4.3); and to refine absolute Hodgeness, adding to Deligne's theorem exactly the rigidification datum (Remark 3.5.3).

The rigidification mechanism passes the four tests *by design*, but passing the tests is necessary, not sufficient. Its central conjecture (RC) is open, and even granting it, the analytic engine has an identified gap: the passage from a closed real (p,p)-current to a *positive* one — obstruction O2, the very step Example 3.2.4 predicts must be where projectivity is spent — is not established. This book does not prove the Hodge Conjecture, and Chapters 44, 48 and 51 state the gap precisely. The four constraints are the standard to which we hold every rival; we hold our own proposal to it too, and it falls short at O2.

3.7 The scorecard

Table 3.1 records, for each mechanism, where it meets each constraint and what remains. The reading is uniform: a mechanism is not "refuted" by a constraint — all five are compatible with all four constraints in the sense of not proving anything false — but each has a constraint at which its progress currently halts, and the table names it. Plate 3.3 renders the same information as a decision procedure: given a proposed mechanism, one runs it through the four tests in turn, and the exit node records the residual open problem, which for the reductions of this book is one of the ledger entries O1–O9 of Chapter 48.

Table 3.1. The four design constraints against the principal mechanisms. Entries say where, or how, each mechanism meets the constraint; the last column names the residual difficulty.

Mechanism	Voisin (projectivity)	Integrality (denominators)	Residual difficulty
Motivated cycles	in the definition ($*_L$, polarisation)	rational by design	Mtd = Alg: produces no cycles
Kuga–Satake / UKS	only in combination; KS alone is Kähler	isogeny, small index	algebraicity of the KS correspondence; UKS
Bloch–Beilinson	correspondences, diagonal	rational by design	addresses $\ker(\mathrm{cl}^p)$, not Im
Arithmetic / Frobenius	choice of arithmetic model	ℓ-adic normalisation	the Tate conjecture; Hodge = Tate
Rigidification (this book)	once, at positivity (Chapter 36)	once, at lattices (Chapter 38)	RC open; positivity gap O2

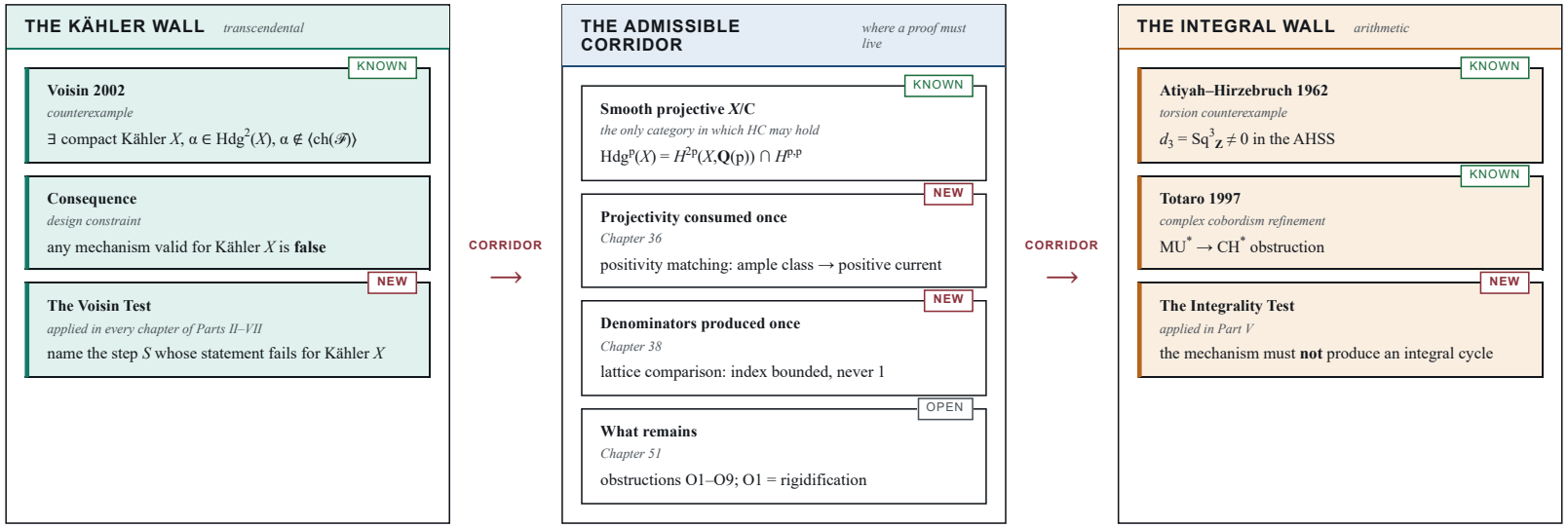
The same five mechanisms, scored against the two *calibration* constraints, are as follows. Every mechanism in the table is compatible with the Coniveau Test (each respects $\tilde{N}^\bullet$ and none re-proves Hodge's refuted formulation) and with the Absolute-Hodge Test's consistency clause (each produces absolutely Hodge output). They differ in how much they *add* beyond absolute Hodgeness: motivated cycles and the arithmetic route explain Principle B but stop at AH; Kuga–Satake adds the reduction to abelian varieties; rigidification proposes the increment AH → Alg as its datum, conditional on RC. This is the content of Plates 3.1 and 3.2.

Remark 3.7.1 (*The tests and the obstruction ledger*). **[NEW]** The four constraints are coarser than the nine-entry obstruction ledger O1–O9 assembled in Chapter 48, and it clarifies both to see how the former refine into the latter. The Voisin Test is carried by the positivity obstructions O2 and O3 (positivity matching and analytic support of the produced current), which are exactly the steps at which projectivity is spent; the Integrality Test is the denominator obstruction O4, the arithmetic of the defect group Ob^{2p} of Definition 3.3.1; the Coniveau Test is implicated in the degeneration obstruction O8, which tracks how filtration level behaves under specialisation and forbids a mechanism from lowering coniveau for free; and the Absolute-Hodge Test is the pair O1, O5 (the existence of a rigidification datum and the fullness of the resulting trace), the categorical increment beyond Deligne isolated in Criterion 3.5.1. The two remaining entries O6, O7 — a motivic t-structure and the Lefschetz standard conjecture B — are deliberately quarantined in Chapter 48 and are *not* consumed on the main line, precisely because O7 would follow from HC and may not be assumed. That four tests should organise a nine-obstruction ledger is not a coincidence: each test is a necessary condition, and the ledger is the finer bookkeeping of the sufficient conditions a single mechanism must meet all at once.

3.8 What this chapter does not do

This chapter formalises constraints; it proves no case of the Hodge Conjecture, and it must not be read as ranking the mechanisms it surveys. Three cautions.

- The four constraints are *necessary conditions* on a correct mechanism, not sufficient ones. A proposal may meet all four and still fail, as this book's own does at O2; passing the tests earns the right to be taken seriously, nothing more.
- The Coniveau and Absolute-Hodge tests (Criteria 3.4.2 and 3.5.1) are methodological criteria, not metatheorems: unlike the Voisin and Integrality tests, they are not forced in full by a single theorem, and we have flagged them `\NEW` as design principles resting on the `\KNOWN` facts of [BO74] and Chapter 2.
- The survey of Section 3.6 is not a survey of *all* approaches — notably it slights the ℓ-adic and p-adic Hodge-theoretic methods, the theory of Shimura varieties, and the recent condensed and prismatic technologies, which appear later (Chapters 16 and 23) — but of the four whose relation to the constraints is cleanest.



WHAT THE CORRIDOR FORBIDS

PROPOSED MECHANISM	SURVIVES VOISIN?	SURVIVES INTEGRALITY?	VERDICT
Purely transcendental / Hodge-theoretic	no — works for Kähler	—	refuted as stated
Purely K-theoretic, integral	—	no	refuted as stated
Absolute Hodge (Deligne 1982)	yes	yes	true but strictly weaker than HC
Motivated cycles (André 1996)	yes	yes	reduces HC, does not prove it
Kuga–Satake / carrier towers	yes	yes	open: no universal carrier
The archimedean prism (this volume)	yes — fails AP1, AP9	yes — Ch. 38	open at O1, O2

KNOWN — theorem of the literature CONDITIONAL — proved here under stated hypotheses NEW — introduced in this volume OPEN — unproved

Plate 3.1. The four design constraints as a filter. A proposed mechanism for the Hodge Conjecture must survive the Voisin Test (fail in the compact Kähler category), the Integrality Test (tolerate denominators), the Coniveau Test (respect Grothendieck's corrected level filtration), and the Absolute-Hodge Test (do strictly more than Deligne's Principle B). Each test is a wall on which a class of naive proofs breaks; the arrows show where the naive harmonic-form mechanism of Example 3.2.4 is stopped.

What it does not show: a mechanism that passes; the tests refute naive proofs but certify none, and a mechanism that consumes neither projectivity nor denominators is refuted before it begins.

THE TWO PROHIBITIONS ARE INDEPENDENT THEOREMS, NOT CONJECTURES

ELEMENT OF THE DIAGRAM	THEOREM OR DESIGN TARGET	CATEGORY / COEFFICIENTS	STATUS
Kähler wall (left axis)	Voisin 2002 — HC false for compact Kähler	Kähler; rational class	KNOWN, false
Integral wall (bottom axis)	Atiyah–Hirzebruch 1962, Totaro 1997 — integral HC false	projective; integral class	KNOWN, false
Orthogonality	Voisin's class rational, Kollár's projective: each invisible to the other test	independence of the two walls	KNOWN
Coniveau ceiling	corrected generalised Hodge conjecture (Grothendieck 1969)	respects the coniveau filtration	KNOWN, upper bound
Absolute-Hodge ceiling	Deligne 1982 — Hodge classes on abelian varieties are absolutely Hodge	absolutely Hodge, **not** algebraic	KNOWN, upper bound
Admissible interior	spend projectivity once (Ch. 36), produce denominators once (Ch. 38)	smooth projective / **C**	OPEN — O1, O2

KNOWN — theorem of the literature CONDITIONAL — proved here under stated hypotheses NEW — introduced in this volume OPEN — unproved

Plate 3.2. The constraint architecture. The two prohibitions — the Kähler wall (Voisin) and the integral wall (Atiyah–Hirzebruch, Kollár) — are placed on orthogonal axes, expressing their independence (Remark 3.3.3); the two calibrations — coniveau and absolute Hodge — are the shadows the conjecture is known to cast. A correct mechanism occupies the interior: it must spend projectivity once and denominators once, and lie below both calibration ceilings.

What it does not show: a mechanism in the admissible interior; the diagram places the two prohibitions and the two ceilings, and no known construction is drawn inside them.

1	**Input** — a claimed mechanism M producing an algebraic cycle from a Hodge class α on smooth projective X.
2	**Test 1 — Voisin** — does every step of M remain valid when X is compact Kähler? If **yes**, then M contradicts Voisin 2002 and is **refuted**. Otherwise identify the failing step S_{proj}.
3	**Test 2 — Integrality** — does M produce a cycle with integral coefficients? If **yes**, then M contradicts Atiyah–Hirzebruch and is **refuted**. Otherwise identify the step S_{den} producing denominators.
4	**Test 3 — Coniveau** — is M compatible with Grothendieck's corrected generalised Hodge conjecture? A mechanism proving the uncorrected form is **refuted** (Grothendieck 1969).
5	**Test 4 — Absolute Hodge** — does M do strictly more work than Deligne 1982? Absolute Hodgeness is strictly weaker than algebraicity; a mechanism that only reproves it has **not** addressed HC.
6	**Output** — if all four are passed, record S_{proj} and S_{den} in the obstruction ledger and proceed to Chapter 48.

THIS VOLUME, RUN THROUGH ITS OWN PROCEDURE

```
input : the archimedean prism Prism(X) = Spec_def R(H*(X,C); F, Fbar)

Test 1 : compact Kahler X satisfies AP2-AP7, AP10
          but FAILS AP1  (no definable model: definability needs an arithmetic model)
          and  FAILS AP9  (no integral Kahler class)
        > S_proj = positivity matching, Ch. 36 / TCRM Step B        [PASS]

Test 2 : lattice comparison in Ch. 18 has bounded but non-unit index
        > S_den  = lattice comparison, Ch. 38                       [PASS]

Test 3 : coniveau filtration handled in Ch. 31; no over-proof [PASS]
Test 4 : absolute Hodgeness recovered as the AP1-AP10 fragment;
          algebraicity needs AP11 (= RC), strictly more             [PASS]

verdict : NOT REFUTED. Open at O1 (rigidification) and O2 (positivity).
          This is not a proof and must not be reported as one.
```

KNOWN — theorem of the literature CONDITIONAL — proved here under stated hypotheses NEW — introduced in this volume OPEN — unproved

Plate 3.3. A decision procedure for auditing a proposed mechanism against the four constraints. The mechanism is run through the tests in sequence; each exit node is either a refutation (it proves a known-false statement) or a residual open problem, which for the reductions of this book is one of the obstruction-ledger entries O1–O9 of Chapter 48.

What it does not show: a proof: an approach that passes all four tests is merely not yet refuted, which is the most that can honestly be claimed for the framework of this volume.

Status of Chapter 3

[NEW] Definitions 3.2.1 and 3.2.2 (the formalisation of consuming projectivity) and the criteria Criteria 3.4.2 and 3.5.1 (the Coniveau and Absolute-Hodge tests as design principles). [KNOWN] [Voi02a; AH62; Kol92] Criteria 3.2.3 and 3.3.2 are logical consequences of Voisin 2002 and of Atiyah–Hirzebruch/Kollár. No case of HC is proved; the survey and scorecard are expository. The book's own mechanism is included and shown to stop at the positivity gap O2.

Notes and References

The idea of disciplining a proof strategy by the counterexamples it must avoid is old — it is implicit in Grothendieck's insistence that the standard conjectures, not the Hodge conjecture, are the right general framework — but the sharp form used here, that Voisin's Kähler counterexample turns "use projectivity" into a checkable demand, is a direct logical consequence of Voisin's counterexample [Voi02a]. We make no separate priority claim for that formulation. The formalisation of Definitions 3.2.1 and 3.2.2 is this book's, and it is deliberately modest: it does no more than make the word "mechanism" precise enough that Criterion 3.2.3 is a theorem rather than a slogan. The underlying counterexample is [Voi02a], discussed in Chapter 2; the reader who wants the Kähler theory in full should consult Voisin's books [Voi03].

The Integrality Test rests on Atiyah–Hirzebruch [AH62], Totaro [Tot97] and Kollár, as in Chapter 2; the packaging into the defect group $Z^{2p}(X)$ and its study through unramified cohomology is the work of Colliot-Thélène and Voisin [CTV12], whose viewpoint we adopt again in Chapter 38. The coniveau material is Grothendieck's: the refutation of Hodge's original conjecture and the corrected generalised conjecture are in the one-page note [Gro69a], and the general philosophy of levels and standard conjectures is in the Bombay lecture [Gro69b]. The reader should keep the two Grothendieck 1969 papers distinct: one refutes Hodge's coniveau conjecture, the other proposes the standard conjectures.

The absolute-Hodge material is Deligne's [Del82], in the Milne notes; Principle B and its use to propagate absolute Hodgeness from CM points are the technical heart of that paper, and André's motivated cycles [And96] are the unconditional Tannakian refinement, giving the inclusions Alg $\subseteq$ Mtd $\subseteq$ AH $\subseteq$ Hdg that structure Section 3.5. The Kuga–Satake construction is [KS67], with Deligne's application to the Weil conjectures for $K3$ surfaces in [Del72]; its modern arithmetic culmination in odd characteristic, the Tate conjecture for $K3$ surfaces over finite fields, is Charles [Cha13] and Madapusi Pera [MP15], and it is the model for the Universal Kuga–Satake programme of Chapters 23 and 49. The Bloch–Beilinson filtration is conjectural; Jannsen's account [Jan90] and his separate semisimplicity theorem [Jan92] are our references, and the transcendental phenomena it organises go back to Mumford's theorem on zero-cycles quoted in Remark 1.6.2.

Two attributions deserve care. First, we have been at pains, following the book's canon, never to state that Deligne proved the Hodge conjecture for abelian varieties; he proved absolute Hodgeness [Del82], and the gap to algebraicity is precisely the fourth constraint. Second, the arithmetic/Frobenius route we have sketched is a philosophy with many contributors — it is the Tate conjecture read as a tool for the Hodge conjecture — and we have attributed to it only what is proved, namely the absolute-Hodge compatibility and the $K3$ Tate theorem; the reduction of Hodge to Tate in general is open and is not claimed here. The framing of the arithmetic model as the analogue of the book's definable prism is developed, with due caution, in Chapters 13 and 16, where the analogy with Bhatt–Scholze prisms [BS22] is explicitly labelled structural rather than literal. The three plates of this chapter were designed to be read together with Table 3.1; the conventions of the figures — semantic colour, greyscale legibility, the "reading the figure" footer — are those fixed for the whole volume.

Chapter 4

The Architecture of This Volume

This chapter gives the mathematical architecture of the volume rather than a paraphrase of its table of contents. Its organising claim is that algebraicity should be formulated as descent along a definable Rees deformation joining the Hodge filtration to the Betti rational structure. It explains the archimedean prism, the associated descent and rigidification problems, the four open conjectures that control the proposed mechanism, and the obstruction ledger that records every known failure point. It also separates the mathematical, computational, and formal registers of the book, and explains the epistemic tags used to prevent conditional reductions from being mistaken for a proof of the Hodge Conjecture. The architecture is new; most of its individual ingredients are classical or established modern theorems. Nothing in this chapter, or in the volume, proves the Hodge Conjecture.

4.1 Introduction

The Hodge Conjecture asks whether a rational cohomology class of type (p,p) on a smooth projective complex variety is represented by an algebraic cycle with rational coefficients. Its familiar formulation places two subspaces of $H^{2p}(X,\mathbb{Q}(p))$ side by side: the image of the cycle class map and the subspace cut out by the Hodge filtration. That formulation is exact, but it is static. It displays the two ends of the problem without providing a space on which passage from one end to the other can be studied. The purpose of this volume is to construct such a space, to formulate descent across it, and to isolate with unusual precision the points at which the construction is known, conditional, or open. The resulting point of view is indexed under , , and .

The one-sentence thesis is:

> *Algebraicity is not a property of a cohomology class; it is a descent datum that the class carries along a prism whose special fibre is the Hodge decomposition and whose generic fibre is the Betti lattice.*

The sentence is deliberately stronger than a metaphor and deliberately weaker than a theorem. It proposes a change in the type of object to be studied. Instead of asking only whether

$$\alpha \in \mathrm{Hdg}^p(X) := H^{2p}(X,\mathbb{Q}(p)) \cap \mathrm{F}^p H^{2p}(X,\mathbb{C})$$

belongs to $\mathrm{Im}(\mathrm{cl}^p)$, we attach to α a lifting problem over a definable deformation of the Hodge filtration. The obstruction to solving that lifting problem is then compared with the failure of α to be algebraic. This comparison is exact only after the axioms and conjectures stated later in the book have been made explicit.

Three warnings govern the whole architecture. First, the rational Hodge Conjecture remains open. Secondly, its integral analogue is false by Atiyah–Hirzebruch (1962) and Totaro (1997) [AH62; Tot97]; therefore any mechanism which silently produces integral cycles is untenable. Thirdly, the analogous assertion for arbitrary compact Kähler manifolds is false by Voisin (2002) [Voi02a]; therefore any mechanism which does not visibly consume projectivity is likewise untenable. These are not qualifications appended to the project after the fact. They are design constraints, called respectively the *Integrality Test* and the *Voisin Test*.

The chapter proceeds from the central deformation to its descent problem, then from descent to rigidification, from rigidification to the four central conjectures, and from those conjectures to the complete obstruction ledger. It next displays the dependency geometry of the eleven parts that follow and separates the three registers in which claims are recorded. The final sections state where projectivity and denominators enter and give a candid account of what the architecture achieves.

A mathematical architecture is useful only if it identifies load-bearing beams and failure modes; a catalogue of aspirations would not suffice.

4.2 The thesis unpacked

Principle 4.2.1 (Algebraicity as descent). *[NEW] Let X be smooth and projective over $\mathbb{C}$, and let $\alpha \in \mathrm{Hdg}^p(X)$. The organising principle of this volume is that the algebraicity of α should be measured by the existence of compatible descent and rigidification data for α on the definable Rees deformation of the Hodge filtration, rather than by an additional condition imposed only on the fibre $\alpha \in H^{2p}(X, \mathbb{Q}(p))$.*

The principle is new as an architecture, not as a theorem about cycles. Rees modules, Hodge filtrations, comparison isomorphisms, period maps, definable analytic geometry, and descent are all established constructions. The new move is to assemble them into one lifting problem whose obstruction groups can be audited. The exact relation with algebraic cycles is developed in Chapters 12, 13, 15 and 18; the equivalence with the four conjectural forms of the problem is proved, as an equivalence of open statements, in Theorem 47.3.1.

To see why the wording "not a property" is useful, compare two modes of description. A property is tested on a point: one asks whether α lies in a prescribed subset. A descent datum is tested on overlaps and compatibilities: one asks whether local lifts glue, whether monodromy preserves them, whether the rational lattice agrees with the complex filtration, and whether the glued object has an algebraic or positive realisation. The second mode records more structure. It can fail in several distinguishable degrees, and these degrees are precisely what the obstruction complex of Definition 18.3.1 is designed to retain.

The word *prism* also carries a precise architectural meaning. A physical prism separates a single beam into visible components and can be traversed in the opposite direction to recombine them. Here the special fibre separates cohomology into Hodge-graded pieces, while the generic fibre remembers the unfiltered complex vector space together with its rational Betti lattice. The total object is a Rees deformation. The optical analogy should not be pressed beyond that statement: the archimedean prism is not a Bhatt–Scholze prism, has no δ-ring, no prime p, and no Frobenius. The comparison with prismatic cohomology is structural and is made explicitly in Chapter 16; see Bhatt–Scholze (2022) [BS22]. The notation is intended to keep the two-fibre geometry visible, not to assert an identity of theories.

The thesis has four clauses. "Algebraicity" refers throughout to rational algebraic cycles modulo homological equivalence. "Descent datum" refers to effective descent on the period site $\mathcal{S}_{\mathrm{per}}$, whose covers are definable surjections admitting local definable sections. "Special fibre" refers to

$$\mathrm{gr}_{\mathrm{F}}^{\bullet} H^*(X, \mathbb{C}) \simeq \bigoplus_{a+b=*} H^{a,b}(X),$$

and "generic fibre" refers to $H^*(X, \mathbb{C})$ equipped with the image of $H^*(X, \mathbb{Z})/\mathrm{tors}$ and hence with its $\mathbb{Q}$-structure. None of these clauses alone distinguishes algebraic classes from arbitrary Hodge classes. The distinction is conjecturally carried by compatibility among them.

Remark 4.2.2. The shift from a static membership question to a dynamic lifting question does not make the Hodge Conjecture formally weaker. A reformulation can be exact and still be no easier to solve. Its value lies in separating obstructions which are conflated in the equality $\mathrm{Im}(\mathrm{cl}^p) = \mathrm{Hdg}^p(X)$. In particular, existence, positivity, support, denominators, and categorical fullness become different entries rather than one undifferentiated failure.

4.3 The archimedean prism

Let H be a finite-dimensional complex vector space with a decreasing filtration $\mathrm{F}^\bullet H$. Its Rees module is

$$\mathrm{Rees}_{\mathrm{F}}(H) := \sum_{a\in\mathbb{Z}} t^{-a}\mathrm{F}^a H \subset H[t,t^{-1}].$$

After the harmless choice of sign convention encoded in Definition 13.5.1, restriction to $t \neq 0$ recovers the constant family with fibre H, while the fibre at $t = 0$ is $\mathrm{gr}_{\mathrm{F}}H$. Applied to the Hodge filtration on $H^*(X,\mathbb{C})$, this is the linear core of the archimedean prism. The Rees construction is classical; the decisive additional requirement is that the total object be formed in the category $\mathbf{An}^{\mathrm{def}}$ of definable complex analytic spaces, developed in Chapter 6.

Construction 4.3.1 (Preview of the prism). [NEW] For a smooth projective complex variety X, the object later denoted $\Delta(X)$ is the definable relative spectrum of the Rees algebra of the Hodge filtration, together with the comparison datum

$$\iota_X : H^*(X,\mathbb{Z})/\mathrm{tors} \longrightarrow H^*(X,\mathbb{C}).$$

Its structural morphism is $\Delta(X) \to \mathbb{A}^1_{\mathrm{def}}$, its special fibre is $\mathrm{gr}_{\mathrm{F}}H^*(X,\mathbb{C})$, and its fibre over 1 is the Betti realisation with its rational lattice. The complete definition and functoriality proof are given in Definition 13.5.1 and Theorem 13.5.2.

The qualifier *definable* is not decorative. In the unrestricted analytic category a holomorphic family may oscillate or accumulate in ways for which no finite stratification, uniform bound, or algebraisation theorem is available. The o-minimal substrate of Chapter 5 forbids such behaviour, and the definable complex geometry of Chapter 6 turns closed definable analytic loci into algebraic loci. Tame uniformisation of period maps in Chapter 7 then places the Hodge-theoretic variation inside this substrate. Thus the prism has an arithmetic or algebraic input encoded through definability; it is not merely the analytic Rees space of an arbitrary filtered vector bundle.

The two fibres play asymmetrical roles. At $t = 0$, the Hodge type of a class is visible because the filtration has been graded. At $t = 1$, rationality and integral comparison are visible because the Betti lattice has not been forgotten. An algebraic cycle supplies a class compatible with both views: its de Rham class lies in F^p, its Betti class is rational, and its formation is stable under the geometric operations used to define descent. An arbitrary rational Hodge class has the same fibrewise signatures. The unresolved problem is whether those signatures extend to the stronger rigidified datum demanded over the total prism.

Example 4.3.2 (A divisor class). Let X be a smooth projective surface and let L be a line bundle. The class $c_1(L) \in H^2(X,\mathbb{Z}(1))$ lies in F^1 and determines a section of the Rees module whose value at 0 belongs to $H^{1,1}(X)$. Because L itself is an algebraic object, pullback along a definable family and gluing along definable covers are inherited from ordinary descent for line bundles. This is the model case in which descent is already effective. The Lefschetz $(1,1)$-theorem (Lefschetz 1924) says conversely that every integral $(1,1)$-class arises this way [Lef24]; the difficult higher codimension problem begins where no line bundle is available.

Example 4.3.3 (A class with identical fibrewise tests). Let X be an abelian fourfold of Weil type and let α be a rational Weil Hodge class in middle degree. Its value on the special fibre has type $(2,2)$ and its value on the generic fibre belongs to the rational Betti structure. Those two tests do not decide whether α is algebraic; the Hodge Conjecture for abelian varieties is open in general, and Deligne's theorem that their Hodge classes are absolutely Hodge does not prove algebraicity (Deligne 1982) [Del82]. The example shows why a prism with only identified fibres would be insufficient. The additional datum must govern transport, monodromy, rationality, and geometric realisation.

Two coordinate models of the architecture

Worked model I: a split abelian surface. [KNOWN] [HMS17] Let

$$E_i = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}), \qquad E_{2i} = \mathbb{C}/(\mathbb{Z} + 2i\mathbb{Z}), \qquad A = E_i \times E_{2i}.$$

With integral homology basis a_1, b_1, a_2, b_2, the period matrix is

$$\Omega = \begin{pmatrix} i & 0 \\ 0 & 2i \end{pmatrix}, \qquad H_1(A,\mathbb{Z}) = \mathbb{Z}a_1 \oplus \mathbb{Z}b_1 \oplus \mathbb{Z}a_2 \oplus \mathbb{Z}b_2.$$

Write α_j, β_j for the dual cohomology basis. The two holomorphic forms are

$$\omega_1 = \alpha_1 + i\beta_1, \qquad \omega_2 = \alpha_2 + 2i\beta_2.$$

Thus the weight-one Hodge numbers are $h^{1,0} = h^{0,1} = 2$, and exterior squaring gives

$$(h^{2,0}, h^{1,1}, h^{0,2}) = (1, 4, 1).$$

The Weil operator on $H^1(A,\mathbb{Q})$, in the ordered basis $(\alpha_1, \beta_1, \alpha_2, \beta_2)$, is the hand-checkable rational matrix

$$C = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -2 & 0 \end{pmatrix}, \qquad C^2 = -I_4.$$

Indeed $C\omega_j = i\omega_j$ for $j = 1, 2$, while conjugation gives $C\overline{\omega}_j = -i\overline{\omega}_j$. The integral classes $\eta_j = \alpha_j \wedge \beta_j$ are of type $(1,1)$. In the lattice $\mathbb{Z}\eta_1 \oplus \mathbb{Z}\eta_2$, cup product has matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

because $\eta_1^2 = \eta_2^2 = 0$ and $\int_A \eta_1 \wedge \eta_2 = 1$. Hence the product polarisation $\theta = \eta_1 + \eta_2$ satisfies $\int_A \theta^2 = 2$. [NEW] This example displays every input of the prism in coordinates: the Betti lattice, the Hodge filtration generated by ω_1, ω_2, the rational Weil operator, and the positive integral class. Chapter 13 forms its bi-Rees object, while Chapter 36 consumes θ at the positivity step.

Worked model II: a rank-three K3-type lattice. [KNOWN] [GH78] Let $L = \mathbb{Z}e \oplus \mathbb{Z}f \oplus \mathbb{Z}g$ with bilinear form

$$Q = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Set

$$\omega = e - f + ig, \qquad H^{2,0} = \mathbb{C}\omega, \qquad H^{0,2} = \mathbb{C}\overline{\omega}, \qquad H^{1,1} = \mathbb{C}(e+f).$$

Direct multiplication gives

$$Q(\omega,\omega) = -2(1)(-1) + 2i^2 = 0, \qquad Q(\omega,\overline{\omega}) = -2(1)(-1) + 2i(-i) = 4.$$

Moreover $Q(e+f,\omega) = 1 - 1 = 0$. Consequently this is a polarised weight-two Hodge structure with

$$(h^{2,0}, h^{1,1}, h^{0,2}) = (1, 1, 1), \qquad H^{1,1} \cap L_{\mathbb{Q}} = \mathbb{Q}(e+f), \qquad Q(e+f, e+f) = -2.$$

The Hodge filtration is represented in the lattice basis by the column matrix

$$F^2 = \operatorname{span}_{\mathbb{C}} \begin{pmatrix} 1 \\ -1 \\ i \end{pmatrix}, \qquad F^1 = \operatorname{span}_{\mathbb{C}} \left\{ \begin{pmatrix} 1 \\ -1 \\ i \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

[NEW] The special Rees fibre records the three one-dimensional Hodge summands, while the fibre over 1 remembers the integral Gram matrix Q. The rational $(1,1)$-vector $e+f$ passes both fibrewise tests. Only a geometric realisation and its descent data can decide whether a class with these coordinates is supplied by a cycle. This is the linear model used by the K3-type carrier discussion in Chapter 23, and its degeneration is consumed by Chapter 24.

4.4 Descent and rigidification

Descent is the demand that data constructed locally on a cover be compatible on double overlaps and satisfy the cocycle condition on triple overlaps. In the present setting the covers belong to the period site introduced in Definition 14.2.2. Its objects are definable families of polarised Hodge structures; a covering family is represented by a definable surjection which admits local definable sections. The effectiveness theorem of Theorem 15.4.2 says that suitable definable families can be recovered from their local pieces. This is a theorem about the ambient category, not a theorem asserting that the local pieces required by a given Hodge class exist.

For a class α, one first asks for local extensions through period charts. On intersections one compares these extensions by transition automorphisms. Failure of equality produces a Cech 1-cocycle with values in the rigidification sheaf; failure of integral compatibility appears one degree later. Schematically one has

$$\{\text{local lifts of } \alpha\} \longrightarrow Z^1(\mathcal{S}_{\mathrm{per}}, \mathrm{Rig}) \longrightarrow H^1(\mathcal{S}_{\mathrm{per}}, \mathrm{Rig}).$$

The point of sheaf cohomology here is not terminological elegance. It distinguishes the absence of a local lift from the failure of local lifts to glue and from the failure of a glued complex object to match a rational lattice.

Definition 4.4.1 (Rigidification, architectural form). [NEW] A *rigidification datum*, in the sense previewed here and defined precisely in Definition 34.2.2, consists of a definable family of polarised sub-Hodge structures over a definable base, containing the given Hodge class in a distinguished fibre, with arithmetic monodromy and a fibrewise Weil operator defined over $\mathbb{Q}$, together with the descent compatibilities required on $\mathcal{S}_{\mathrm{per}}$. It refines the statement that the class remains of Hodge type.

Rigidification is stronger than horizontality and stronger than membership in a Hodge locus. The Cattani–Deligne–Kaplan theorem says that the locus on which a flat integral class remains of Hodge type is a countable union of algebraic subvarieties (Cattani–Deligne–Kaplan 1995) [CDK95]. Bakker–Klingler–Tsimerman gave a tame proof using definability (2020) [BKT20]. Neither theorem constructs an algebraic cycle representing the class. A rigidification datum adds rational group-theoretic and descent constraints intended to bridge that gap; whether every Hodge class admits one is precisely open.

Proposition 4.4.2 (Descent is not existence). [NEW] *Effective descent for definable families does not imply that a rigidification datum exists for every Hodge class.*

Proof. An effective descent theorem has the logical form: if local objects and isomorphisms satisfying a cocycle condition are given, then there exists a global object realising them. The Rigidification Conjecture asks for the local objects and the required rational compatibility in the first place. The antecedent of effective descent is therefore not supplied by the theorem. The distinction is made concrete by the boundary map in Theorem 18.4.1: a non-zero obstruction class records precisely that the input needed for descent cannot be chosen compatibly. □

Necessity: the cocycle hypothesis cannot be dropped. [NEW] Cover a point by three labelled copies U_1, U_2, U_3, put the one-dimensional vector space $\mathbb{Q}$ on each copy, and prescribe transition matrices

$$g_{12} = (1), \qquad g_{23} = (1), \qquad g_{13} = (-1).$$

All three maps are isomorphisms, but on the triple overlap $g_{23}g_{12} = (1) \neq (-1) = g_{13}$. If a global line with chosen local identifications existed, composing those identifications would force the cocycle equality, a contradiction. Thus local objects and pairwise isomorphisms do not suffice without compatibility. Chapter 15 proves effectivity once compatibility is present, and Chapter 18 turns its failure for rigidifications into the boundary class used by Chapter 34.

The term *rigidification* is meant literally: one replaces a point of a large analytic or Hodge-theoretic moduli problem by a datum with fewer automorphisms and arithmetic monodromy. In the motivic register, the same operation appears as fullness of a realisation functor: a Hodge-theoretic endomorphism should lift to a motivic one. In the analytic register, it appears as concentration of a positive current onto analytic support. The convergence of these descriptions is the reason rigidification, rather than mere definability, is load-bearing.

4.5 The four conjectural faces

The architecture organises four open conjectures. They arise in different languages and are not presented as independent evidence for four unrelated programmes. Their purpose is to expose equivalent forms of the same missing step and thereby allow methods from arithmetic groups, motives, currents, and abelian varieties to test one another.

Tag	Name	Architectural content
RC	Rigidification Conjecture	\OPEN Every rational Hodge class admits a definable rigidification with arithmetic monodromy and rational Weil operator; see Conjecture 34.3.1.
THC	Trace–Hodge Conjecture	\OPEN The trace–cycle map from endomorphisms of traceable motives to Hodge classes is surjective, equivalently the Hodge realisation is full; see Definition 29.4.1 and Theorem 30.3.1.
CP_p	Concentration Principle	\OPEN A positive closed (p,p)-current representing a *big* rational Hodge class can be deformed within its cohomology class to one with analytic support; see Conjecture 45.2.3.
UKS	Universal Kuga–Satake	\OPEN Every polarised rational Hodge structure of even weight with a non-zero Hodge class occurs functorially, with Mumford–Tate control, as a summand in the cohomology of an abelian variety; see Chapters 23 and 49.

The Rigidification Conjecture is the central face. It says that the local Hodge-theoretic datum can be made arithmetic and descendable. The Trace–Hodge Conjecture is its categorical shadow: if every Hodge morphism comes from the category **TrM**, then the monodromy-compatible object needed for rigidification is available. The Concentration Principle is its analytic shadow: after positivity has been supplied, the diffuse representative must become geometrically supported. The Universal Kuga–Satake statement is a reduction principle: it seeks to place arbitrary even-weight Hodge structures inside the tensor geometry of abelian varieties, where group-theoretic tools are strongest.

Each formulation has a different temptation which the architecture blocks. For RC, definability may be confused with arithmeticity; they are not the same. For THC, injectivity of a trace map may be confused with fullness; they are not the same. Injectivity comes from the faithful Hodge realisation on homological motives; Jannsen's semisimplicity theorem (1992) describes only the numerical quotient and supplies neither injectivity nor fullness [Jan92]. For CP_p, the existence of a closed (p,p)-form may

be confused with positivity; this is Obstruction O2. For UKS, the classical Kuga–Satake construction for K3 type may be extrapolated without proof to every even weight; that extrapolation is exactly the open content.

Warning 4.5.1. `\OPEN` None of RC, THC, CP_p, or UKS is proved in this volume. The master theorem identifies their logical relation to the Hodge Conjecture; an equivalence of open statements is not a proof of any one of them.

4.6 The master equivalence in advance

The culmination of Part VII is best stated now, because the preceding parts are organised around its arrows. The exact theorem includes the twelve prism axioms, the reduction required for UKS, and the distinction between the rational statement and integral matching. Its architectural content is the following.

Metatheorem 4.6.1 (Master equivalence, informal preview). *[NEW] Subject to the precise formulations fixed in Chapters 23, 29, 34 and 45, the Hodge Conjecture for smooth projective complex varieties is equivalent to the Rigidification Conjecture, to the Trace–Hodge Conjecture, and to the Concentration Principle. The Universal Kuga–Satake Conjecture implies the same statement after the reduction and functoriality hypotheses isolated in Chapter 49. The exact unconditional theorem about these conjectures, including both directions of every asserted equivalence, is Theorem 47.3.1.*

Proof. This chapter records only the dependency. The proof is given in Theorem 47.3.1: the trace route factors through fullness and rigidification; the analytic route factors through the twistor–current rigidification machine and concentration; and the reverse implications use cycle classes to construct the corresponding rigidified, trace, or supported objects. The Kuga–Satake arrow additionally uses the reduction theorem of Chapter 49. No implication is being deduced here independently of those chapters. □

Necessity of the Kuga–Satake qualifier. [KNOWN] [KS67; Del82] Let A be an abelian fourfold of Weil type and let $\alpha \in H^4(A,\mathbb{Q}) \cap H^{2,2}(A)$ be a Weil class. Placing α in abelian cohomology has already been achieved, yet its algebraicity is open in general. Deligne's theorem makes α absolutely Hodge, not algebraic. Hence a universal carrier statement without the abelian endpoint would not close the final arrow. This explicit object is why Chapter 23 states UKS as open and why Chapter 49 records a reduction rather than an unconditional additional equivalence.

The visual core is

$$\begin{array}{ccccc} \mathrm{RC} & \Longleftrightarrow & \mathrm{CP}_p & & \\ \Updownarrow & & \Updownarrow & & \\ \mathrm{THC} & \Longleftrightarrow & \mathrm{HC} & \Longleftarrow & \mathrm{UKS}, \end{array}$$

where the last arrow is labelled by the reduction of Chapter 49. The vertical arrow on the left is the passage between a geometric rigidification and categorical fullness. The top horizontal arrow is the analytic bridge built in Chapters 44 and 45. The bottom horizontal arrow is the trace route of Section 47.6. The diagram is a map of obligations, not a commutative diagram of constructed functors at every vertex.

Why advance an equivalence before building its terms? Because it changes how the reader should evaluate local results. A theorem on definable period maps does not by itself advance directly to a cycle; it verifies a substrate requirement. A semisimplicity theorem does not provide fullness; it controls one side of a categorical map. A compactness theorem for currents does not provide positivity; it becomes useful only after the positive cone has been entered. The master equivalence prevents these genuine achievements from being over-read.

4.7 The obstruction ledger

The obstruction ledger is the negative image of the master theorem. It records the places where a proposed proof can fail and the mathematical home of each failure. The completeness assertion is relative to the twelve prism axioms of Chapter 46; it does not claim that every conceivable approach to the Hodge Conjecture must use this vocabulary.

Entry	Obstruction	Where it lives	Architectural role
O1	Existence of a rigidification datum for one Hodge class	$H^1(\mathcal{S}_{\mathrm{per}}, \mathrm{Rig})$	`\OPEN` The Rigidification Conjecture itself.
O2	Matching a rigidification with a positive current	$H^1(\mathcal{S}_{\mathrm{per}}, \mathrm{Pos})$	`\OPEN` The gap in Step B of the analytic machine.
O3	Forcing analytic support; Bishop compactness requires a mass bound	Complex analytic geometry of currents	**[KNOWN]** [Kle68] Discharged as an independent entry: the polarisation reduces its mass bound to O2.
O4	Bounding the index of the produced rational lattice	$H^2(\mathrm{Ob}^\bullet(X))$	**[NEW]** Discharged modulo O1 by Corollary 38.7.1; the denominator mechanism is completely identified.
O5	Fullness of Hodge realisation on $\mathbf{TrM}(\mathbb{C})$	Rigid tensor categories	**[KNOWN]** Discharged as an independent entry: the master theorem identifies it with O1.
O6	Existence of a motivic t-structure for the strongest derived form	Motivic homological algebra	**[KNOWN]** Discharged from the main implication: the master theorem does not use it.
O7	The Lefschetz standard conjecture B	Algebraic correspondences	**[KNOWN]** [Kle68] Discharged as an admissible premise: it lies downstream of HC and is quarantined rather than assumed.
O8	Lifting cycles from a special fibre to a general fibre	Degeneration theory	**[KNOWN]** Discharged from the fixed-fibre equivalence; it remains a refinement for families.
O9	Making CDK/BKT finiteness and height bounds effective	O-minimal counting and computation	**[KNOWN]** Discharged from the qualitative equivalence; it remains a quantitative refinement.

Metatheorem 4.7.1 (Completeness relative to the prism axioms). **[NEW]** *Relative to axioms* AP1–AP12 *of Chapter 46, every failure in the master construction factors through one of* O1–O9*; no tenth obstruction is required. Moreover* O5 *implies* O1*, while* O1–O4 *together imply the rational Hodge Conjecture. After the reductions of Theorem 48.6.1, only* O1 *and* O2 *remain open as independent obstructions;* O3–O9 *are discharged in the precise ledger sense.*

Proof. This is the theorem proved in Theorem 48.6.1. Filter the construction by local existence, descent, positivity, support, lattice comparison, categorical lifting, auxiliary motivic structure, degeneration, and effectivity. Taking the homotopy fibre at each successive arrow produces, in order, the nine failure objects displayed above; exactness of the resulting tower shows that a failed composite has a first failed stage and hence contributes to one of those nine objects.

It remains to reduce the list. The master equivalence identifies categorical fullness O5 with rigidification O1. Axiom AP4 supplies the mass control needed after positivity, so O3 adds no independent difficulty beyond O2. Corollary 38.7.1 identifies the unique denominator and discharges O4 modulo O1. The motivic t-structure O6 is not used; O7 is downstream of HC by Kleiman and therefore cannot be an input; and O8, O9 refine families and effectivity without obstructing the fixed-fibre qualitative equivalence. Thus precisely O1 and O2 survive as independent open entries. The assertion remains relative to the chosen axioms and does not rule out an obstruction formulated outside this architecture. □

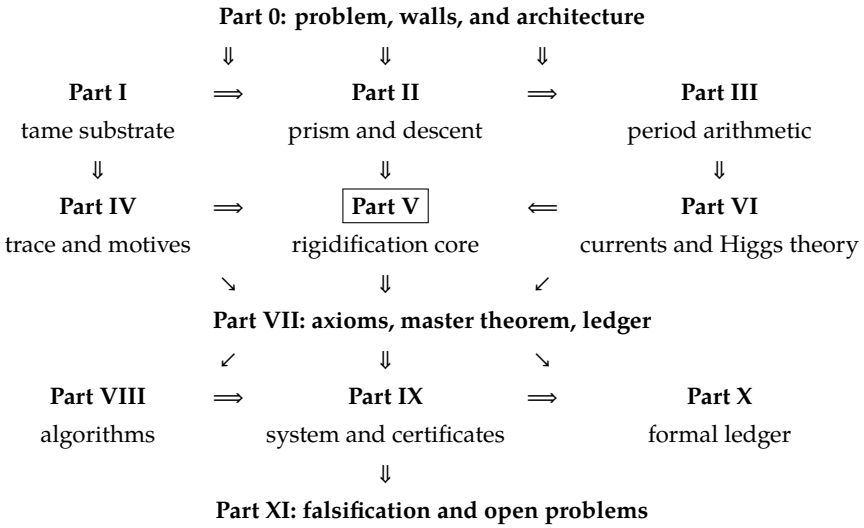
Necessity of the relative hypothesis. **[KNOWN]** [Voi02a] The compact Kähler counterexamples of Voisin provide the explicit sharpness test. Such a manifold has a real Hodge structure and a Weil operator, but need not have the definable arithmetic model or integral ample class required by AP1 and AP9. If those axioms are dropped, the nine-stage factorisation above is no longer exhaustive and the rational Kähler analogue is false. The word "relative" in Metatheorem 4.7.1 therefore cannot be removed.

The ledger has two functions. Mathematically, it makes reductions composable: a theorem eliminating O3 can be used without pretending to eliminate O2. Epistemically, it prevents a familiar pattern of false proof in which an argument passes from a Hodge representative to a positive current, or from a numerical relation to an exact period relation, without recording the missing implication. Computationally, it tells the system of Parts VIII and IX what kind of certificate is being sought.

The entries have different logical statuses. Only O1 and O2 remain open as independent entries. O5 is another name for the O1 load under the categorical equivalence, and O3 reduces to O2 after polarisation supplies the required mass bound. Chapter 38 discharges O4 by locating and quantifying the unique denominator, modulo the already open O1. O6 is deliberately absent from the main line; O7 is quarantined downstream; and O8, O9 are family and effectivity refinements. Calling these seven entries "discharged" means that none contributes an additional independent gap. It does not assert universal vanishing of every associated group.

4.8 The dependency geometry of the volume

The volume is not linearly dependent in the order of its chapters. Its logical shape is a fan: a tame analytic substrate supports a prism and a period-theoretic route; motivic and analytic engines approach the same rigidification problem from opposite sides; a master theorem compares them; and computational and formal systems audit examples and proof obligations without being used as premises for the mathematical equivalence.



Part I, beginning with Chapter 5, supplies o-minimality, definable complex geometry, tame uniformisation, definable Chow, and the tame site. Its output is a category stable enough for the Rees deformation and rigid enough for algebraisation. Part II constructs $\Delta(X)$, defines descent, and packages failure in $\mathrm{Ob}^{\bullet}(X)$. Part III studies Mumford–Tate groups, Hodge loci, atypical intersections, absolute Hodge classes, Kuga–Satake constructions, and degenerations. These parts survive independently of the truth of RC.

Part IV builds the motivic engine around traceable motives and the trace–cycle map. It obtains an exact reformulation but not a supply of cycles. Part V formulates rigidification and isolates the unique projectivity and denominator steps. Part VI develops the analytic engine: positive currents, Bishop compactness, non-abelian Hodge theory, Higgs lifts, mixed Hodge modules, and the twistor–current rigidification machine TCRM. The central open passage is from a closed Hodge representative to a positive current with controlled support.

Part VII is the logical hinge. It states the twelve axioms, proves the master equivalence of conjectures, checks the ledger, analyses the abelian reduction, and states the precise gap. Parts VIII and IX are not a numerical proof of the Hodge Conjecture. They form POLARIS-HC, a system for exact algebra, Picard–Fuchs equations, certified periods, relation discovery, cycle search, and certificate management. Its trust ladder runs from heuristic output to machine-checked proof, and a candidate relation found by PSLQ or LLL is never promoted merely because it is numerically convincing.

Part X asks what can presently be formalised. It records the negative fact that the full Hodge Conjecture cannot yet be stated in Lean 4 with the existing mathematical libraries, because several foundational objects are missing. The system HELM stores typed proof obligations, while CONSILIUM separates proposing, verifying, falsifying, bibliographic checking, and refereeing. Part XI turns the whole book against itself through falsification tests and a catalogue of open problems.

Proposition 4.8.1 (Modularity of the architecture). *[NEW] If the Rigidification Conjecture is false, the tame geometry of Part I, the period theory of Part III, the motivic reformulation of Part IV, the analytic results preceding the open positivity step in Part VI, and the computational and formal audit systems of Parts VIII–X remain mathematically meaningful. What fails is the chain of implications through Parts V–VII that would identify every Hodge class with an algebraic cycle.*

Proof. The cited parts are constructed from established inputs without using RC as a premise. Their theorems concern definability, period maps, categories, currents, algorithms, or proof records. The

implication to algebraic cycles first requires the open lift recorded as O1 and its analytic companions. Removing that lift therefore disconnects the master chain but does not invalidate the independent constructions. This dependency audit is developed in Chapter 69. □

Sharpness of the modular boundary. [KNOWN] [Wei79] For the same abelian fourfold of Weil type, the exact front end can store the defining equations, the period modules can compute certified enclosures, and the motivic chapter can form the trace–cycle target whether or not the distinguished Weil class is algebraic. What cannot be emitted without RC is a certified algebraic cycle mapping to that class. Thus the independent modules survive while the cycle-producing composite does not. Chapters 52 and 55 consume this separation in software, and Chapter 69 consumes it in the survival analysis.

4.9 Three registers

The same mathematical project is pursued in three registers, which must not be conflated. The *mathematical register* contains definitions, theorems, conjectures, reductions, and counterexamples. Its currency is proof. A claim enters this register as `\KNOWN` only when it has a proof or a verified source; it enters as `\COND` only with named hypotheses; and it remains `\OPEN` when no proof is known. The prism, the period site, the obstruction complex, and the equivalence theorem belong here.

The *computational register* contains exact symbolic calculations, certified numerical enclosures, candidate period relations, searched-for cycles, and machine-readable artefacts. Its currency is a certificate with an explicit trust level. A floating-point period is not a cohomology class; a small residual is not a proof of a rational relation; a polynomial returned by a search is not a verified cycle until its geometric and cohomological properties have been checked. Parts VIII and IX therefore distinguish heuristic, certified-numeric, exact-symbolic, and machine-checked levels.

The *formal register* contains objects and proofs that have been represented in a proof assistant. Its currency is a kernel-checked term relative to a declared foundation and imported library. Formalisation is not a stronger adjective attached to an informal argument. One may possess a correct published theorem whose prerequisites are absent from the library, or a fully formal numerical inequality which is irrelevant to the geometric claim. Part X records this gap instead of hiding it.

Definition 4.9.1 (Register-preserving assertion). [NEW] An assertion is *register-preserving* if its conclusion is stated in the same register as its certified premises, or if every bridge to another register is named and separately verified. Thus a certified numerical enclosure may support a certified exclusion bound, but it becomes an exact algebraic relation only through an exact or formally verified bridge.

Proposition 4.9.2 (No silent promotion). [NEW] *In the architecture of this volume, no computational or formal artefact is used to promote a mathematical statement tagged* `\OPEN` *to* `\KNOWN` *unless a register-preserving proof of the bridge is supplied.*

Proof. This is a design invariant enforced by the trust ladder of Chapter 62 and the typed obligations of Chapter 66. Each artefact records its register and trust level; bridge obligations are separate nodes in the dependency graph. Consequently a missing bridge remains visible and carries its previous epistemic status. □

Necessity of a numerical bridge. [KNOWN] [FBA99] The continued-fraction convergent $99/70$ gives

$$\left|70\sqrt{2} - 99\right| \approx 0.0050506,$$

although $70\sqrt{2} - 99 \neq 0$. Later convergents make the residual arbitrarily smaller without producing an integer relation between 1 and $\sqrt{2}$. Therefore a small residual, even one enclosed rigorously, certifies only an inequality at a stated coefficient bound. An exact symbolic bridge is indispensable. Chapters 55 and 56 consume this example when separating interval enclosure from integer-relation discovery, and Chapter 62 encodes the separation in the trust ladder.

The three registers interact productively. Mathematics specifies exact targets for computation. Computation discovers examples, rejects implausible claims, and supplies certified local data. Formalisation reveals hidden dependencies and ambiguous definitions. But none is a substitute for the others. In particular, a formal proof of a conditional theorem remains conditional, and a billion-digit period computation remains numerical.

4.10 Epistemic tags and the logic of the book

Every substantial formal assertion carries one of four tags. `\KNOWN` means that the statement is an established theorem or is proved in the text. `\COND` means that the statement is proved from explicitly named conjectures or axioms not presently known. `\NEW` means that the definition, construction, or theorem is introduced in this volume; it does not mean that the statement is speculative, since a new theorem may have an unconditional proof. `\OPEN` means that no proof is claimed. The tags answer two different questions: provenance and logical status. A new definition can be perfectly rigorous, while an old conjecture remains open.

The tags exist because long conditional arguments are visually dangerous. A reader may remember the conclusion but forget the hypothesis after several chapters. In work surrounding the Hodge Conjecture this danger is acute: standard conjectures, fullness statements, positivity assumptions, and variational statements can each make an argument look complete. The tags force the missing premise to recur at the point of use. They are therefore a mathematical device for maintaining a dependency invariant, not editorial ornament.

Criterion 4.10.1 (Epistemic admissibility). *[NEW] A theorem-sized claim is admissible in this volume only if it has exactly one epistemic tag, all non-trivial hypotheses are stated, and either a proof is supplied or a verified attribution with author and year is given. An implication whose premise is open must be tagged `\COND` even if the deduction from that premise is elementary.*

Proof. The criterion is constitutive of the proof ledger rather than a theorem about external mathematics. Its enforcement is mechanical at the level of source review and semantic at the level of hypotheses. The distinction between `\NEW` and `\OPEN` is essential: the master equivalence is a new theorem about open conjectures, whereas the conjectures themselves remain open. □

Necessity of explicit hypotheses. [KNOWN] [Kle68] Standard Conjecture B would provide algebraic inverse Lefschetz operators and shorten the trace argument, but Kleiman proves HC $\Rightarrow$ B. Deleting the hypothesis from a theorem proved using B would therefore disguise a conditional route as an unconditional one. Chapter 46 consumes the criterion when assigning statuses to the prism axioms, Chapter 47 consumes it in the noncircularity audit, and Chapter 66 represents the same dependency as a typed obligation.

One consequence is that "HC is equivalent to RC" and "RC is true" have visibly different statuses. Another is that the Cattani–Deligne–Kaplan theorem, despite its extraordinary strength, is not described as a partial proof that individual Hodge classes are algebraic. It proves algebraicity of the parameter locus on which a class remains Hodge; it does not produce a cycle on the fibres. Similarly, Deligne's absolute Hodgeness theorem for abelian varieties is recorded as [KNOWN] [Del82] without being allowed to mutate into the false statement that the Hodge Conjecture is known for abelian varieties.

4.11 Where projectivity is consumed

The architecture is constrained by Voisin's compact Kähler counterexample. A smooth projective variety has an integral ample class, whereas a compact Kähler manifold need only have a real Kähler class and need not possess an algebraic or even definable model. The volume localises these distinctions rather than invoking projectivity diffusely in every chapter.

Principle 4.11.1 (Localisation of projectivity). *[NEW] In the main analytic route, projectivity is consumed exactly at the positivity-matching step of Theorem 36.2.3: an ample integral class is used to transport a rigidified*

Hodge datum to a positive closed (p,p)-current of the required rational type. Earlier tame constructions prepare the category in which this step can be stated, but they do not themselves prove the positivity match.

Necessity of the ample integral class. **[KNOWN]** [Voi02a] On $X = \mathbb{P}^1 \times \mathbb{P}^1$, with fibre classes u, v, one has

$$u^2 = v^2 = 0, \qquad u \cdot v = 1, \qquad [\,\cdot\,]_{\langle u,v\rangle} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The class $u + v = c_1(\mathcal{O}(1,1))$ is integral and ample, and $(u+v)^2 = 2$; it is exactly the kind of rational positivity certificate used by Theorem 36.2.3. A general compact Kähler manifold need only supply a real Kähler class, and Voisin's counterexample shows that replacing the integral ample input by that weaker datum cannot preserve the conclusion. Chapter 36 consumes the distinction in the projectivity firewall, and Chapter 44 consumes it in the current machine.

The statement should be read together with a subtlety. Definability enters earlier and is obtained from algebraic input; a general compact Kähler manifold has no canonical definable model of the kind used here. This is a categorical barrier, verified in Chapters 6 and 10. The unique *positive* use of projectivity in the proof-shaped analytic mechanism, however, is the ample-class step of Chapters 36 and 44. Distinguishing these roles keeps the Voisin Test sharp: definability restricts the domain of the framework, while integral ampleness supplies the positivity that the analytic machine needs.

After positivity, Bishop's theorem and Chow-type algebraisation can turn analytic support with controlled mass into algebraic support; Bishop's analyticity criterion is an established theorem (Bishop 1964) [Bis64]. But a rational Hodge class supplies a closed real (p,p)-form, not a positive one. The move into the positive cone is O2 and is open. The book does not conceal that gap behind the phrase "choose a positive representative".

Denominators are likewise localised. The comparison of the glued complex datum with the Betti lattice can require passage to a finite-index sublattice, and the resulting rational cycle carries a denominator. This is the lattice-comparison step of Theorem 38.5.2 and the degree-two obstruction O4. Since the integral Hodge Conjecture is false, bounded denominators are not an imperfection to be removed globally; they are a necessary feature of any viable rational mechanism.

4.12 What this chapter does not do

This chapter does not construct an algebraic cycle from a general Hodge class. It does not prove the Rigidification Conjecture, the Trace–Hodge Conjecture, the Concentration Principle, or Universal Kuga–Satake. It does not provide the missing positive representative in the twistor–current machine. Chapter 38 does control and discharge the denominator mechanism, but only modulo the already open rigidification entry. Consequently it does not prove the Hodge Conjecture.

What the chapter does achieve is a typed map of the problem. It specifies the deformation on which descent is to be studied, distinguishes effective descent from existence of descent data, relates four conjectural faces, and gives a finite obstruction ledger relative to explicit axioms. It also ensures that known tame and period-theoretic theorems can be used at their full strength without being asked to produce algebraic cycles they do not produce.

The architecture may fail in at least three informative ways. The prism may be too coarse to distinguish algebraic from non-algebraic Hodge classes. Rigidification may not exist for every class, in which case RC is false and the central chain breaks. Or rigidification may exist but fail to yield positivity, leaving O2 open; the support entry O3 introduces no independent gap after the polarisation reduction of Chapter 48. Each failure would refute a specific component rather than dissolve the whole book into an untestable programme.

Status of Chapter 4

[NEW] The architectural formulation of algebraicity as descent, the three-register discipline, and the integrated obstruction map are new organising structures. [KNOWN] [Lef24; CDK95; BKT20; Bis64; Kin71; Voi02a; AH62; Tot97] The filtration construction used here, Lefschetz $(1,1)$, CDK algebraicity of Hodge loci, BKT definability, Bishop–King analyticity, and the two counterexample walls are established results. [OPEN] The four central conjectures remain open. In the ledger, only O1 and O2 remain open as independent obstructions; O3–O9 are discharged in the sense of Theorem 48.6.1, with O4 discharged specifically by Corollary 38.7.1. No claim of a proof of the Hodge Conjecture is made.

Notes and References

The historical source of the problem is Hodge's theory of harmonic integrals and his formulation of the conjecture in the 1950 International Congress address; see Hodge (1941, 1952) [Hod41; Hod52]. The Lefschetz $(1,1)$-theorem is the indispensable model for the successful passage from Hodge type to algebraic geometry in codimension one; see Lefschetz (1924) [Lef24].

The two walls are theorem-level constraints, not philosophical cautions. Atiyah–Hirzebruch (1962) and Totaro (1997) provide failures of integral algebraicity [AH62; Tot97]. Voisin (2002) constructs a compact Kähler counterexample to the extension of the rational Hodge statement [Voi02a]. The localisation of projectivity and denominators in this volume is designed to make these counterexamples visible in the dependency graph.

The Rees deformation belongs to standard filtered algebra and underlies many forms of Hodge-to-de Rham interpolation. Simpson's work on Higgs bundles and local systems gives the nearby λ-connection picture used later in the analytic engine; see Simpson (1992) [Sim92]. The word "prism" is used structurally. For the arithmetic theory of prisms and prismatic cohomology, which is categorically different from the present construction, see Bhatt–Scholze (2022) [BS22].

The algebraicity of Hodge loci is due to Cattani–Deligne–Kaplan (1995) [CDK95]. Bakker–Klingler–Tsimerman (2020) place the relevant period geometry in the o-minimal category and obtain a tame proof [BKT20]. These theorems motivate, but do not establish, the rigidification conjecture.

The categorical route uses the semisimplicity of numerical motives proved by Jannsen (1992) [Jan92]; it supplies a canonical semisimple numerical quotient. Injectivity of Hodge realisation is instead definitional after passing to homological equivalence, and neither fact implies fullness. The distinction between absolute Hodge classes and algebraic classes is especially important for abelian varieties; Deligne (1982) proves the former property, not the Hodge Conjecture [Del82].

The obstruction ledger, the master equivalence, and the three-register protocol are internal contributions of this volume. Their merit must be judged by whether they expose genuine proof obligations and survive the falsification tests of Chapters 68 and 69. They should not be read as a claim that naming an obstruction removes it. The central mathematical work left open is exactly the construction or refutation of rigidification and the positivity and support bridges attached to it.

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